

# Technical Validation Supplement

Measurement of  $\nu_\mu + \bar{\nu}_\mu$  Charged-Current Single Pion Production on Argon in the NuMI Beam

Companion to the analysis note

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## Abstract

This supplement holds the validation and diagnostic material for the  $\text{CC}\pi^\pm$  NuMI cross-section measurement: the studies that establish what the measurement can and cannot resolve, the estimators that were tried and rejected, and the closure tests that qualify the extraction. It is separated from the analysis note so that the note can state the analysis and its results while the studies that justify its choices remain available and citable at full length.

The two documents are checked for numerical consistency where current results are duplicated (`report/check_tables.py`, `xsec_analyzer/check_staleness.py`); every number here is from the release named on the title page unless a table says otherwise. The history of the analysis, the corrections that shaped it and the values before and after, is in the change log and is not repeated here.

The central results of this supplement are negative ones, and they set the scope of the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near  $0.34 \text{ GeV}/c$  and carries little information about the initial momentum above roughly  $0.4 \text{ GeV}/c$ . Every alternative estimator tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — fails to recover it, and a truth-level golden selection performs no better than the classifier, which places the limitation in the physics rather than in the estimator. This is what reduces  $p_\pi$  to two bins and  $W_{\pi p}$  to an integrated cross section.

# 1 What this document is for

The analysis note says what was measured and why the choices were made. This supplement holds the studies behind those choices, so that a reader who wants to check one of them can, without the note having to carry them. Nothing here is needed to follow the note, and every study here is pointed at from the place in the note where its conclusion is used.

It is organised by question rather than by when the work was done.

- **Estimators and binning:** why the pion momentum is reported in two regions and what was tried to do better; how every binning was chosen and re-tested; the proton-tagged observables and their additional smearing; the muon angle in its two parameterisations.
- **Validation of the extraction:** the identity closure, what the closure  $\chi^2$  does and does not test, the ensemble study, and the consistency between configurations and with the sibling  $\nu_e$  measurement.
- **Control regions:** the definitions, the frozen procedure under which they were opened, the transfer factors and their uncertainties.
- **Selection and infrastructure:** the selection optimisation, cuts considered and rejected, and the automated guards on the pipeline.
- **Physics background:** the production mechanisms, nuclear effects and prior measurements, at greater length than the note allows.

Terms of art are defined in the Glossary at the end, which is shared with the note.

## Part I

# Estimators and binning

## 2 Pion momentum: the range estimator and why it fails<sup>[INCLUSIVE]</sup>

### 2.1 The pion-momentum bias and why $p_\pi$ is the worst-conditioned observable

The  $p_\pi$  response is the least diagonal of any observable in the analysis. This section quantifies why, because the cause is physical and sets a floor on what the measurement can say about the pion momentum spectrum.

#### 2.1.1 Two separate effects, only one of them corrected.

The reconstructed pion momentum stored in the ntuple, `candidate_pion_mom_reco`, is `trk_range_muon_mom_v`: a range-based momentum evaluated under the *muon* mass hypothesis and applied to a pion track. Two distinct things follow.

First, the mass hypothesis itself biases the estimate low, because for a given range a pion carries more momentum than a muon. The size is  $\sim 17\text{--}19\%$  from the CSDA range scaling, and it is essentially independent of momentum. This effect *is* corrected, though not in the branch: the reco bin definitions in `ccpi_ppi_bin_config_opt.txt` apply a kinetic-energy-preserving  $\mu \rightarrow \pi$  mass swap,  $E_\mu \rightarrow T \rightarrow E_\pi \rightarrow p_\pi$ , before binning. It works; in the lowest true bin  $\langle p^{\text{reco}}/p^{\text{true}} \rangle$  goes from 0.888 with the raw branch to 0.976 with the correction.

Second, and not corrected by anything, charged pions interact hadronically before coming to rest. The reconstructed track then ends at the interaction point rather than the stopping point, so the visible range, and any momentum inferred from it, is truncated. This grows rapidly with momentum and comes to dominate.

### 2.1.2 The reconstructed momentum saturates.

Table 1 and Figure 1 make the consequence plain: as the true momentum rises by a factor of four and a half, the reconstructed momentum barely moves, saturating at  $\approx 0.34$  GeV/ $c$ . Above that the estimator carries almost no information about how energetic the pion actually was.

Table 1: Pion momentum reconstruction versus truth, selected signal, Run-1 FHC. “raw” is the `candidate_pion_mom_reco` branch; “corr.” additionally applies the  $\mu \rightarrow \pi$  mass swap used by the reco bin definitions. The mass correction removes the bias at threshold; nothing removes the saturation above it.

| true $p_\pi$ [GeV/ $c$ ] | $N$ | $\langle p^{\text{true}} \rangle$ | $\langle p^{\text{corr}} \rangle$ | raw/true | corr./true |
|--------------------------|-----|-----------------------------------|-----------------------------------|----------|------------|
| 0.175–0.250              | 528 | 0.210                             | 0.205                             | 0.888    | 0.976      |
| 0.250–0.300              | 270 | 0.276                             | 0.238                             | 0.790    | 0.863      |
| 0.300–0.350              | 254 | 0.325                             | 0.253                             | 0.716    | 0.779      |
| 0.350–0.400              | 265 | 0.374                             | 0.270                             | 0.664    | 0.721      |
| 0.400–0.500              | 403 | 0.443                             | 0.279                             | 0.581    | 0.629      |
| 0.500–0.600              | 252 | 0.548                             | 0.307                             | 0.520    | 0.560      |
| 0.600–0.800              | 272 | 0.688                             | 0.326                             | 0.441    | 0.473      |
| 0.800–1.200              | 164 | 0.951                             | 0.336                             | 0.329    | 0.353      |

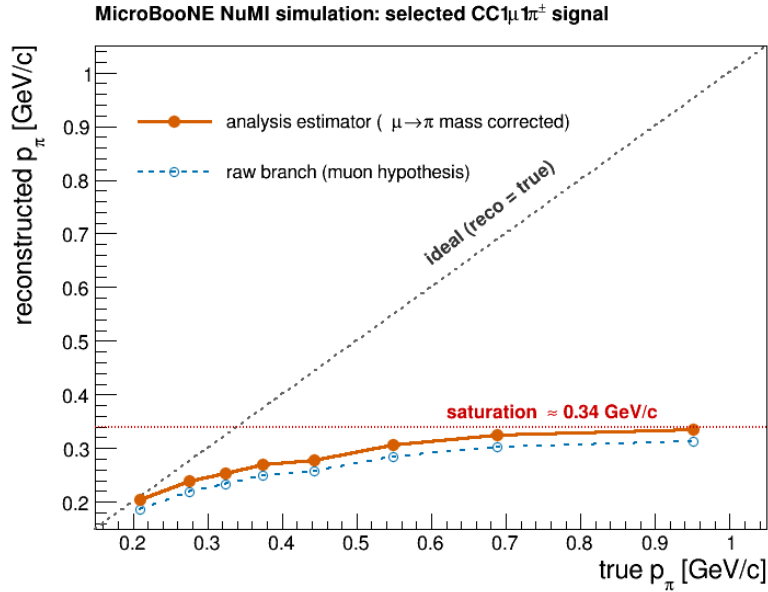


Figure 1: Mean reconstructed against mean true pion momentum, selected signal. The analysis estimator (filled,  $\mu \rightarrow \pi$  mass corrected) tracks truth at threshold but flattens out near 0.34 GeV/ $c$ : a true 0.95 GeV/ $c$  pion reconstructs at the same place as a true 0.55 GeV/ $c$  one. The raw `candidate_pion_mom_reco` branch (open) sits systematically below it by the uncorrected mass-hypothesis offset. The saturation is caused by hadronic interaction truncating the track range, and no range-based estimator can recover it.

### 2.1.3 What that does to the response matrix.

Migrating the five analysis bins with the corrected estimator gives a nearly lower-triangular matrix: the diagonal fraction falls from 94.3% in the first true bin to 36.5, 23.1, 10.4 and finally

6.2% in the last. For the highest true bin ( $p_\pi > 0.55 \text{ GeV}/c$ ) only 6.2% of events are reconstructed in the corresponding reco bin while 32.6% fall all the way into the *lowest* one. Every true bin above the first therefore deposits most of its events into reco bin 1, which is exactly the “pile-up into the first bin” seen in Fig. 18 of the analysis note.

#### 2.1.4 How the BNB CC1 $\pi^\pm$ analysis handles the same problem.

The published BNB CC1 $\pi^\pm$ Xp analysis (internal note v0.91) faces an identical estimator, it too takes “the momentum of pion candidates from range”, and its treatment is worth reproducing here, because it is the recipe this analysis followed.

That analysis does three things we do not. First, it defines a *golden pion*: one that comes to rest and leaves a Bragg peak, as opposed to one that scattered or interacted. Second, it trains a dedicated *golden-pion BDT* and applies it as an extra cut for the pion-momentum differential cross section only, leaving the total and angular results on the generic selection. The BDT separates golden from non-golden pions using track “wiggleness” (the standard deviation of the angular difference between directions of adjacent points along the fitted track, so a direct handle on scattering), the number of reconstructed descendant particles, the track–shower score, and the Bragg likelihood ratio. The cost and benefit are quoted explicitly: efficiency falls from 20.3% to 9.5% and purity is essentially unchanged (51.9  $\rightarrow$  49.5%), but the golden fraction of the selected signal rises from 36.3% to 67.7%. They apply the same logic to the muon momentum, restricting that cross section to contained muons (8.4% efficiency, 56.1% purity) so it uses range only.

Third, and most directly comparable, their binning is chosen against an explicit criterion: each bin must hold at least 100 predicted MC events *and* the diagonal element of the smearing matrix must exceed 0.68. The resulting pion-momentum edges are 0.10, 0.16, 0.19, 0.22, 0.60 GeV/ $c$  with an overflow bin, fine structure only where the estimator still tracks truth, then a single wide bin from 0.22 to 0.60 and everything above 0.60 in overflow.

Set against that criterion our binning does not survive: the diagonal fractions measured above are 94.3, 36.5, 23.1, 10.4 and 6.2%, so *one* of five bins passes 0.68 and the top three fail it by a wide margin. Our edges (0.175–0.550 in five bins, open above) put four boundaries inside precisely the region where the reconstructed momentum has saturated.

#### 2.1.5 Consequences, and what would improve it.

This is not a bias on the extracted cross section; the response matrix contains it and the unfolding inverts it, but it is the reason the  $p_\pi$  result leans harder on the regularisation than any other observable. With 6% diagonal strength in the top bin, the measurement above  $\sim 0.5 \text{ GeV}/c$  is effectively inferred from the response model rather than measured.

The BNB note therefore supplies a tested recipe, and *all four* of its BDT inputs are already available to us. The Bragg likelihood ratios (`trk_bragg_pion`, `_mip`, `_mu`, `_p`) and the track–shower score are in the ntuples; the multi-pion pion-ID BDT in this analysis already trains on exactly those, and the descendant counts are `pfp_trk_daughters_v` / `pfp_shr_daughters_v`. Wiggleness is there too, under a different name: the PeLEE ntuples carry `trk_avg_deflection_stdev_v` (with companions `_mean_v` and `_separation_mean_v`), which is precisely the standard deviation of the angular deflection between adjacent track points. It is fully populated and is carried forward by `ProcessNTuples`.

#### 2.1.6 Wiggleness works, but with the opposite sign to the naive expectation.

Testing it directly on truth-matched charged pions in the Run-1 FHC overlay, calling a pion “golden-like” when the mass-corrected range momentum recovers the true momentum to within 20%, and “interacting” when it recovers less than 60%, gives a clear separation, but golden pions are the *wigglier* ones, by about a factor two in the mean (0.0048 against 0.0027). That is not a

momentum artefact: it survives binning in track length, with the ratio staying at 0.51–0.63 from 5 cm to 400 cm.

The sign makes physical sense on reflection. A pion that stops travels its full range and scatters violently through its final slow centimetres, so the deflection spread averaged along the track is large. A pion that interacts hadronically is cut short while still fast and never reaches that wiggly end, so its track is straighter. High wiggleness is thus a signature of *having stopped*, exactly the population for which a range-based momentum is meaningful.

As a single cut it is already useful: on this sample the golden fraction is 39.9% before any cut, and requiring `trk_avg_deflection_stdev_v` > 0.001 keeps 86.8% of golden pions while rejecting 48.8% of interacting ones, lifting the golden fraction to 53.0%. A tighter cut at 0.002 gives 57.8% efficiency and 55.7% golden. For reference the BNB’s four-variable BDT reaches 67.7% golden at roughly half the generic-selection efficiency, so one variable recovers a good part of the gain and the remaining three, all of which we have, should close much of the rest.

Three options follow from this, in increasing order of effort: rebin  $p_\pi$  against the BNB’s own criterion (diagonal > 0.68), which on the numbers above means one resolved bin at threshold plus a wide open bin; add a golden-pion cut for the  $p_\pi$  cross section built from the variables already available, accepting the factor-two efficiency loss the BNB analysis accepts; or reproduce their BDT in full. None changes the total or angular results, which do not use this estimator. The first is the adopted scheme (§2.1.6); the classifier was trained and tested and found not to change the conclusion (§2.1.7).

### Adopted response: a two-bin $p_\pi$ scheme

The released binning is two bins, [0.175, 0.205] GeV/ $c$  and everything above, which raises the migration diagonals from 96/35/21/10/6% (full analysis sample; 94.3/36.5/23.1/10.4/6.2% on Run 1 alone, Table 1) to **91/76%**, clearing the 68% criterion in *both* bins. (These are measured on the analysis sample; the golden-pion study of Sec. 2.1.7 uses a looser track sample whose diagonals are lower, and the two sets are not interchangeable.)

The boundary is not a free choice dressed up as an optimisation. The worst diagonal is always the upper bin and it falls monotonically as the boundary rises, so the binding constraint is the *width* of the first bin rather than the diagonal itself:

| boundary     | first-bin width              | diagonal, bin 1 | diagonal, bin 2 |
|--------------|------------------------------|-----------------|-----------------|
| 0.185        | 10 MeV/ $c$                  | 79.7%           | 85.4%           |
| 0.195        | 20 MeV/ $c$                  | 88.8%           | 80.0%           |
| 0.200        | 25 MeV/ $c$                  | 90.4%           | 77.7%           |
| <b>0.205</b> | <b>30 MeV/<math>c</math></b> | <b>91.0%</b>    | <b>75.5%</b>    |
| 0.220        | 45 MeV/ $c$                  | 93.3%           | 69.1%           |
| 0.250        | 75 MeV/ $c$                  | 95.6%           | 56.1%           |
| 0.300        | 125 MeV/ $c$                 | 96.3%           | 39.0%           |

Both bins clear the criterion anywhere between roughly 0.19 and 0.22, so the choice within that window is governed by bin width rather than by the criterion. 0.205 is the narrowest first bin with precedent: 30 MeV/ $c$  matches the narrowest bin used in the BNB CC1 $\pi$  analysis. Going narrower trades first-bin diagonal for upper-bin diagonal and at 10 MeV/ $c$  the first bin itself falls to 80% on only 325 events, which is not a useful cross-section bin; going wider, 0.220 still passes but leaves the upper bin at 69%, with no margin. An exhaustive scan over all edge sets (`macros/ppi_binning_scan.C`) returns 0.205 as the best first boundary for every number of bins tried, and for the proton-tagged selection as well.

A three-bin variant was built and measured before being discarded. The best achievable (same 30 MeV/ $c$  minimum width, edges 0.205 and 0.265) is 91/51/52%, against 91/76% for two bins: the third bin costs 25 points on the worst bin and drops it well below the criterion. Two

30 MeV/ $c$  bins followed by an open bin (0.205, 0.235) is worse still at 91/39/62%, and note that it degrades the *upper* bin too, from 76% to 62%: narrowing the top bin’s reco range pushes genuinely high-momentum events out of their own bin. Four bins reach only 91/39/39/42%. Above  $\sim 0.25$  GeV/ $c$  the reconstructed momentum barely varies, so a second boundary there separates almost nothing and merely divides one poorly-measured region into two equally poor halves. The same ordering holds under a golden-pion cut (67/52/46%), so it is not an artefact of the selection.

### 2.1.7 Closure.

The two-bin scheme is carried through the full chain, universe construction, Wiener–SVD unfolding and systematics, and closes in all three configurations (Table 2); a five-bin extraction on the same sample closes too, at  $\chi^2 \approx 2/5$ , which is the point: closure cannot expose a response that carries no shape information, only the migration criterion can.

Table 2: Two-bin  $p_\pi$  result for all three beam configurations,  $d\sigma/dp_\pi$  in  $10^{-38}$  cm<sup>2</sup>/(GeV/ $c$ )/Ar. Bin 1 is [0.175, 0.205] GeV/ $c$ , bin 2 everything above. All curves are  $A_C$ -smeared, so they live in the same space as the measurement. Generated from the release (`report/tools/ppi2bin_tables.py`: closure sidecars and `curves_incl_*_ppi2bin.tsv`); the uncertainty row is the total.

|                     | FHC                   |            |                       | RHC                   |            |                       | Combined              |            |                       |
|---------------------|-----------------------|------------|-----------------------|-----------------------|------------|-----------------------|-----------------------|------------|-----------------------|
|                     | bin 1                 | bin 2      | $\sigma_{\text{int}}$ | bin 1                 | bin 2      | $\sigma_{\text{int}}$ | bin 1                 | bin 2      | $\sigma_{\text{int}}$ |
| unfolded data       | 1.96                  | 0.96       | 0.822                 | 1.79                  | 0.87       | 0.748                 | 2.06                  | 0.90       | 0.775                 |
| uncertainty         | $\pm 1.36$            | $\pm 0.37$ |                       | $\pm 1.12$            | $\pm 0.34$ |                       | $\pm 1.26$            | $\pm 0.34$ |                       |
| $A_C$ truth         | 2.34                  | 0.95       |                       | 1.71                  | 0.89       |                       | 2.17                  | 0.90       |                       |
| uB tune             | 2.29                  | 0.93       |                       | 1.72                  | 0.88       |                       | 2.15                  | 0.89       |                       |
| GENIE               | 1.82                  | 0.75       |                       | 1.29                  | 0.63       |                       | 1.67                  | 0.67       |                       |
| GiBUU               | 3.28                  | 0.94       |                       | 2.27                  | 0.82       |                       | 3.03                  | 0.85       |                       |
| NEUT                | 3.09                  | 1.18       |                       | 2.24                  | 1.02       |                       | 2.87                  | 1.07       |                       |
| NuWro               | 2.98                  | 1.12       |                       | 2.11                  | 0.96       |                       | 2.74                  | 1.01       |                       |
| $\chi^2/\text{ndf}$ | 0.11/2 ( $p = 0.95$ ) |            |                       | 0.01/2 ( $p = 0.99$ ) |            |                       | 0.01/2 ( $p = 1.00$ ) |            |                       |

On the current release all three configurations close, with  $\chi^2$  of 0.11, 0.015 and 0.01 on two bins (the RHC value rounds either way in the note’s tables) and unfolded-to-truth integral ratios of 0.997, 0.981 and 0.989 (width-weighted unfolded over  $A_C$ -smeared truth from the released curves, `curves_incl_*_ppi2bin.tsv`; the  $\chi^2$  from Tables 17–19 of the analysis note). The four generators bracket the data in both bins of every configuration, NEUT and GiBUU highest and GENIE lowest, the same ordering as in the other observables; the released  $A_C$  for FHC is  $\begin{pmatrix} 0.53 & 0.01 \\ -0.51 & 0.87 \end{pmatrix}$  (`A_C_incl_FHC5_ppi2bin.tsv`), whose off-diagonal element is the reminder that the two bins are not independent after merging. The total fractional uncertainty is 50% (FHC), the largest of the inclusive observables, because the two bins each carry a large share of the flux normalisation with little statistical averaging. Figure 3 shows  $A_C$  for all three configurations. RHC regularises a little harder than FHC in the first bin (diagonal 0.38 against 0.53, off-diagonal  $-0.37$  against  $-0.51$ ), and the combined extraction borrows the most (diagonal 0.53, off-diagonal  $-0.58$ ); the second-bin diagonals are 0.87–0.88 in all three. Despite the different conditioning the three extractions close in normalisation within 2% (the ratios 0.997, 0.981 and 0.989 above), so there is no common normalisation offset associated with how hard  $A_C$  smears.

At 91/76% both bins clear the 68% criterion, so the two-bin scheme makes the inclusive  $p_\pi$  result usable on the standard test rather than merely honest about its limitations. It remains a coarse measurement: two bins cannot resolve a shape, and above  $\sim 0.34$  GeV/ $c$  the reconstructed momentum saturates, so the upper bin is an integral over a wide range rather than a differential

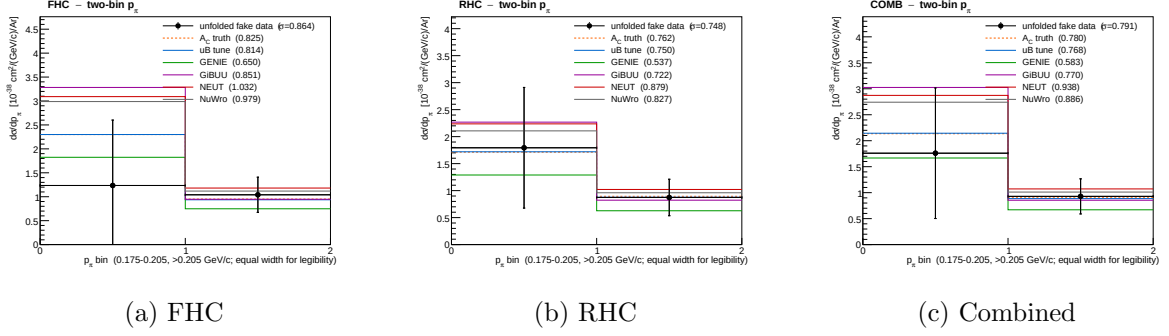


Figure 2: The two-bin  $p_\pi$  cross section, the graphical form of Tab. 2. Unfolded fake data (points, with the full systematic uncertainty) against the  $A_C$ -smeared truth, the uB tune and the four generators; every curve is  $A_C$ -smeared, so all of them live in the same space as the measurement. **The two bins are drawn with equal width for legibility:** they are 0.030 and 0.795 GeV/ $c$  wide, so to scale the first would occupy 4% of the axis. Area on these panels is therefore not proportional to cross section, and the integrated  $\sigma$  is printed on each panel instead. The generators bracket the data in every bin of every configuration; GENIE is generally the lowest prediction and NEUT and GiBUU the highest, and which generator is closest changes from bin to bin.

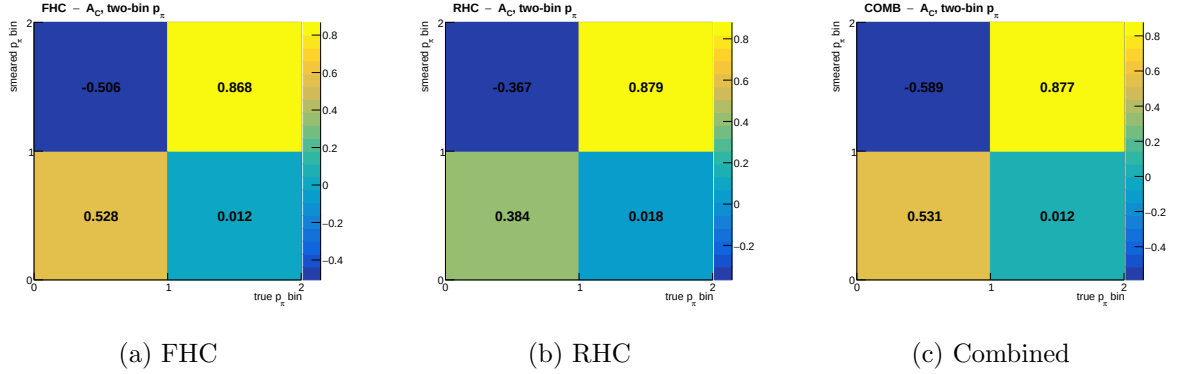


Figure 3: The additional-smearing matrix  $A_C$  for the two-bin  $p_\pi$  scheme, in the convention  $\hat{x}_i = \sum_j (A_C)_{ij} t_j$  so that a column is a true bin and a row is the smeared bin it feeds.  $A_C$  is not normalisation-preserving, so its rows do not sum to unity. The negative entry (true bin 1 feeding smeared bin 2, drawn top-left since the smeared axis runs upward) is the regularisation borrowing between the two bins; it is why the truth curves in Fig. 2 must be  $A_C$ -smeared before being compared with the measurement rather than plotted raw.

point. What the scheme buys is a result whose response is defensible, not a competitive shape measurement.

### A golden-pion classifier was trained and tested: it works, and it is not enough

Following the BNB recipe, a golden-pion BDT was trained and evaluated (`macros/golden_pion.train.C`, `macros/golden_pion.compare.C`). The truth label is `mc_end_p < 0.01 GeV/c`, the pion ranged out, joined to reconstruction by matching `backtracked_p` to `mc_p`, which pairs 100% of candidate tracks with no ambiguity. The inputs are the three deflection variables, the four Bragg likelihood ratios, the track–shower score and the descendant count. Training used 126 518 labelled tracks from Runs 1, 2 and 4; the split is by *run*, so all numbers below are on Run 2, which the classifier never saw.

#### 2.1.8 As a classifier it reproduces the BNB benchmark.

ROC integral 0.758. At 44.8% efficiency it lifts the golden fraction from 43.1% to 64.9%, against the BNB’s  $36.3 \rightarrow 67.7\%$  at roughly 47% efficiency. The variable ranking is dominated by the deflection variables (`wmean`, then `wstd`), with the Bragg ratios contributing essentially nothing, worth noting, since Bragg was expected to carry the stopping signature.

#### 2.1.9 But it does not fix the binning, and cannot.

Table 3 gives the migration diagonal in the five analysis bins as the cut is tightened. The first bin is already fine and stays fine; no other bin reaches 0.68 at any working point, even at 15% efficiency.

Table 3: Golden-pion BDT: efficiency, achieved golden fraction, and the  $p_\pi$  migration diagonal per true bin, on the held-out run. The BNB criterion for a usable bin is a diagonal above 0.68.

| BDT cut                                 | eff.  | golden frac. | b1   | b2   | b3   | b4   | b5   |
|-----------------------------------------|-------|--------------|------|------|------|------|------|
| none                                    | 100%  | 43.1%        | 99.2 | 23.7 | 13.2 | 6.4  | 5.5  |
| $> -0.05$                               | 57.0% | 58.1%        | 99.1 | 29.4 | 19.7 | 11.2 | 7.1  |
| $> 0.00$                                | 44.8% | 64.9%        | 99.1 | 33.1 | 24.5 | 15.4 | 9.1  |
| $> +0.10$                               | 24.6% | 79.4%        | 99.1 | 42.9 | 36.8 | 29.3 | 15.7 |
| $> +0.15$                               | 15.2% | 84.5%        | 99.2 | 46.9 | 42.2 | 36.1 | 19.5 |
| <i>perfect (truth) golden selection</i> |       |              | 98.9 | 36.1 | 24.1 | 14.2 | 8.9  |

#### 2.1.10 These diagonals are measured on a different sample from the analysis ones, and the two must not be mixed.

The table above is built on the labelled *track* sample: every truth-matched pion track passing loose quality requirements (purity  $> 0.5$ , track score  $> 0.5$ , length  $> 5$  cm), from events of any topology, 126 518 tracks in total. The migration diagonals quoted elsewhere in this note, 96/35/21/10/6% for five bins and 91/76% for two, are measured on the *analysis* sample: events passing the full  $CC1\mu1\pi^\pm$  selection and satisfying the signal definition, about 10 700 per beam mode. The analysis sample has the better response, because the selection removes exactly the events whose pion is poorly reconstructed. On the track sample the same five-bin scheme reads 99/24/13/6/6% and the two-bin scheme 98/62%.

Those track-sample figures are not the analysis response, and the 62% must not be read as the inclusive  $p_\pi$  measurement failing the usable-bin criterion. It does not: on the analysis sample



the upper bin sits at 76%. The numbers in this table are correct for what they measure, and the conclusions drawn *within* it, which compare rows of the same sample, are unaffected.

The last row is the decisive one, and it must be read against the *efficiency* column rather than against the tightest cut. Perfect truth-level golden selection keeps 43.1% of tracks and gives 36.1, 24.1, 14.2, 8.9% in bins 2–5; the classifier at the matching 44.8% efficiency gives 33.1, 24.5, 15.4, 9.1%. They agree bin by bin. The classifier is therefore already performing as well as perfect knowledge of which pions stopped, and no better classifier exists to be built.

### 2.1.11 Multiple Coulomb scattering was tried as a pion momentum estimator, and as a classifier input.

Range saturates because the pion interacts before stopping, which truncates the track. Multiple-Coulomb-scattering (MCS) momentum is inferred from the scattering along whatever track exists and is not truncated by an interaction, so it is the natural candidate to replace range. The CC0 $\pi$  analysis uses exactly this pairing, range for contained tracks and MCS for exiting ones. Measured on 122 878 truth-matched pion tracks (`trk_mcs_muon_mom_v`, filled for 99.9% of them):

| true $p_\pi$ [GeV/c] | $\langle \text{len} \rangle$ | $\langle \text{range/true} \rangle$ | RMS  | $\langle \text{MCS/true} \rangle$ | RMS  |
|----------------------|------------------------------|-------------------------------------|------|-----------------------------------|------|
| 0.175–0.250          | 25 cm                        | 0.834                               | 0.16 | 4.335                             | 14.1 |
| 0.320–0.420          | 46 cm                        | 0.629                               | 0.23 | 2.438                             | 7.2  |
| 0.550–0.800          | 71 cm                        | 0.446                               | 0.18 | 1.423                             | 3.5  |
| 0.800–1.500          | 84 cm                        | <b>0.319</b>                        | 0.16 | <b>1.011</b>                      | 2.1  |

The trade is stark. In the top bin range is biased to 0.319, which is the saturation, while MCS is essentially unbiased at 1.011; but MCS carries a 210% RMS there and 1400% in the lowest bin. It is unusable as a per-event estimator. The reason is structural and explains why the CC0 $\pi$  recipe does not transfer: they apply MCS to *exiting* tracks, which are long by construction, whereas our problematic pions are contained but interacting, 25–84 cm long, and MCS needs of order a metre.

The same note uses the *disagreement* between the two estimators as a track-quality BDT input rather than as an estimator. That is well motivated here: a pion that interacts has a truncated range but an untruncated MCS, so the fractional difference  $(p_{\text{range}} - p_{\text{MCS}})/p_{\text{range}}$  should flag precisely the events that break the response. Adding it to the golden-pion classifier, it ranks *second of ten* in importance, behind only the mean deflection. It nonetheless changes nothing:

| classifier inputs         | ROC integral | diagonals at $\sim 44\%$ efficiency |
|---------------------------|--------------|-------------------------------------|
| baseline (nine variables) | 0.758        | 99.1/33.1/24.5/15.4/9.1             |
| + range–MCS difference    | 0.757        | 99.0/34.8/24.8/14.8/9.1             |

The interpretation is that the new variable is highly informative but not *additionally* informative: it is strongly correlated with the deflection variables already present, since both measure whether the track scattered heavily and ended abruptly. High importance rank with unchanged ROC is the signature of a redundant input. This is an independent confirmation of the ceiling argument above: the classifier already extracts what golden-ness has to offer, and the limit is the physics of the label rather than the quality of the classifier.

One tempting variant was deliberately not pursued. Retraining on the label “range is accurate”,  $|p_{\text{corr}}/p_{\text{true}} - 1| < 0.2$ , would target the quantity we actually care about and would certainly raise the diagonals. It would also be selecting on the answer: keeping events where the estimator happens to be right sculpts the response by construction and inflates the diagonal without measuring anything, the same trap documented for the tight BDT working point above. The golden label is used precisely because it is a physical property of the pion, independent of how well it was reconstructed.

### 2.1.12 A caveat on the tighter working points.

The  $> 0.15$  row looks better than perfect truth selection (46.9, 42.2, 36.1, 19.5%), which is impossible for an honest classifier and is worth understanding rather than quoting. It arises because a hard cut on the classifier preferentially keeps *long* tracks: inside the top true bin, the surviving tracks have a mean length of 120.8 cm against 78.2 cm for all pions, and a mean reconstructed momentum of 0.406 against 0.307 GeV/c. The cut is thus partly a cut on the reconstructed momentum itself, which raises the diagonal by construction while sculpting the spectrum. The apparent gain is circular and the tight working points should not be used. This is the same failure mode anticipated for track length as a direct input; it enters here through the deflection variables, which correlate with length, even though length itself was excluded from training.

### 2.1.13 Why, and what is left.

Golden pions still reconstruct at  $\langle p^{\text{corr}}/p^{\text{true}} \rangle = 0.73$  on this sample. Interaction is therefore not the only problem: the momentum conversion is itself biased. A single flat rescaling does *not* repair it, applying the best global factor drops the first bin from 98.9 to 49.1% and leaves 0/5 bins passing; because the bias is momentum-dependent, exactly as the saturation curve of Figure 1 shows.

Coarser binning helps but does not rescue it either. Collapsing to two bins (0.175, 0.230,  $\infty$ ) gives 98.7 and 58.3% with perfect golden selection: better, still short of 0.68. Only a single bin; that is, no differential measurement at all, trivially passes.

### 2.1.14 Conclusion.

The comparison is worth stating plainly rather than filing as a partial success. The classifier does what it was built to do and matches the published benchmark, but the  $p_\pi$  differential measurement above  $\sim 0.23$  GeV/c cannot be made to meet the BNB’s own binning criterion with a range-based estimator, with or without a golden-pion cut. The BNB analysis passes that criterion because its resolved bins (0.10–0.16, 0.16–0.19, 0.19–0.22) all sit *below* our 0.175 GeV/c threshold, in the region where range still tracks momentum; above 0.22 it uses a single wide bin and an overflow. The realistic options for this analysis are therefore to lower the pion threshold toward 0.10 GeV/c and resolve the region where the estimator works, to adopt a two-bin-plus-overflow  $p_\pi$  result and say so, or to replace the range-based estimator altogether. The golden-pion BDT is a useful ingredient in any of those, but it is not by itself the fix.

**Scope of this conclusion.** The failure described in this subsection concerns attempts to recover a finer multi-bin  $p_\pi$  spectrum, and its numbers are for the loose truth-matched track sample. It does not invalidate the adopted two-bin analysis-sample result (§2.1.6, Table 2), whose migration diagonals are 91% and 76%.

### 2.1.15 Scope of $p_\pi$ .

Both candidate routes to a finer  $p_\pi$  measurement were tried; as the preceding subsection and the following paragraph show, the golden-pion classifier does not repair the response even at perfect truth-level selection, whereas the proton tag does not change the response appreciably either way. At the two-bin scheme the inclusive selection already satisfies the binning criterion, so  $p_\pi$  need not be withdrawn from it; what it cannot support is a five-bin *shape* measurement. The other inclusive observables are unaffected: the angular variables and  $p_\mu$  retain their shape information, and the total cross section does not use this estimator at all.

### 2.1.16 The proton-tagged route.

The other candidate route to a sharper  $p_\pi$  response is the proton tag. It has been measured for FHC, RHC and the combined exposure: at the same two-bin edges the proton-tagged response is indistinguishable from the inclusive one, marginally worse rather than better. Applying the same migration-diagonal test gives

| selection            | config. | $N_{\text{sig}}$ | bin 1 | bin 1 [0.175, 0.205] | bin 2 (> 0.205) | both > 0.68? |
|----------------------|---------|------------------|-------|----------------------|-----------------|--------------|
| inclusive CC1mu1piXp | FHC     | 1056             |       | 91.0%                | 75.5%           | yes          |
| <b>proton-tagged</b> | FHC     | 469              |       | <b>88.9%</b>         | <b>74.2%</b>    | <b>yes</b>   |
| CC1mu1pi1p           | RHC     | 621              |       | <b>89.4%</b>         | <b>75.1%</b>    | <b>yes</b>   |
|                      | comb.   | 1090             |       | <b>89.2%</b>         | <b>74.7%</b>    | <b>yes</b>   |

measured on the numuMC overlays named in the corresponding `file_properties` lists, selected signal, with the reconstructed momentum mass-corrected exactly as the bin config does it. All four rows clear the criterion in both bins, and the proton-tagged values sit about two points *below* the inclusive ones rather than above them. The three proton-tagged configurations agree to better than a percentage point, so the response is a property of the selection rather than of a particular exposure, but the proton tag does not improve it.

That is the expected result. The pion-momentum response is set by whether the pion ranges out or interacts hadronically before stopping, which is a property of the pion's own passage through the argon. Requiring a reconstructed proton at the vertex changes the event sample, the purity and the efficiency, but it does not change what happens to the pion downstream, so there is no mechanism by which it would sharpen the momentum response, and the measurement says so directly.

The consequence is that the proton-tagged  $p_\pi$  measurement stands on its own merits, as the pion-momentum cross section of an exclusive final state that also yields  $W$  and TKI, and not as a repair of the inclusive one. The inclusive  $p_\pi$  at two bins is equally defensible on the response test.

Table 4: Two-bin  $p_\pi$  on the proton-tagged CC1mu1pi1p selection, all three beam configurations.  $d\sigma/dp_\pi$  in  $10^{-38} \text{ cm}^2/(\text{GeV}/c)/\text{Ar}$ ; bin 1 is [0.175, 0.205] GeV/c, bin 2 everything above. All curves are  $A_C$ -smeared. The proton-tagged `xsec` configs declare only the tune and the fake data, so no generator rows appear. Regenerated from the release (`report/tools/ppi2bin.tables.py`); the uncertainty row is the total.

|                       | FHC                   |             | RHC                   |             | Combined              |             |
|-----------------------|-----------------------|-------------|-----------------------|-------------|-----------------------|-------------|
|                       | bin 1                 | bin 2       | bin 1                 | bin 2       | bin 1                 | bin 2       |
| unfolded data         | 1.086                 | 0.586       | 0.386                 | 0.525       | 0.810                 | 0.551       |
| uncertainty           | $\pm 0.836$           | $\pm 0.250$ | $\pm 0.796$           | $\pm 0.222$ | $\pm 0.851$           | $\pm 0.221$ |
| $A_C$ truth           | 1.145                 | 0.598       | 1.056                 | 0.458       | 1.243                 | 0.518       |
| uB tune               | 1.139                 | 0.591       | 1.135                 | 0.469       | 1.287                 | 0.521       |
| unfolded/truth        | 0.95                  | 0.98        | 0.37                  | 1.15        | 0.65                  | 1.06        |
| $\sigma_{\text{int}}$ | 0.498                 |             | 0.429                 |             | 0.462                 |             |
| $\chi^2/\text{ndf}$   | 0.01/2 ( $p = 1.00$ ) |             | 1.21/2 ( $p = 0.55$ ) |             | 0.40/2 ( $p = 0.82$ ) |             |

All three extractions close, with  $\chi^2$  of 0.01, 1.21 and 0.40 on two bins on the current release. The  $p_\pi$  differential cross section satisfies the field's binning criterion in every bin of every beam configuration here, as it does on the inclusive selection; the two are equivalent on that test, and the proton-tagged version is reported because it belongs to the exclusive final state, not because it is required to make  $p_\pi$  measurable.

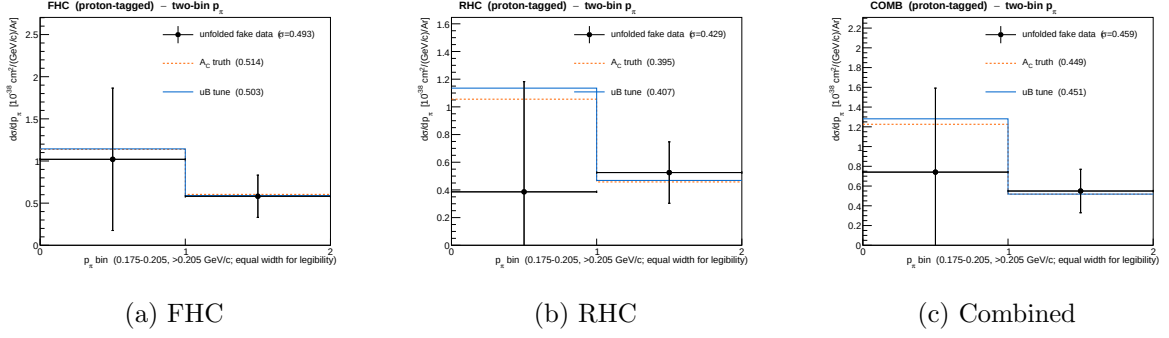


Figure 4: The two-bin  $p_\pi$  cross section on the proton-tagged CC1mu1pi1p selection, the graphical form of Tab. 4. Same edges as the inclusive result of Fig. 2 so the two are directly comparable, and the bins are again drawn equal-width.

### 2.1.17 The least-populated bin.

The RHC first bin unfolds to 0.37 times its  $A_C$ -smeared truth ( $0.386 \pm 0.796$  against 1.056) with an unremarkable closure  $\chi^2$  of 1.21/2: a  $0.8\sigma$  fluctuation on a bin whose uncertainty is twice its content. Independent throws move it by more than  $\pm 1\sigma$ . The bin holds too few events to say anything, and any real-data value in it should be read with its uncertainty and not its central value.

One caveat remains. The 0.205 GeV/c boundary was independently preferred by the migration-level edge scan run on the proton-tagged selection (`macros/ppi_binning_scan.C`, FHC; `logs/binning_scan/1p_fhc.txt`) as well as by the inclusive scan, so it is not merely inherited. It was not, however, re-optimised through a separate universe, covariance and unfolding production for every candidate boundary, which would need its own universe construction per configuration ( $\sim 1$ – $2$  h each); that is not expected to change the conclusion, since both bins clear the criterion with margin in all three proton-tagged configurations.

## 3 Rejected: extending the pion threshold below 0.175 GeV/c

[INCLUSIVE]

*The extension described here was tested and rejected; it is not part of the measurement, whose threshold is  $p_\pi > 0.175$  GeV/c.*

**The low-momentum bin, as a test.** The extension was implemented for  $p_\pi$ : a  $[0.113, 0.175]$  GeV/c bin is prepended to the pion-momentum binning, lowering the measured phase space to the  $\nu_e$  threshold. The truth-level signal below 0.175 GeV/c is recovered from the stored signal-component branches (the reconstruction reproduces MC\_Signal above 0.175 GeV/c to 0.4%, the residual being the unstored kaon veto); 3323 true signal events populate the new bin. Its selection efficiency is 2.2%; an order of magnitude below the 15–17% of the standard bins (Table 5), as anticipated from the reconstruction turn-on. Despite this, the Wiener-SVD extraction remains well behaved with the bin included: the  $M_A^{\text{RES}}$  fake-data closure gives  $\chi^2/\text{ndf} = 1.41/6$  ( $p = 0.97$ ) and the flux term is unchanged (17% reco, 18.4% unfolded, amplification 1.08 $\times$ ). The bin therefore adds to the measurement, at the price of a large per-bin uncertainty driven by the low efficiency.

The generator predictions have been regenerated on the six-bin scheme (the ppi histogram uses  $p_\pi > 0.113$  while the other observables retain  $p_\pi > 0.175$ , mirroring the analysis). GENIE, NuWro and NEUT all provide the low bin; GiBUU could not be re-binned as its generator-level event record was not retained, so it is omitted from the extended  $p_\pi$  comparison. Against the

(fake) data the generators give  $\chi^2/\text{ndf} = 11.1/6$  (GENIE,  $p = 0.09$ ),  $7.0/6$  (NuWro,  $p = 0.33$ ) and  $6.3/6$  (NEUT,  $p = 0.39$ ): NEUT and NuWro describe the extended spectrum well, while GENIE is mildly disfavoured, consistent with the known GENIE over-prediction of low-momentum  $\text{CC}\pi^\pm$  production.

Table 5: Selection efficiency per true  $p_\pi$  bin with the low-momentum bin added. The new  $[0.113, 0.175]$  GeV/c bin sits at 2.2%, an order of magnitude below the standard bins.

| True $p_\pi$ [GeV/c]     | $N_{\text{true}}$ | $N_{\text{reco}}$ | Efficiency  |
|--------------------------|-------------------|-------------------|-------------|
| <b>0.113–0.175</b> (new) | 3323              | 74                | <b>2.2%</b> |
| 0.175–0.250              | 3368              | 526               | 15.6%       |
| 0.250–0.320              | 2190              | 367               | 16.8%       |
| 0.320–0.420              | 2831              | 479               | 16.9%       |
| 0.420–0.550              | 2499              | 410               | 16.4%       |
| 0.550–1.000              | 2933              | 429               | 14.6%       |

## 4 Binning Optimisation, Systematic Breakdown, and Threshold Validation [INCLUSIVE]

This section records the binning optimisation on which the released binnings are based. Its table gives the binnings of the optimisation stage ( $p_\pi$  five bins,  $\theta_{\mu\pi}$  seven,  $\cos\theta_\pi$  five), which the response and  $A_C$  tests of §2.1 and §4.3 then coarsened or re-binned; the released binnings are those of the analysis note’s appendix.

### 4.1 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC  $\nu_\mu$   $1\mu 1\pi^\pm$  topology with the momentum/angle cuts removed):

| Threshold             | below                  | above                  | turn-on                   |
|-----------------------|------------------------|------------------------|---------------------------|
| $p_\mu > 0.15$ GeV/c  | $\sim 0\text{--}1.5\%$ | $\sim 9\text{--}12\%$  | $0.145 \rightarrow 0.155$ |
| $p_\pi > 0.175$ GeV/c | $\sim 0\text{--}3\%$   | $\sim 12\text{--}16\%$ | $0.165 \rightarrow 0.185$ |

Both are correctly placed, lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco–true  $\approx 0$  at threshold,  $-0.15$  GeV/c at 0.40) from the  $\mu$ -vs- $\pi$  mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the  $p_\pi$  reco bins.

### 4.2 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold  $\geq 45$  selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

| Observable        | bins | Total before | Total after  |
|-------------------|------|--------------|--------------|
| $p_\mu$           | 22→7 | 33.2%        | <b>25.0%</b> |
| $p_\pi$           | 5→5  | 27.7%        | 27.0%        |
| $\cos\theta_\mu$  | 12→5 | 31.3%        | <b>26.7%</b> |
| $\cos\theta_\pi$  | 12→5 | 48.5%        | <b>33.9%</b> |
| $\theta_{\mu\pi}$ | 9→7  | 28.4%        | <b>23.5%</b> |

### 4.3 Re-check of every released binning

The binnings of the optimisation stage were coarsened where  $A_C$  smearing left no resolved information ( $\cos\theta_\pi$  5  $\rightarrow$  4,  $\theta_{\mu\pi}$  7  $\rightarrow$  5). Because the normalisation, the weights, the beam-off treatment, the fake-data throw and the detector variations that condition the response were all settled afterwards, every released binning was then re-tested, for whether finer or more bins are admissible. Two stages were used. First a pre-screen on the processed files (`macros/binning_screen.C`): for the released binning and finer candidates of each observable, the column-normalised migration diagonal per true bin, the efficiency per bin and the selected events per reconstructed bin, with the per-run scales and the central-value weight, and the 0.68 diagonal criterion applied to every bin. Second, the candidates that passed were built and unfolded in full (universes, covariance, closure) and compared with the released extraction on the  $A_C$  row sums, the  $A_C$  conditioning ( $s_{\min}/s_1$ ), the data-statistical term, the per-bin total uncertainties and the closure ratios.

Table 6: (Binning re-check on the release preceding the  $\cos\theta_\pi$  decision: “released” means the binning in use when the test was run; the forward-split five-bin  $\cos\theta_\pi$  row is the released binning.) Pre-screen: smallest column-normalised migration diagonal over the true bins of the released binning and of each finer candidate, FHC / RHC. The two-bin  $p_\pi$  row is evaluated here with a lower edge on the first reconstructed bin, which the released configuration does not have (the framework’s own diagonals are 0.91 and 0.76); the finer  $p_\pi$  candidates fail in their upper bins regardless.

| Observable        | Binning            | min. diagonal FHC / RHC   | criterion                          |
|-------------------|--------------------|---------------------------|------------------------------------|
| $p_\mu$           | released 7         | 0.33 / 0.31               | fails already (MCS resolution; §4) |
|                   | 8, 9 bins          | 0.17 / 0.18, 0.22 / 0.22  | fail                               |
| $\cos\theta_\mu$  | released 5         | 0.68 / 0.68               | at the limit                       |
|                   | 6, 7 bins          | 0.64 / 0.63, 0.61 / 0.61  | fail                               |
| $\theta_\mu$      | released 5         | 0.67 / 0.65               | at the limit                       |
|                   | 7 bins             | 0.63 / 0.62               | fail                               |
| $\cos\theta_\pi$  | released 4         | 0.81 / 0.83               | pass                               |
|                   | 5 (backward split) | 0.77 / 0.77               | pass                               |
|                   | 5 (forward split)  | 0.74 / 0.76               | pass                               |
|                   | 6 bins             | 0.73 / 0.75               | pass                               |
| $\theta_{\mu\pi}$ | released 5         | 0.79 / 0.79               | pass                               |
|                   | 6, 7 bins          | 0.74 / 0.73, 0.75 / 0.73  | pass                               |
| $p_\pi$           | released 2         | 0.58 / 0.56 (see caption) | —                                  |
|                   | 3, 4 bins          | 0.39 / 0.40, 0.29 / 0.29  | fail                               |

Table 7: (Binning re-check on the release preceding the  $\cos\theta_\pi$  decision: “released” means the binning in use when the test was run; the forward-split five-bin  $\cos\theta_\pi$  row is the released binning.) Full extraction of the candidates that passed the pre-screen, FHC / RHC: range of the  $A_C$  row sums,  $A_C$  conditioning  $s_{\min}/s_1$ , bin-averaged data-statistical term, range of the per-bin total uncertainty, and range of the per-bin closure ratio to the  $A_C$ -smeared truth. Backward split: edges  $-1, -0.3, 0.2, 0.5, 0.75, 1$ ; forward split:  $-1, -0.1, 0.35, 0.6, 0.8, 1$ ; six bins:  $-1, -0.4, 0, 0.3, 0.55, 0.75, 1$ ;  $\theta_{\mu\pi}$  six bins:  $0, 0.55, 0.75, 0.95, 1.3, 1.85, 2.6$  rad; seven bins:  $0, 0.5, 0.7, 0.9, 1.15, 1.45, 1.9, 2.6$  rad.

| Binning                  | $A_C$ row sums        | $s_{\min}/s_1$ | data stat    | per-bin unc.   | closure               |
|--------------------------|-----------------------|----------------|--------------|----------------|-----------------------|
| $\cos\theta_\pi$ rel. 4  | 0.71–0.82 / 0.79–0.86 | 0.27 / 0.24    | 12.0 / 11.5% | 30–49 / 32–55% | 0.95–1.09 / 0.94–1.07 |
| backward split 5         | 0.65–0.76 / 0.63–0.80 | 0.20 / 0.27    | 16.0 / 15.2% | 29–73 / 34–91% | 0.83–1.08 / 0.87–1.03 |
| forward split 5          | 0.68–0.77 / 0.70–0.84 | 0.46 / 0.38    | 12.7 / 12.3% | 30–53 / 33–48% | 0.89–1.07 / 0.90–1.05 |
| 6 bins                   | 0.73–0.84 / 0.77–0.89 | 0.18 / 0.15    | 19.0 / 16.3% | 32–91 / 32–54% | 0.66–1.13 / 0.81–1.23 |
| $\theta_{\mu\pi}$ rel. 5 | 0.52–1.03 / 0.58–1.15 | 0.008 / 0.002  | 8.2 / 8.4%   | 35–40 / 30–39% | 0.95–1.06 / 1.03–1.08 |
| 6 bins                   | 0.56–1.29 / 0.58–1.36 | 0.002 / 0.000  | 8.4 / 10.2%  | 34–40 / 30–41% | 0.95–1.06 / 1.03–1.11 |
| 7 bins                   | 0.63–1.41 / 0.65–1.06 | 0.000 / 0.000  | 11.5 / 11.2% | 37–40 / 32–46% | 0.85–1.04 / 0.87–1.14 |

Conclusions.  $p_\mu$ ,  $\cos\theta_\mu$ ,  $\theta_\mu$  and  $p_\pi$  cannot go finer: their released binnings already sit at or below the diagonal criterion, set by the MCS momentum resolution and the angular resolution against the neutrino direction, and every finer candidate fails it.  $\theta_{\mu\pi}$  admits six or seven bins by the diagonal and closes acceptably, but the additional bins carry no resolved information: the smallest singular value of  $A_C$ , already 0.008 of the largest at five bins, goes to zero, and the row sums overshoot to 1.3–1.4 in the bins the regularisation has to fill in, so the released five bins stand.  $\cos\theta_\pi$  is the one observable with room, and only in the forward hemisphere where the events are: splitting the backward region produces a bin of 73–91% uncertainty that closes at 0.83–0.87, and six bins do the same, whereas the forward split (edges  $-1, -0.1, 0.35, 0.6, 0.8, 1$ ) keeps every diagonal above 0.74, keeps the per-bin uncertainties in the released range, raises the data-statistical term by only 0.7% and *improves* the conditioning of  $A_C$  ( $s_{\min}/s_1$  from 0.27 to 0.46 in FHC, 0.24 to 0.38 in RHC) while closing to 0.89–1.07. It is the only change this re-check supports, and it is the released binning: the generator predictions are computed at these edges from the event-level files (the unchanged observables reproduce the previous prediction files to machine precision, which also verifies the generator chain end to end), and the D’Agostini cross-check is run on it. No control-region distribution uses  $\cos\theta_\pi$ , so the frozen sideband binning is unaffected. The proton-tagged  $\cos\theta_\pi$  keeps its four bins. The released binnings of the proton-tagged observables were re-tested on the maximin scan and against a relaxed 0.50 diagonal criterion as well as the nominal 0.68: twenty-six candidate binnings pass the relaxed diagonal and every one of them fails on the conditioning of  $A_C$ . The single exception is a two-bin  $W_{\text{had}}$  (0.58/0.54), which is better conditioned than the released six-bin scheme and was nevertheless not adopted, since two bins buy that conditioning by discarding the shape  $W_{\text{had}}$  is measured for. No released proton-tagged binning changes. The one binning that did change is the new  $\theta_p$ , whose edges were taken from the RHC chain rather than the FHC one after the scan was run on both: at the same four bins they condition three to four times better in both usable configurations.

## 5 Hadronic mass, model comparison, and estimator surveys

[PROTON-TAGGED]

### 5.1 The $W$ binnings are finer than the resolution

$W_{\pi p}$  shows the least model tension of the proton-tagged observables. The cause is not physics.

The bin edges were chosen for equal population, six bins of roughly equal content, without reference to the resolution. Measured on selected signal, FHC Runs 1/2/4:

|                            | $W_{\pi p}$    | $W_{\text{had}}$ |
|----------------------------|----------------|------------------|
| residual RMS [GeV/ $c^2$ ] | 0.142          | 0.283            |
| robust half-width (16–84%) | 0.101          | 0.177            |
| bias (mean reco – true)    | – <b>0.111</b> | –0.032           |
| narrowest analysis bin     | <b>0.040</b>   | 0.100            |
| widest analysis bin        | 1.430          | 1.300            |
| width ratio                | <b>35.7</b>    | 13.0             |

The second and third  $W_{\pi p}$  bins are 40 MeV/ $c^2$  wide against a resolution of 101–142 MeV/ $c^2$ : they are two to three times narrower than the detector can resolve. The bias alone, –111 MeV/ $c^2$ , is larger than three of the six bin widths. The migration diagonals follow directly, 87.6/22.3/11.7/12.0/10.0/2.3%: only the first bin is measured, and the top bin at 2.3% is essentially unmeasured.

This is the same failure as the five-bin  $p_\pi$  scheme, and for the same underlying reason.  $W_{\pi p}$  is reconstructed from the pion and proton momenta, so it inherits the saturation of the range-based pion momentum described in §2.1; its resolution cannot be better than that of its inputs.



An exhaustive edge scan, the same procedure and criterion used for  $p_\pi$ , gives:

| scheme                     | diagonals                     | verdict            |
|----------------------------|-------------------------------|--------------------|
| current, six bins          | 87.6/22.3/11.7/12.0/10.0/2.3% | 1 of 6 usable      |
| best two bins (split 1.15) | <b>83.2/77.9%</b>             | <b>both usable</b> |
| best three bins            | 83.2/49.7/45.9%               | fails              |
| best four bins             | 83.2/45.0/28.2/28.2%          | fails              |

As with  $p_\pi$ , two bins is not merely the best option but the only one that satisfies the criterion, and the first bin is thin (137 selected signal events, against a floor of 100). The implication for the model comparison is direct: a low generator  $\chi^2$  at six bins would reflect a binning that cannot resolve the differences between generators, not agreement between them. The same test should be applied to  $W_{\text{had}}$ , whose upper bins are also narrower than its 177 MeV/ $c^2$  half-width, before either  $W$  result is quoted as a shape measurement. A finer edge scan of the six-bin  $W_{\text{had}}$  has not been run. The two-bin alternative was extracted and is better conditioned (0.58/0.54) but was not adopted, because it discards the shape the observable is measured for; the six-bin scheme is retained with that caveat.

**Adopted scheme: two bins split at 1.15 GeV/ $c^2$ .** The boundary is more tightly constrained than it was for  $p_\pi$ , where anything between roughly 0.19 and 0.22 worked. Here only one choice satisfies both the diagonal criterion and the 100-event floor:

| boundary [GeV/ $c^2$ ] | bin 1        | bin 2        | $N(\text{bin 1})$ |                       |
|------------------------|--------------|--------------|-------------------|-----------------------|
| 1.13                   | 76.5%        | 88.0%        | 51                | below the event floor |
| <b>1.15</b>            | <b>83.2%</b> | <b>77.9%</b> | <b>137</b>        | <b>adopted</b>        |
| 1.17                   | 86.8%        | 66.8%        | 266               | bin 2 just fails      |
| 1.19                   | 87.6%        | 56.0%        | 468               | fails                 |
| 1.23                   | 91.7%        | 37.6%        | 1114              | fails                 |

Moving the boundary up buys population in the first bin at the direct expense of the second: the two diagonals trade against each other steeply, and the window in which both exceed 0.68 is only about  $\pm 0.01$  GeV/ $c^2$  wide. That narrowness is itself a consequence of the resolution being comparable to the full width of the peak region.

Two consequences should be carried with the result. First, **the first bin is thin**, 137 selected signal events against a floor of 100, so its statistical uncertainty will be large once systematics are folded in; the  $p_\pi$  first bin, with 469 events, carries 77% (Table 4). Second, if that proves too sparse in practice, 1.17 is the natural fallback: it doubles the population for a bin-2 diagonal of 66.8%, marginally below threshold, which is a defensible trade to state explicitly rather than a criterion to fail silently.

The scheme is implemented in `configs/ccpi1p_Wpipr_bin_config_2bin.txt`, with the reasoning recorded alongside it in a companion `.README`. Running it through the extraction settled the question (§5.2): the two-bin scheme meets the migration criterion but leaves only about 137 selected signal events in the first bin, which the Wiener filter then suppresses, driving  $A_C$  towards rank one for statistical rather than algebraic reasons. Widening the first bin restores its population at the cost of the upper-bin diagonal, and no scanned edge satisfies both requirements at once.

The apparent  $W_{\pi p}$  agreement therefore does *not* survive a binning the detector can resolve, and the differential shape is withdrawn: at this exposure the proton-tagged sample supports an integrated cross section.  $W_{\text{had}}$  carries the same limitation, since it depends on the same pion momentum.



## 5.2 What limits $W_{\pi p}$ , and what can be done about it

The two-bin  $W_{\pi p}$  scheme of §5.1 was built and run. It does not work, for a reason distinct from the six-bin failure, and the investigation that followed settles what the pion momentum can and cannot deliver.

**The two-bin extraction fails on statistics.** The *response* behaves exactly as designed: the framework returns migration diagonals of 0.802 and 0.773, matching the truth-level scan, and the closure is acceptable at  $\chi^2 = 1.17/2$  ( $p = 0.56$ ). The *extraction* nonetheless carries almost no shape information, because  $A_C$  collapses:

(*Matrices from the pass in which the two-bin  $W_{\pi p}$  scheme was run, written with true bins as rows; the two  $p_\pi$  rows are the same-pass comparison, and the released inclusive  $p_\pi$  matrix is that of Fig. 3.*)

| two-bin scheme                    | $A_C$                                                           | $s_2/s_1$    |
|-----------------------------------|-----------------------------------------------------------------|--------------|
| $W_{\pi p}$ , first bin $N = 137$ | $\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$ | <b>0.031</b> |
| $p_\pi$ proton-tagged, $N = 469$  | $\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$ | 0.577        |
| $p_\pi$ inclusive, $N = 1056$     | $\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$ | 0.528        |

The top-left entry is 0.018:  $A_C$  passes under two per cent of the first bin and takes  $-0.41$  from the second. This is a rank-one collapse of the kind described in §7.1, but its cause is statistical rather than a defective regulariser. With 137 events the first bin has such poor signal-to-noise that the Wiener filter suppresses that mode almost entirely, and the ordering across the three schemes,  $137 \rightarrow 0.031$ ,  $469 \rightarrow 0.577$ ,  $1056 \rightarrow 0.528$ , follows population rather than anything geometric.

$W_{\pi p}$  is therefore caught between two requirements. Bins narrow enough to satisfy the response criterion leave the first bin too sparse to unfold; bins wide enough for a stable unfold (1.19 gives 468 events, matching the  $p_\pi$  case that works) fail the criterion at 56%.

**No binning escapes it.** The conclusion does not rest on the one scheme that was tried. Scanning *every* two- and three-bin scheme with edges between 1.12 and 1.70 GeV/ $c^2$  (FHC Run1+2+4, 3505 selected signal events) finds none that keeps both a migration diagonal above 0.68 and adequate statistics in every bin:

| two-bin edge  | $N_1/N_2$ | diag <sub>1</sub> / diag <sub>2</sub> |                      |
|---------------|-----------|---------------------------------------|----------------------|
| 1.15          | 137/3367  | 0.832/0.779                           | first bin too sparse |
| 1.18          | 351/3153  | 0.866/0.611                           |                      |
| 1.24          | 1298/2206 | 0.930/0.347                           | upper bin unusable   |
| 1.27 (median) | 1744/1760 | 0.951/ <b>0.261</b>                   |                      |
| 1.45          | 2966/538  | 0.993/0.030                           |                      |

The trend exposes the mechanism. Reconstructed  $W_{\pi p}$  is biased about 111 MeV low, inheriting the saturation of the range-based pion momentum (§2.1), so raising the edge far enough to balance the statistics drains the upper bin by downward migration: diag<sub>1</sub> climbs from 0.786 to 0.999 across the scan while diag<sub>2</sub> collapses from 0.927 to 0.004. There is no edge at which both bins are simultaneously populated and diagonal, and no three-bin combination does better — the third-bin diagonal never exceeds 0.347.

**At the current exposure  $W_{\pi p}$  does not support a shape measurement** and is quoted as an integrated cross section unless the pion momentum improves. That integrated number is a one-bin extraction rather than a regularised unfolding: with a single bin there is nothing to unfold, since the response is  $1 \times 1$ ,  $A_C$  is trivially unity and the regularisation has no effect. It is reported through the total flux-averaged cross-section path of Sec. 11.4 of the analysis note, and integrated over the full range it is the same quantity as the total

proton-tagged cross section. The corresponding integrated generator predictions are built by `generator_predictions/newg4/make_wpipr_1bin.C`, summing the per-bin cross sections of the six-bin flux-averaged predictions.

**Six-bin  $W_{\pi p}$  closure, diagnostic only.** For completeness the six-bin extraction was carried through the full chain in all three configurations (detector variations included). It closes against the  $A_C$ -smeared truth, but that closure cannot restore shape information the response does not contain (the top bin is 2% diagonal), so these numbers are a check of the machinery and *not* a result, and they do not appear in the analysis note’s result tables. The six-bin generator predictions (GENIE 0.501, NuWro 0.646, GiBUU 0.735, NEUT  $0.783 \times 10^{-38}$  cm<sup>2</sup>/Ar, FHC), exist in the release for the same diagnostic purpose.

Table 8: Six-bin  $W_{\pi p}$  fake-data closure, diagnostic only (withdrawn as a differential result).  $\sigma_{\text{int}}$  in  $10^{-38}$  cm<sup>2</sup>/Ar; columns as in the note’s proton-tagged closure tables.

| Config   | $\sigma_{\text{int}}$ | unf./truth | $\chi^2/\text{ndf}$ |
|----------|-----------------------|------------|---------------------|
| FHC      | 0.466                 | 0.97       | 0.92/6              |
| RHC      | 0.305                 | 0.88       | 2.56/6              |
| Combined | 0.359                 | 0.85       | 1.12/6              |

**The limitation is the pion momentum, and only the pion momentum.** Correlating the  $W_{\pi p}$  residual against each input residual isolates the cause:

| input                | $r$           | $W$ residual RMS when...           |              |
|----------------------|---------------|------------------------------------|--------------|
| pion momentum        | <b>+0.726</b> | all events                         | 0.142        |
| proton momentum      | +0.012        | pion momentum well reconstructed   | <b>0.061</b> |
| pion $\cos \theta$   | −0.029        | proton momentum well reconstructed | 0.130        |
| proton $\cos \theta$ | −0.032        | both well reconstructed            | 0.038        |

Fixing the pion momentum alone takes the resolution from 0.142 to 0.061, a factor 2.3; fixing the proton buys almost nothing, and the angles are irrelevant. A single defect propagates through the whole analysis: the hadronic interaction that truncates the pion track forces  $p_\pi$  to two bins, and forces  $W_{\pi p}$  out of a shape measurement altogether.

**Reparameterising to  $W^2$  does not help.** A monotonic transformation with mapped edges leaves the migration matrix unchanged, which was verified directly: splitting at 1.15 GeV/ $c^2$  in  $W$  and at  $1.15^2 = 1.3225$  GeV<sup>2</sup>/ $c^4$  in  $W^2$  gives diagonals of 83.2% and 77.9% in both cases, on identical events. Resolution and bin width scale together,  $\sigma(W^2) = 2W\sigma(W)$ , so  $\sigma/\text{width}$  is invariant at about 2 either way. This is the same lesson as  $\theta_\mu$  versus  $\cos \theta_\mu$  (§7.1).

### 5.3 Survey of pion momentum estimators without a magnetic field

Four approaches were measured on truth-matched pion tracks. All are limited by the same mechanism, that each measures what the pion did *before* it interacted:

| estimator             | $\langle \text{reco}/\text{true} \rangle$ , top bin | RMS  | note                                     |
|-----------------------|-----------------------------------------------------|------|------------------------------------------|
| range (analysis)      | 0.319                                               | 0.16 | saturates near 0.34 GeV/ $c$             |
| calorimetric energy   | 0.383                                               | 0.31 | also truncated, and $2\times$ noisier    |
| MCS                   | 1.011                                               | 2.1  | unbiased, unusably noisy on short tracks |
| golden-pion selection | 0.327                                               | 0.18 | no gain above 0.8 GeV/ $c$               |

**Calorimetry does not escape the truncation.** Deposited charge stops at the interaction point just as range does, so  $\langle \text{calo}/\text{true} \rangle$  falls from 1.136 to 0.383 across the momentum range while the RMS is three to five times that of range (1.007 to 0.313, against 0.157 to 0.158). It is the same measurement with more noise.

**The golden-pion selection does not help where it is needed.** Splitting the sample by the truth label shows that golden pions are not well reconstructed either, and that their advantage disappears exactly where it is needed:

| true $p_\pi$ [GeV/ $c$ ] | golden fraction | golden $\langle \text{range}/\text{true} \rangle$ | non-golden   |
|--------------------------|-----------------|---------------------------------------------------|--------------|
| 0.175–0.250              | 77.8%           | 0.873                                             | 0.699        |
| 0.320–0.420              | 44.1%           | 0.705                                             | 0.570        |
| 0.550–0.800              | 21.4%           | 0.507                                             | 0.429        |
| 0.800–1.500              | 17.2%           | <b>0.327</b>                                      | <b>0.317</b> |

Above 0.8 GeV/ $c$  the golden and non-golden values are identical to within one per cent. The reason is physical: a 1 GeV/ $c$  pion has a CSDA range of several metres in argon and cannot stop inside a 2.5 m TPC unless it has already lost most of its energy through processes that truncate or break the track. “Golden” at high momentum therefore selects a pathological subsample rather than a clean one, which is why the classifier of §2.1.7 reaches a ceiling: above  $\sim 0.5$  GeV/ $c$  there is no clean subsample to find.

**A multivariate regression fixes the bias but redistributes the resolution.** A BDTG regression on  $p_\pi^{\text{true}}$ , using range, MCS, calorimetry, length, the three deflection variables, the four Bragg likelihoods, track score and daughter count, was trained on Runs 1 and 4 and evaluated on the held-out Run 2. It largely removes the saturation in the mean,  $\langle \text{pred}/\text{true} \rangle$  running 1.196 to 0.722 against 0.835 to 0.319 for range, at comparable RMS. Rebuilding  $W_{\pi p}$  with it:

| pion estimator         | $W$ bias       | $W$ RMS      |
|------------------------|----------------|--------------|
| range (analysis)       | −0.127         | 0.144        |
| BDTG regression        | − <b>0.005</b> | 0.119        |
| true $p_\pi$ (ceiling) | +0.003         | <b>0.046</b> |

The bias falls by a factor of 25, which matters independently of binning: a  $-127$  MeV/ $c^2$  shift distorts the whole  $W$  distribution. The resolution improves only from 0.144 to 0.119 against a ceiling of 0.046, so the regression recovers about a quarter of what is available.

Its effect on the binning is a redistribution rather than a rescue. The six-bin diagonals move from 98/30/19/19/13/6% to 63/38/33/43/52/56%: the top bin improves nearly tenfold, but the first bin degrades from 98% to 63%, and in the two-bin scheme the worst bin falls from 73% to 58%. That degradation is regression to the mean, the estimator being pulled toward the sample centre, and it damages the low- $W$  region where the resonance physics sits. The same pattern appears in  $p_\pi$  itself, where the first bin falls from 99.2% to 60.6% while the top bin rises from 0.5% to 33.2%.

Two objections stand against adopting it. The estimator would be MC-trained and sits upstream of everything, so its model dependence becomes a systematic on every result rather than on one cut. And the trade it offers moves measurability out of the region of interest. It is not adopted here.

**What would actually work.** The ceiling measurements above bound what is available: a perfect pion momentum would give  $W_{\pi p}$  a resolution of 0.046, a factor of three better than the regression achieves and enough to support several bins. The information is carried away by the

interaction products, so the approach that attacks the cause is to recover their energy: the pion and its daughters treated as a mini-calorimeter, using the per-PFP calorimetric energies and daughter links already present in the input files. For the proton-tagged sample there is a second, independent handle in transverse kinematic imbalance, where the muon and proton constrain the pion transverse momentum without using the pion’s own reconstruction at all, at the cost of Fermi-motion smearing of order 250 MeV/ $c$ . The transverse-imbalance route has not been tried; the daughter-recovery route was tested next.

**Recovering the interaction products does not work either.** The ceiling measurements above locate the missing information in the secondaries produced when the pion interacts, so the natural remedy is to treat the pion and its products as a single calorimeter. This was tested. PeLEE carries no parent index, so association is geometric, by PFP start point relative to the pion track end, optionally restricted to the generation immediately below the pion in the Pandora hierarchy. The hierarchy restriction matters: at a 40 cm radius the unrestricted sum collects 0.371 GeV in the lowest momentum bin, where only 0.034 GeV is missing, so it is dominated by unrelated activity.

With the hierarchy cut applied, on Run 1:

| true $p_\pi$ [GeV/ $c$ ] | $\langle \text{KE missing} \rangle$ | $\langle \sum E \text{ of daughters within } R \rangle$ [GeV] |       |       |              |
|--------------------------|-------------------------------------|---------------------------------------------------------------|-------|-------|--------------|
|                          |                                     | 5 cm                                                          | 10 cm | 20 cm | 40 cm        |
| 0.175–0.420              | 0.034                               | 0.023                                                         | 0.026 | 0.029 | 0.037        |
| 0.420–0.800              | 0.246                               | 0.040                                                         | 0.048 | 0.055 | 0.062        |
| 0.800–1.500              | <b>0.659</b>                        | 0.056                                                         | 0.074 | 0.091 | <b>0.104</b> |

“KE missing” is the pion’s true kinetic energy less the calorimetric energy of its own track. In the top momentum bin 659 MeV is unaccounted for and the visible daughters supply 104 MeV, of which about 37 MeV is the baseline seen even in the lowest bin where almost nothing is missing. Genuine recovery is therefore of order 10% at high momentum and 25% in the middle bin.

*It also does not change the binning assessment*, which was tested directly rather than inferred from the bias. Adding the daughter energy improves  $\langle \text{reco/true} \rangle$  everywhere, the top bin moving from 0.373 to 0.470 and the first bin becoming nearly unbiased at 0.953, and it roughly triples the upper migration diagonals. Neither is enough: the best upper diagonal rises from 5.5% to 15.7%, against the 68% required. In the adopted two-bin scheme it is a wash and arguably a loss, the worst bin improving from 61.4% to 71.0% while the first degrades from 97.7% to 82.4%, the same regression-to-the-mean behaviour seen with the multivariate estimator. Since both bins already clear the criterion with plain range, nothing is rescued and the daughter association would be imported as a new systematic for no gain. **The two-bin choice for  $p_\pi$  is therefore robust against every estimator tested**, which is a stronger statement than the binning study alone could make.

That recovery is far too little to change the measurement, and the reason is physical rather than algorithmic. Pion absorption on argon,  $\pi A \rightarrow NN \dots$ , deposits much of the energy in nucleons, roughly half of which are neutrons and therefore invisible in liquid argon; the charged products are frequently short and fall below the threshold to be reconstructed as separate particles. The energy is not merely unassociated, it is largely undetected. Applying the daughter sum would move  $\langle \text{reco/true} \rangle$  in the top bin from 0.32 to about 0.42, still less than half of truth.

**Where this leaves the pion momentum.** Five approaches were measured: range, calorimetry, MCS, golden-pion tagging and a multivariate regression, together with recovery of the interaction products. Each is limited by the same fact, that a charged pion which interacts before stopping destroys the information a non-magnetised detector uses to infer its momentum, and that a substantial part of the energy leaves the detector unseen. The regression is the only one that

materially improves anything, removing the  $W_{\pi p}$  bias, and it does so at the cost of model dependence and of the low- $W$  region.

One avenue remains untried and is independent of the pion’s own reconstruction: for the proton-tagged sample, transverse kinematic imbalance constrains the pion transverse momentum from the muon and proton alone. It is limited by Fermi motion, of order 250 MeV/ $c$ , which is comparable to the present errors rather than better than them, so it is worth measuring but should not be expected to transform the measurement. Failing that, the honest conclusion is that  $p_\pi$  is a two-bin measurement and  $W_{\pi p}$  an integrated one at this exposure, and that both are limited by physics rather than by analysis choices.

## 5.4 Cross-check methods

*Provenance.* The D’Agostini cross-check (Table 10 and the numbers around it) is evaluated on the release. The closure figures of the other methods were evaluated on an earlier fake-data throw; the comparison they support is a property of the operators and does not depend on the throw.

Seven additional methods are run in parallel:

Table 9: Unfolding methods.

| Label                     | Description                                        |
|---------------------------|----------------------------------------------------|
| <code>wiener_svd</code>   | Wiener-SVD (primary result)                        |
| <code>bin_by_bin</code>   | Efficiency correction only; no migration unfolding |
| <code>ibu</code>          | D’Agostini iterative Bayesian, 4 iterations        |
| <code>tikhonov</code>     | Tikhonov-regularised least-squares, $\tau = 1.0$   |
| <code>roo_bayes</code>    | RooUnfoldBayes, 4 iterations                       |
| <code>roo_svd</code>      | RooUnfoldSvd                                       |
| <code>roo_invert</code>   | RooUnfoldInvert (direct matrix inversion)          |
| <code>roo_binbybin</code> | RooUnfoldBinByBin                                  |

The D’Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix  $A_C$ , its unfolded total is free of the (non-norm-preserving)  $A_C$  suppression and is stable against the iteration count. Table 10 compares the two integrated totals for all three configurations: the D’Agostini result sits above the Wiener-SVD total by 31% (FHC), 29% (RHC) and 23% (Combined), equivalently  $A_C$  suppresses the quoted Wiener-SVD integral by 23%, 23% and 19%, and it varies by  $\leq 0.3\%$  across 1, 2 and 4 iterations. The D’Agostini integral is flat across the five observables to 1.2% (FHC, 1.017–1.028), 0.9% (RHC, 0.930–0.939) and 0.5% (Combined, 0.972–0.976), against Wiener-SVD spreads of 24%, 35% and 34%, directly confirming that the observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 20 of the analysis note) is the  $A_C$  smearing rather than a physical or acceptance effect.

Table 10: Wiener-SVD vs. D’Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five inclusive observables ( $p_\mu$ , two-bin  $p_\pi$ ,  $\cos\theta_\mu$ ,  $\cos\theta_\pi$ ,  $\theta_{\mu\pi}$ ;  $10^{-38}$  cm<sup>2</sup>/Ar), on the current release. D’Agostini carries no  $A_C$  smearing, so it recovers a 23–31% higher total, and it is flat across observables (spread 1.2% FHC, 0.9% RHC, 0.5% combined) where the Wiener-SVD integrals spread by 24–35%, confirming that the observable-to-observable variation of  $\sigma_{\text{int}}$  in Table 20 of the analysis note is the  $A_C$  smearing rather than a physical or normalisation effect. The last column is the change of the D’Agostini integral between 1 and 4 iterations.

| Config   | $\langle\sigma\rangle_{\text{WSVD}}$ | $\langle\sigma\rangle_{\text{DAG}}$ | DAG/WSVD | DAG 1→4 iter |
|----------|--------------------------------------|-------------------------------------|----------|--------------|
| FHC      | 0.782                                | 1.022                               | 1.31     | $\leq 0.3\%$ |
| RHC      | 0.723                                | 0.933                               | 1.29     | $\leq 0.3\%$ |
| Combined | 0.790                                | 0.974                               | 1.23     | $\leq 0.1\%$ |

## 6 Proton-tagged selection, binning and additional smearing<sub>[PROTON-TAGGED]</sub>

The proton-tagged cut-flow, its per-observable binning study and its  $A_C$  factors are collected here rather than in the analysis note. The note reports the extracted proton-tagged cross sections and the reasons that sample is secondary scope; the material below is the supporting detail.

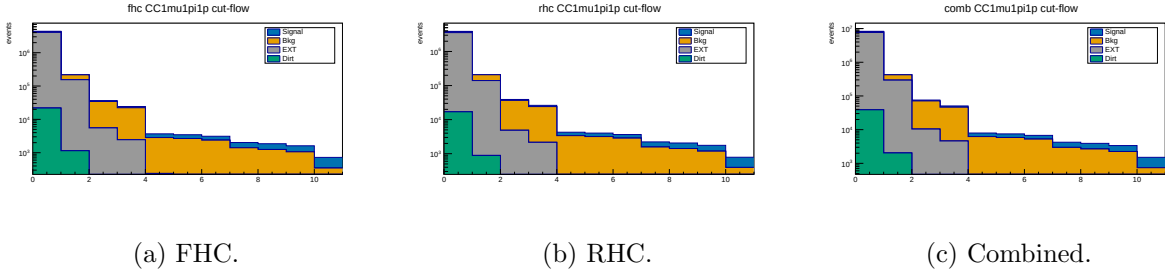


Figure 5: Proton-tagged (CC1mu1pi1p) cut-flow yields at full exposure (log scale; MC-prediction stack, Okabe-Ito: signal blue, background orange, EXT grey, dirt green). The last stage, IdentProton, is the leading-proton identification that defines the subsample.

Table 11: Proton-tagged (CC1mu1pi1p) cut-flow yields at full exposure for FHC, RHC and combined (MC prediction; data blind). Stages None–Nonprotons are the inherited inclusive selection with the proton-tagged signal definition; the final IdentProton stage adds the leading-proton identification. Signal/Bkg from the  $\nu_\mu$  overlays POT-scaled per run, reprocessed with the proton-tagged selection and the full central-value weight under the safe-weight rule (`reprocess_cutflow_cf_1p.sh`); EXT from the per-run beam-off samples scaled by the run gate ratios; dirt from the GENIE NuMI dirt overlay (`macros/cutflow_yields_1p.C`). The final row agrees with the proton-tagged cut-and-count check of the analysis note (Table 22 of the analysis note). Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

| Cut                                                                     | Signal     | Bkg MC     | EXT        | Dirt       | Pred.        | Eff. [%]    | Pur. [%]    |
|-------------------------------------------------------------------------|------------|------------|------------|------------|--------------|-------------|-------------|
| <i>FHC (Runs 1,2,4,5; <math>7.766 \times 10^{20}</math> POT)</i>        |            |            |            |            |              |             |             |
| None                                                                    | 2 888      | 237 145    | 3 256 689  | 19 171     | 3 515 893    | 100.0       | 0.08        |
| InFiducialVol                                                           | 2 424      | 54 365     | 122 860    | 1 005      | 180 654      | 83.9        | 1.3         |
| Topological                                                             | 1 917      | 25 340     | 4 299      | 109        | 31 665       | 66.4        | 6.1         |
| MuonCandidate                                                           | 1 739      | 17 435     | 1 870      | 73         | 21 117       | 60.2        | 8.2         |
| ContainedPion                                                           | 729        | 2 289      | 173        | 14         | 3 204        | 25.2        | 22.7        |
| MuonIn3Planes                                                           | 694        | 2 171      | 144        | 9.4        | 3 018        | 24.0        | 23.0        |
| PionIn3Planes                                                           | 645        | 1 966      | 113        | 3.9        | 2 728        | 22.3        | 23.6        |
| ShowerCut                                                               | 539        | 1 140      | 83         | 3.3        | 1 766        | 18.7        | 30.5        |
| OpeningAngle                                                            | 535        | 1 065      | 32         | 1.1        | 1 633        | 18.5        | 32.8        |
| Nonprotons                                                              | 484        | 910        | 23         | 0.7        | 1 417        | 16.7        | 34.1        |
| <b>IdentProton</b>                                                      | <b>331</b> | <b>298</b> | <b>6.8</b> | <b>0.2</b> | <b>637</b>   | <b>11.5</b> | <b>52.0</b> |
| <i>RHC (Runs 1,2,3,4a,4b,4c; <math>1.108 \times 10^{21}</math> POT)</i> |            |            |            |            |              |             |             |
| None                                                                    | 3 427      | 288 891    | 3 560 975  | 16 958     | 3 870 250    | 100.0       | 0.09        |
| InFiducialVol                                                           | 2 886      | 67 416     | 139 122    | 889        | 210 312      | 84.2        | 1.4         |
| Topological                                                             | 2 318      | 31 870     | 4 767      | 97         | 39 052       | 67.7        | 5.9         |
| MuonCandidate                                                           | 2 111      | 21 962     | 2 090      | 65         | 26 227       | 61.6        | 8.0         |
| ContainedPion                                                           | 882        | 3 181      | 191        | 12         | 4 267        | 25.7        | 20.7        |
| MuonIn3Planes                                                           | 842        | 3 015      | 158        | 8.3        | 4 023        | 24.6        | 20.9        |
| PionIn3Planes                                                           | 784        | 2 739      | 121        | 3.4        | 3 648        | 22.9        | 21.5        |
| ShowerCut                                                               | 647        | 1 480      | 86         | 2.9        | 2 216        | 18.9        | 29.2        |
| OpeningAngle                                                            | 643        | 1 384      | 33         | 0.9        | 2 060        | 18.8        | 31.2        |
| Nonprotons                                                              | 573        | 1 162      | 23         | 0.6        | 1 759        | 16.7        | 32.6        |
| <b>IdentProton</b>                                                      | <b>391</b> | <b>388</b> | <b>7.1</b> | <b>0.2</b> | <b>786</b>   | <b>11.4</b> | <b>49.8</b> |
| <i>Combined FHC+RHC (<math>1.885 \times 10^{21}</math> POT)</i>         |            |            |            |            |              |             |             |
| None                                                                    | 6 315      | 526 036    | 6 817 663  | 36 129     | 7 386 144    | 100.0       | 0.09        |
| InFiducialVol                                                           | 5 310      | 121 781    | 261 981    | 1 894      | 390 966      | 84.1        | 1.4         |
| Topological                                                             | 4 235      | 57 210     | 9 066      | 206        | 70 717       | 67.1        | 6.0         |
| MuonCandidate                                                           | 3 849      | 39 396     | 3 960      | 138        | 47 344       | 61.0        | 8.1         |
| ContainedPion                                                           | 1 611      | 5 470      | 364        | 26         | 7 471        | 25.5        | 21.6        |
| MuonIn3Planes                                                           | 1 536      | 5 186      | 302        | 18         | 7 041        | 24.3        | 21.8        |
| PionIn3Planes                                                           | 1 429      | 4 705      | 234        | 7.3        | 6 376        | 22.6        | 22.4        |
| ShowerCut                                                               | 1 185      | 2 621      | 169        | 6.2        | 3 982        | 18.8        | 29.8        |
| OpeningAngle                                                            | 1 178      | 2 448      | 65         | 2.0        | 3 693        | 18.6        | 31.9        |
| Nonprotons                                                              | 1 057      | 2 072      | 46         | 1.4        | 3 176        | 16.7        | 33.3        |
| <b>IdentProton</b>                                                      | <b>723</b> | <b>686</b> | <b>14</b>  | <b>0.5</b> | <b>1 423</b> | <b>11.4</b> | <b>50.8</b> |



Table 12: Released binning of the eight W/TKI and angular proton-tagged observables, read from the data release (`curves_1p_*.tsv`). The two mass variables keep six equal-population bins chosen from the validated true-signal distributions, with the last bin open at truth level (the upper edge listed is the plotting boundary); the four TKI variables use the coarsened two-bin scheme adopted after the six-bin scheme was found to sit inside the resolution (§5.1).  $W_{\pi p}$  brackets the  $\Delta(1232)$  peak but its differential shape is withdrawn. The five inherited kinematic observables ( $p_\mu$ ,  $p_\pi$ ,  $\cos\theta_\mu$ ,  $\cos\theta_\pi$ ,  $\theta_{\mu\pi}$ ) use the inclusive-analysis binning of Appendix A of the analysis note unchanged.

| Observable       | Definition                                 | Bins           | Released bin edges                                 |
|------------------|--------------------------------------------|----------------|----------------------------------------------------|
| $W_{\pi p}$      | $M(\pi p)$ , direct                        | 6 (withdrawn)  | 1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9 GeV/ $c^2$ |
| $W_{\text{had}}$ | calorimetric                               | 6              | 0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74 GeV/ $c^2$   |
| $\delta p_T$     | $ \vec{p}_T^\mu + \vec{p}_T^{\text{had}} $ | 2              | 0, 0.3, 2.5 GeV/ $c$                               |
| $\delta\alpha_T$ | TKI angle                                  | 2              | 0, 120, 180°                                       |
| $\delta\phi_T$   | TKI angle                                  | 2              | 0, 50, 180°                                        |
| $p_n$            | inferred nucleon momentum                  | 2              | 0, 0.375, 2 GeV/ $c$                               |
| $\theta_p$       | proton polar angle                         | 4 <sup>†</sup> | 0, 0.659, 1.068, 1.539, $\pi$ rad                  |
| $\theta_{\pi p}$ | $\pi$ - $p$ opening angle                  | 3 <sup>†</sup> | 0, 1.193, 2.010, $\pi$ rad                         |

<sup>†</sup> Proposed secondary scope.  $\theta_p$ 's edges are derived on the RHC chain.

## 6.1 Reconstruction of the proton-tagged observables from input quantities

All six observables are built from the three candidate tracks (muon, charged pion, proton) using only reconstructed quantities stored in the input ntuples: the track direction cosines  $\hat{n}$ , the muon-hypothesis range and multiple-scattering momenta and the proton-hypothesis calorimetric kinetic energy.

*Candidate three-momenta.* The muon momentum magnitude is range-based when the muon track ends inside the containment volume (Sec. 7.1 of the analysis note), and multiple-Coulomb-scattering based otherwise. The charged pion, being minimum-ionising, starts from the muon-hypothesis range momentum, converted to the pion hypothesis at fixed kinetic energy in the reconstructed-bin definition exactly as in the inclusive analysis (Sec. 8 of the analysis note); the same two-bin configuration formula is used. The proton momentum follows from its calorimetric kinetic energy under the proton hypothesis,  $p_p = \sqrt{T_p^2 + 2m_p T_p}$  with  $T_p$  the calorimetric kinetic energy. Each three-momentum is  $\vec{p}_i = p_i \hat{n}_i$ , and the energies are  $E_\mu = \sqrt{p_\mu^2 + m_\mu^2}$ ,  $E_\pi = \sqrt{p_\pi^2 + m_\pi^2}$ ,  $E_p = T_p + m_p$  (masses  $m_\mu = 105.7$ ,  $m_\pi = 139.6$ ,  $m_p = 938.3$  MeV).

*Directly measured mass  $W_{\pi p}$ .* The invariant mass of the pion-proton four-vector sum, using no neutrino-direction assumption:

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}.$$

*Transverse kinematic imbalance.* The TKI observables treat the muon against the visible hadronic system  $\vec{p}_{\text{had}} = \vec{p}_\pi + \vec{p}_p$ ,  $E_{\text{had}} = E_\pi + E_p$ . With the components transverse to the beam ( $\hat{z}$ ) written  $\vec{p}_T^\mu$  and  $\vec{p}_T^{\text{had}}$ ,

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T = \arccos \frac{-\vec{p}_T^\mu \cdot (\vec{p}_T^\mu + \vec{p}_T^{\text{had}})}{|\vec{p}_T^\mu| \delta p_T}, \quad \delta\phi_T = \arccos \frac{-\vec{p}_T^\mu \cdot \vec{p}_T^{\text{had}}}{|\vec{p}_T^\mu| |\vec{p}_T^{\text{had}}|},$$

both angles in degrees folded onto  $[0, 180]$ . For a stationary nucleon with no final-state interactions  $\delta p_T \rightarrow 0$ ; Fermi motion broadens it and FSI drive  $\delta\alpha_T \rightarrow 180^\circ$ .

*Calorimetric mass  $W_{\text{had}}$  and struck-nucleon momentum  $p_n$ .* The calorimetric available energy is  $E_{\text{cal}} = E_\mu + (E_{\text{had}} - m_p) + E_b$  with the argon removal energy  $E_b = 24.78$  MeV, giving the

momentum transfer  $Q^2 = -[(0, 0, E_{\text{cal}}, E_{\text{cal}}) - p^\mu]^2$  and

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

(returned as 0 where reconstruction fluctuations make the argument negative). The nucleon-momentum proxy  $p_n$  combines the transverse imbalance with a longitudinal component  $p_L$  from energy-momentum balance,  $p_n = \sqrt{\delta p_T^2 + p_L^2}$ , using the MINERvA prescription [27] as implemented in the framework’s transverse-kinematics tool (option 1):

$$p_L = \frac{R}{2} - \frac{M_{A'}^2 + \delta p_T^2}{2R}, \quad R = M_A + p_z^\mu + p_z^{\text{had}} - E_\mu - E_{\text{had}},$$

with the target mass  $M_A = 22m_n + 18m_p - 0.34381$  GeV (argon-40), the residual mass  $M_{A'} = M_A - m_n + E_b$  and  $E_b = 24.78$  MeV;  $z$  is the beam axis (Sec. 4.2 of the analysis note). The tool was written for a single-proton hadronic system: it receives the summed three-momentum  $\vec{p}_\pi + \vec{p}_p$  and energy  $E_{\text{had}} = E_\pi + E_p$  in the proton slot and forms the hadronic kinetic energy as  $E_{\text{had}} - m_p$ , which is why  $m_p$  appears in  $E_{\text{cal}}$  and  $W_{\text{had}}$  above.  $\delta p_T$ ,  $\delta \alpha_T$  and  $\delta \phi_T$  depend only on the transverse momenta and are exact for the  $\mu + (\pi p)$  system;  $p_L$ ,  $p_n$  and  $W_{\text{had}}$  inherit the single-nucleon residual-mass and removal-energy assumptions of a quasi-elastic-oriented prescription and are therefore *proxies*, not the initial-state nucleon momentum. The generator predictions apply the identical formulas to the generator final states (Sec. 6.2), so the comparison is like for like; a truth-level closure of this prescription against a pion-production-specific definition, and a quantification of the bias from the residual-mass assumption, have not been performed and are listed in Sec. 15.

**Two exposure conventions in the processed files.** One normalisation point is specific to this subsample and worth stating. The processed ntuples use two POT conventions: the files behind the inclusive selection store the *total* run MC POT duplicated in every file of a multi-file run, whereas the proton-tagged files store each file’s *own* POT. Normalising by the individual file POT would be correct for the first set but would scale every split of a multi-file run to the full run exposure in the second, inflating the simulated background by the split multiplicity ( $\sim 3\times$  for RHC) and driving the background-subtracted spectrum negative; normalising by the summed per-file POT has the mirror problem. The analysis therefore normalises each run by the sum of the *distinct* per-file POT values within that run, which reduces to the correct run total under either convention; FHC, with one MC file per run, is unaffected by the choice.

## 6.2 Additional smearing in the proton-tagged observables

Because the Wiener-SVD additional-smearing matrix  $A_C$  is not norm-preserving, each observable’s predictions are rescaled by a different factor when they are brought into the measurement space. Table 13 gives that factor, measured as the ratio of the  $A_C$ -smeared to the truth-level generator integral, for the six proton-tagged observables. Its dominant behaviour is set by the response and the binning; the exact factor is prediction-dependent, and for the tested generators GENIE and GiBUU agree to a few per cent throughout.

On the current release the six proton-tagged observables are suppressed by similar amounts, 0.61–0.79 (Table 13): the two-bin TKI variables and  $W_{\pi p}$  sit at 0.70–0.79 and the six-bin  $W_{\text{had}}$ , the most affected, at 0.61. Sharpness alone does not drive the suppression:  $W_{\pi p}$  is the sharpest distribution measured here (the ratio of its largest to smallest bin is  $\sim 36$ ) yet is suppressed no more than the TKI variables, because  $A_C$  removes structure that the *response* cannot resolve and  $W_{\pi p}$  is built from directly measured pion and proton momenta with equal-population bins that track the  $\Delta(1232)$  peak. No observable is enhanced by  $A_C$  (in the released configurations every prediction-dependent integral factor is below unity — an empirical property of these extractions,

not a mathematical requirement on  $A_C$ , which is a regularisation operator and can suppress or enhance; an amplification would signal a mis-specified regularisation, §7.1). The suppression is applied identically to data, tune and generators, so model comparisons read off any panel are made on equal footing.

Table 13: Mean  $A_C$  row sum for the proton-tagged observables (FHC), from the released matrices `A_C_1p_FHC5_*.tsv`, release; equal to the ratio of the  $A_C$ -smeared to the truth-level integral for a flat prediction. The inclusive observables are given for scale.

| <sup>†</sup> From the current extraction; these observables are not yet in the data release.                                                                                                                           |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Observable                                                                                                                                                                                                             | mean $A_C$ row sum |
| $p_n$                                                                                                                                                                                                                  | 0.79               |
| $\theta_p^\dagger$                                                                                                                                                                                                     | 0.73               |
| $\theta_{\pi p}^\dagger$                                                                                                                                                                                               | 0.61               |
| $\delta\alpha_T$                                                                                                                                                                                                       | 0.79               |
| $W_{\pi p}$                                                                                                                                                                                                            | 0.75               |
| $\delta\phi_T$                                                                                                                                                                                                         | 0.72               |
| $\delta p_T$                                                                                                                                                                                                           | 0.70               |
| $W_{\text{had}}$                                                                                                                                                                                                       | 0.61               |
| <i>Computed from the released matrices (<code>A_C_1p_FHC5_*.tsv</code>). The values span 0.61–0.79, a range of 0.18, so differences in smearing cannot order the generator <math>\chi^2</math> across observables.</i> |                    |

## 7 Angular observables: the $\cos\theta_\mu$ versus $\theta_\mu$ forensics [INCLUSIVE]

### 7.1 Angular variable: $\cos\theta_\mu$ versus $\theta_\mu$

The muon-angle distribution is measured in both  $\cos\theta_\mu$  and  $\theta_\mu$ . The two span the same full range but are different partitions of it: the  $\theta_\mu$  edges (0.43, 0.62, 0.84, 1.17 rad) are not the arccos images of the  $\cos\theta_\mu$  edges (0.45, 0.64, 0.86, 1.10 rad; Table 14 of the analysis note), and they behave very differently under the Wiener–SVD additional smearing  $A_C$ , and the choice matters for how the measurement compares to models.

In  $\cos\theta_\mu$  the signal is sharply forward-peaked: the last two bins ( $\cos\theta_\mu > 0.8$ ) hold most of the rate, and  $A_C$  suppresses the integral of any truth-level prediction strongly (row sums 0.17–0.39 in FHC on the current release, against 0.66–0.91 for  $p_\mu$ ). Because the same  $A_C$  is applied to the data, the tune and every generator, they are compared on equal footing.

The suppression is not a property of the shape of the truth distribution. Across the observables the  $A_C$  integral factor shows no correlation with peakedness:  $W_{\pi p}$  is the sharpest distribution in the analysis (peak-to-minimum ratio 36) yet its factor is 0.75,  $\delta p_T$  is comparatively flat (ratio 10) at 0.70, and the six-bin  $W_{\text{had}}$ , neither sharp nor flat, is the most suppressed at 0.61 (Table 13); the inclusive  $\cos\theta_\mu$  factors sit far below all of these (0.17–0.39 against 0.66–0.91 for  $p_\mu$ ) because its response is the worst conditioned, not because its peak is the sharpest. Within  $\cos\theta_\mu$  itself NEUT and NuWro are the *flattest* of the five truth-level curves (peak-to-minimum 3.8 and 4.3, against 16.3 for the tune) and are smeared no differently from the others: on the release no smeared generator bin is negative and both sit above the tune in every bin.  $A_C$  is the Wiener filter acting on the SVD modes of the response: it removes whatever the data cannot statistically constrain, which is set by the conditioning of the response for that observable and binning, not by the shape of the truth distribution, and it is not norm-preserving, so it can suppress or enhance.

## Why the regularisation operator must respect the bin widths

Wiener–SVD regularises with a matrix  $C$ ; this analysis uses the second-derivative form, built on the *density* with the actual bin widths. The reason is best shown by what the plain tridiagonal stencil  $(1, -2, 1)$  does on these binnings. That stencil is a second derivative only on a uniform grid; the released binnings are strongly non-uniform, with width ratios up to 14:1 (Table 15), so on them it is not a second derivative at all. It is near-singular, its row sums are the  $10^{-6}$  regularisation floor, and its near-null direction is the constant vector rather than anything physical. The Wiener filter is then free to retain that direction alone, which is a rank-1  $A_C$  that passes normalisation and discards shape.

The FHC  $\theta_\mu$  extraction is the clearest case, because its bin widths span a factor of 10.4. Unfolded with the uniform stencil, the unfolded data, the  $A_C$ -smeared truth, the MicroBooNE tune and all four generators share the same normalised shape to four decimal places (Table 14): writing the four generator predictions as the columns of an input matrix and their  $A_C$ -smeared counterparts as the columns of the output matrix, the inputs are full rank ( $s_2/s_1 = 5.0 \times 10^{-2}$ ) while the outputs give  $s_2/s_1 = 2.8 \times 10^{-6}$ . Every shape degree of freedom is annihilated, each model is mapped onto one common curve and only its normalisation survives; any agreement with the data is then true by construction, and a model comparison made on such an extraction is meaningless.  $\cos \theta_\mu$ , the next most non-uniform binning at 14.5, is the other observable strongly affected; the remaining observables sit at width ratios of 3–6 and are affected far less.

Table 14: FHC  $\theta_\mu$  unfolded with a *uniform-grid* second-derivative stencil: normalised bin contents. Five independent inputs, the measurement and four generator predictions plus the tune, collapse onto one shape, the signature of a rank-1  $A_C$ .

|                      | bin 1  | bin 2  | bin 3  | bin 4  | bin 5  |
|----------------------|--------|--------|--------|--------|--------|
| Unfolded data        | 0.1485 | 0.3361 | 0.2899 | 0.1934 | 0.0321 |
| $A_C$ -smeared truth | 0.1484 | 0.3359 | 0.2901 | 0.1933 | 0.0323 |
| MicroBooNE tune      | 0.1484 | 0.3359 | 0.2901 | 0.1934 | 0.0322 |
| NEUT                 | 0.1484 | 0.3359 | 0.2901 | 0.1934 | 0.0322 |

**The operator used.** Both derivative regularisation matrices difference the density on the actual grid: each column is divided by its bin width and the stencil uses the true spacings between bin centres, with a one-sided first-derivative stencil at the ends. (A Dirichlet closure was rejected: it asserts the distribution vanishes at the edges, which is false here,  $\cos \theta_\mu$  peaks at its upper edge.) With no widths supplied the code reduces to the uniform stencil. The widths are passed in by the unfolding application, which holds the slice definitions that the cross-section extractor does not. The effect on the conditioning of  $A_C$ , measured as  $s_2/s_1$  of the smearing matrix itself, is given in Table 15: every inclusive observable improves and none is degraded except combined  $\cos \theta_\mu$ , marginally; in the proton-tagged family combined  $W_{\pi p}$  (0.022) and  $\delta\phi_T$  (0.083) would be as compromised as FHC  $\theta_\mu$  under the uniform stencil.

**$\theta_\mu$  on the release.** The fake data are Poisson throws from the MicroBooNE tune, so the tune is the model that *ought* to fit best, and it does (FHC  $\theta_\mu$ ,  $\chi^2$  on five bins: tune 2.14,  $p = 0.83$ ; GENIE 5.55; GiBUU 6.07; NEUT 8.46; NuWro 9.10; closure 1.52,  $p = 0.91$ ). Under the uniform stencil every model gave 2.8–4.7, indistinguishable, because every model had been mapped onto the same curve. Both parameterisations of the muon angle are released and should be read together; the  $\theta_\mu$   $\chi^2$  are systematically the larger (analysis note, Table 14 of the analysis note).

Table 15:  $A_C$  conditioning  $s_2/s_1$  with the uniform-grid stencil (“uniform”) and with the width-aware operator (“width-aware”), measured on one fake-data pass at the binnings in use when the test was run (the  $\cos\theta_\pi$  row is a five-bin scheme since replaced); the width-aware values of the release follow from the released matrices. With the bin-width ratio of each observable. A value approaching zero means the smearing matrix has collapsed onto a single mode and the extraction carries no shape information.

| observable                                   | width<br>ratio | FHC          |             | RHC     |             | COMB                                       |
|----------------------------------------------|----------------|--------------|-------------|---------|-------------|--------------------------------------------|
|                                              |                | uniform      | width-aware | uniform | width-aware | uniform $\rightarrow$ width-aware          |
| $p_\mu$                                      | 6.3            | 0.34         | 0.90        | 0.31    | 0.94        | 0.19 $\rightarrow$ 0.68                    |
| $\cos\theta_\mu$                             | 14.5           | 0.46         | 0.57        | 0.45    | 0.55        | 0.42 $\rightarrow$ 0.39                    |
| $\cos\theta_\pi$                             | 3.6            | 0.67         | 0.71        | 0.55    | 0.79        | 0.55 $\rightarrow$ 0.74                    |
| $\theta_{\mu\pi}$                            | 3.0            | 0.42         | 0.85        | 0.47    | 0.65        | 0.42 $\rightarrow$ 0.46                    |
| $\theta_\mu$                                 | 10.4           | <b>0.015</b> | <b>0.51</b> | 0.22    | 0.62        | <b>0.045 <math>\rightarrow</math> 0.38</b> |
| $p_\pi$ (2 bins)                             | —              | 0.46         | 0.53        | 0.41    | 0.44        | 0.29 $\rightarrow$ 0.40                    |
| <i>proton-tagged, combined configuration</i> |                |              |             |         |             |                                            |
| $W_{\pi p}$                                  | —              |              |             |         |             | <b>0.022 <math>\rightarrow</math> 0.60</b> |
| $W_{\text{had}}$                             | —              |              |             |         |             | 0.23 $\rightarrow$ 0.86                    |
| $\delta p_T$                                 | —              |              |             |         |             | 0.25 $\rightarrow$ 0.93                    |
| $\delta\alpha_T$                             | —              |              |             |         |             | 0.24 $\rightarrow$ 0.48                    |
| $\delta\phi_T$                               | —              |              |             |         |             | <b>0.083 <math>\rightarrow</math> 0.85</b> |
| $p_n$                                        | —              |              |             |         |             | 0.21 $\rightarrow$ 0.88                    |

## 7.2 Total flux-averaged cross section and systematic breakdown

The measurement is the one-bin extraction of Sec. 11.4 of the analysis note on the Poisson-thrown fake data (one true bin over the full  $\cos\theta_\mu$  range, no overflow, Wiener filter off,  $A_C \equiv 1$ ; `data.release/total_xsec.tsv`), with its full universe-propagated uncertainty; at unblinding it becomes the real beam-on extraction. It agrees with all four generators within  $0.8\sigma$ : standalone GENIE sits  $0.5\text{--}0.8\sigma$  low, while GiBUU, NEUT and NuWro lie within  $0.8\sigma$ .

Table 16: Total flux-averaged cross section from the one-bin extraction (Sec. 11.4 of the analysis note;  $10^{-38} \text{ cm}^2/\text{Ar}$ ), compared with the flux-averaged generator predictions. The measurement uncertainty is the complete configured one-bin covariance (flux, cross section, re-interaction, detector, POT, targets and statistics; MCS-momentum, beamline-geometry, beam-off gate-ratio and Run-2 stand-in terms are not propagated, Sec. 15 of the analysis note). `data.release/total_xsec.tsv`. Values in parentheses give each generator’s deviation from the measurement in units of the measurement uncertainty. The generator predictions are integrated over the full binning with open upper bins (a closed upper edge on  $p_\mu$  or  $p_\pi$  would discard 4–8% of each prediction).

| Config   | Measurement     | GENIE                 | GiBUU                 | NEUT                  | NuWro                 |
|----------|-----------------|-----------------------|-----------------------|-----------------------|-----------------------|
| FHC      | $1.03 \pm 0.40$ | 0.82 (+0.5 $\sigma$ ) | 1.13 (−0.2 $\sigma$ ) | 1.32 (−0.7 $\sigma$ ) | 1.25 (−0.6 $\sigma$ ) |
| RHC      | $0.93 \pm 0.35$ | 0.67 (+0.8 $\sigma$ ) | 0.94 (−0.0 $\sigma$ ) | 1.10 (−0.5 $\sigma$ ) | 1.04 (−0.3 $\sigma$ ) |
| Combined | $0.98 \pm 0.37$ | 0.74 (+0.6 $\sigma$ ) | 1.03 (−0.1 $\sigma$ ) | 1.20 (−0.6 $\sigma$ ) | 1.14 (−0.4 $\sigma$ ) |

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 17–19. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the fake-data POT of each configuration. The RHC detector term is the largest single-configuration entry, reaching 34.7% on  $p_\pi$  and 35.0% on  $\cos\theta_\mu$ .

Table 17: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%), regenerated from the systematics dumps of the release (logs/systdump); “Prediction total” excludes the data statistics.

| Source                         | $p_\mu$     | $p_\pi$     | $\cos \theta_\mu$ | $\cos \theta_\pi$ | $\theta_{\mu\pi}$ | $\theta_\mu$ |
|--------------------------------|-------------|-------------|-------------------|-------------------|-------------------|--------------|
| <b>Prediction total</b>        | <b>39.9</b> | <b>66.9</b> | <b>41.0</b>       | <b>39.0</b>       | <b>37.1</b>       | <b>29.7</b>  |
| Cross section (GENIE)          | 10.4        | 22.5        | 10.5              | 11.4              | 10.6              | 7.8          |
| Flux (PPFX)                    | 28.1        | 42.3        | 27.7              | 27.1              | 28.8              | 23.8         |
| Detector                       | 23.6        | 41.4        | 27.1              | 23.2              | 19.9              | 14.9         |
| Reinteraction                  | 4.4         | 16.5        | 3.7               | 5.1               | 4.1               | 3.4          |
| MC stat                        | 4.5         | 8.9         | 4.7               | 4.2               | 2.8               | 2.7          |
| EXT stat                       | 0.7         | 3.4         | 1.6               | 1.2               | 1.0               | 1.0          |
| Data stat                      | 14.5        | 27.9        | 15.1              | 13.6              | 8.9               | 8.7          |
| POT + targets                  | 3.4         | 5.5         | 3.6               | 3.4               | 3.6               | 2.9          |
| <b>Total (incl. data stat)</b> | <b>42.7</b> | <b>73.0</b> | <b>43.8</b>       | <b>41.5</b>       | <b>38.3</b>       | <b>31.1</b>  |

Table 18: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 17.

| Source                         | $p_\mu$     | $p_\pi$     | $\cos \theta_\mu$ | $\cos \theta_\pi$ | $\theta_{\mu\pi}$ | $\theta_\mu$ |
|--------------------------------|-------------|-------------|-------------------|-------------------|-------------------|--------------|
| <b>Prediction total</b>        | <b>37.7</b> | <b>48.6</b> | <b>46.3</b>       | <b>38.5</b>       | <b>32.6</b>       | <b>36.1</b>  |
| Cross section (GENIE)          | 12.0        | 14.4        | 9.1               | 10.8              | 10.7              | 10.7         |
| Flux (PPFX)                    | 27.6        | 28.4        | 25.8              | 25.2              | 25.2              | 25.6         |
| Detector                       | 21.5        | 34.7        | 35.0              | 25.5              | 16.6              | 22.4         |
| Reinteraction                  | 4.2         | 8.6         | 4.1               | 5.1               | 3.7               | 3.6          |
| MC stat                        | 3.8         | 3.7         | 4.8               | 3.6               | 2.4               | 2.5          |
| EXT stat                       | 1.0         | 1.2         | 1.2               | 1.0               | 0.6               | 0.7          |
| Data stat                      | 13.0        | 13.0        | 16.7              | 12.3              | 8.4               | 9.0          |
| POT + targets                  | 3.8         | 4.0         | 3.6               | 3.5               | 3.5               | 3.5          |
| <b>Total (incl. data stat)</b> | <b>40.5</b> | <b>50.5</b> | <b>49.5</b>       | <b>40.5</b>       | <b>33.7</b>       | <b>37.2</b>  |

Table 19: Combined FHC+RHC systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 17.

| Source                         | $p_\mu$     | $p_\pi$     | $\cos \theta_\mu$ | $\cos \theta_\pi$ | $\theta_{\mu\pi}$ | $\theta_\mu$ |
|--------------------------------|-------------|-------------|-------------------|-------------------|-------------------|--------------|
| <b>Prediction total</b>        | <b>35.4</b> | <b>52.0</b> | <b>37.8</b>       | <b>37.5</b>       | <b>33.6</b>       | <b>35.7</b>  |
| Cross section (GENIE)          | 10.3        | 17.2        | 10.7              | 10.7              | 10.1              | 9.5          |
| Flux (PPFX)                    | 24.5        | 32.4        | 25.9              | 25.7              | 25.8              | 26.2         |
| Detector                       | 21.9        | 33.8        | 24.3              | 23.0              | 18.0              | 21.4         |
| Reinteraction                  | 4.5         | 11.9        | 3.9               | 5.1               | 3.5               | 3.8          |
| MC stat                        | 4.0         | 4.1         | 2.9               | 2.9               | 1.9               | 2.5          |
| EXT stat                       | 0.7         | 1.4         | 0.9               | 0.8               | 0.5               | 0.9          |
| Data stat                      | 13.2        | 13.5        | 9.5               | 9.7               | 6.6               | 8.4          |
| POT + targets                  | 3.2         | 4.4         | 3.5               | 3.5               | 3.4               | 3.4          |
| <b>Total (incl. data stat)</b> | <b>38.0</b> | <b>53.9</b> | <b>39.1</b>       | <b>38.8</b>       | <b>34.2</b>       | <b>36.9</b>  |

## Part II

# Validation of the extraction

## 8 Identity closure: unfolding the prediction with itself [SHARED]

The fake-data closure tests reported in the body compare an unfolded Poisson throw against the truth it was thrown from. They therefore test the whole chain at once, and a modest  $\chi^2$  can hide a genuine inconsistency inside the unfolding itself, because the statistical fluctuation of the throw is the dominant term.

The test here removes that fluctuation. Wiener–SVD returns  $\hat{x} = Ur$  for a measured reco vector  $r$ , and defines the additional smearing matrix as  $A_C = UR$ , where  $R$  is the response (smearacceptance) matrix. Feeding an *exact* prediction  $r = Rt$  back through the chain must therefore return

$$\hat{x} = URt = A_C t \quad (1)$$



identically, for *any* true vector  $t$ . Rather than pick one  $t$ , we test the matrix identity directly:

$$\|UR - A_C\|_\infty \stackrel{!}{=} 0. \quad (2)$$

This is stronger than a single-vector test and independent of the data. It is sensitive to the failure mode a fake-data test cannot see: the unfolding matrix and the smearing matrix being built from different responses, or from responses evaluated in different universes, which would bias every extracted cross section by a fixed factor while leaving the closure  $\chi^2$  untouched.

The check is implemented inside `UnfolderNuMI` and runs on every extraction, printing a `[IDENTITY]` line. The release has 51 extractions (six inclusive observables and eleven proton-tagged ones, each in three beam configurations) and every one of them passes the same check. The worst deviation over the 51 extractions of the release is  $4.4 \times 10^{-6}$  relative, on FHC  $W_{\pi p}$ ; the inclusive observables are all below  $2.5 \times 10^{-7}$ . These are the rounding errors of the singular-value decomposition and the matrix products that follow it, not physical effects: they remain five orders of magnitude below the smallest systematic uncertainty in the analysis.

The values are larger than the  $10^{-10}$  level seen before the regularisation matrices were made width-aware, and the reason is arithmetic rather than physical. The width-aware operator carries factors of  $1/w_j$  that span decades across a binning with a 14:1 width ratio, so the matrix products accumulate more floating-point error.  $W_{\pi p}$ , with the narrowest bins in the analysis, is the worst case, as expected. Nothing about the identity is weaker; only the arithmetic that verifies it is noisier. The unfolding construction is therefore exactly self-consistent, and every deviation seen in the fake-data closure tests is attributable to the statistical throw and to the response model, not to the unfolding implementation.

## 9 Why the results are presented as they are

[INCLUSIVE]

### 9.1 The two-bin $p_\pi$ result

Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ $c$ : a pion of true momentum 0.95 GeV/ $c$  reconstructs at almost the same value as one of 0.55 GeV/ $c$ . The two bins have migration diagonals of 91% and 76% (column-normalised  $M$ , Sec. 10.3 of the analysis note) and are sound, but the upper bin integrates over a very wide physical interval and is not a differential point in the usual sense. The true upper bin is open-ended,  $p_\pi > 0.205$  GeV/ $c$  (the true-bin definition places no upper edge on it); the 1 GeV/ $c$  that appears in the data release and in Appendix A of the analysis note is only the finite boundary adopted to assign the bin a width for plotting, not a kinematic limit. The released  $d\sigma/dp_\pi$  values are therefore bin-averaged densities over very different intervals (0.030 and 0.795 GeV/ $c$ ), and the panel draws the two at equal visual width, flagged on the axis, so the plotted area does not represent cross section. The primary statement of the result is the pair of bin-integrated partial cross sections (Table 20); the density representation is secondary, since it depends on the conventional plotting width of the open bin. These are the partial cross sections,  $\sigma_i = (d\sigma/dp_\pi)_i \Delta p_{\pi,i}$ , which do not depend on the choice of plotting boundary for the lower bin and depend on it for the upper bin only through the physical content above 1 GeV/ $c$ , which the open true bin already includes. For the open upper region the released density and the conventional width of 0.795 GeV/ $c$  form a matched representation of the extracted bin integral: a user should compare bin-integrated predictions after applying  $A_C$ , not reinterpret the density over a different finite upper boundary, since changing the boundary while holding the released density fixed would change the product:

Table 20: Two-bin  $p_\pi$  result as bin-integrated partial cross sections ( $10^{-38}$  cm<sup>2</sup>/Ar), computed from the released `curves_incl.*_ppi2bin.tsv` as density  $\times$  physical width; the uncertainty is the total uncertainty from the released covariance (the same fractional uncertainties as Tables 17 of the analysis note–19). Both bins are  $A_C$ -smeared like every other unfolded quantity, so a prediction must be smeared by the released  $A_C$  before comparison.

| Region                                        | FHC               | RHC               | Combined          |
|-----------------------------------------------|-------------------|-------------------|-------------------|
| $\sigma(0.175 < p_\pi < 0.205 \text{ GeV}/c)$ | $0.037 \pm 0.041$ | $0.054 \pm 0.034$ | $0.053 \pm 0.038$ |
| $\sigma(p_\pi > 0.205 \text{ GeV}/c)$         | $0.827 \pm 0.292$ | $0.694 \pm 0.268$ | $0.738 \pm 0.269$ |

The estimator survey establishing that this is a physical information limit rather than an unrefined estimator is in the technical supplement.

## 9.2 What the closure $\chi^2$ values mean

The closure  $\chi^2$  quoted for each extraction compares the unfolded result with the fake-data truth using the complete configured systematic covariance, and the resulting  $p$ -values cluster near unity (median 0.98 across all fifty-one extractions; ten lie below 0.9 and the lowest is 0.20). That should not be read as excellent agreement with the model. For a self-closure with correctly-sized uncertainties the  $p$ -values would be roughly uniform on  $[0, 1]$ , and they are not.

The reason is that the measurement and the truth it is being compared against share their systematics: both come from the same generator, the same flux and the same exposure, so the flux, cross-section and exposure terms are common-mode and largely cancel in the difference while still inflating the covariance used to judge it. Recomputing the same comparison with the statistical covariance alone raises  $\chi^2/\text{ndf}$  from 0.02–0.2 to 0.5–0.8, much nearer the expected value (a study on one release; the released closure values are those of Tables 17 of the analysis note–19, and it is the effect rather than the values that matters here):

| Extraction                     | full   | prediction-only | statistics-only |
|--------------------------------|--------|-----------------|-----------------|
| inclusive FHC $p_\mu$          | 0.14/7 | 0.22/7          | 1.01/7          |
| inclusive comb $p_\mu$         | 1.43/7 | 2.87/7          | 4.71/7          |
| proton-tagged FHC $p_\mu$      | 1.01/7 | 2.20/7          | 3.02/7          |
| proton-tagged comb $W_{\pi p}$ | 0.48/6 | 0.77/6          | 4.88/6          |

The closure  $\chi^2$  is therefore a check that the extraction does not *disagree* with its input beyond the quoted band, which it passes, and not a measure of how well the band is sized. A test of the latter needs either a covariance restricted to the terms that differ between measurement and reference, or an ensemble of throws. The restricted covariance is not used for the released closure tables, whose  $p$ -values should be read with that limitation in mind; the sizing is tested instead with the 32-member statistical ensemble of the next subsection for three representative FHC extractions.

## 10 Consistency between configurations and with the sibling $\nu_e$ measurement [INCLUSIVE]

### 10.1 Consistency with the sibling $\nu_e$ CC $1\pi$ measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion  $\nu_e$  CC  $1\pi$  analysis shares the same cross-section-extraction framework, beam and reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CC $\pi$  programme should reconcile.

*Consistent by construction.* Both use the Wiener–SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly



one  $\pi^\pm$ , zero  $\pi^0$  or heavier mesons, any number of protons, charged current), and the same systematic suite, PPFx flux hadronic universes (the 12 beamline-geometry unisims the  $\nu_e$  analysis carries are evaluated but not in this analysis’s released covariance, Sec. 15 of the analysis note), the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% / 100% POT / target / dirt normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4  $2.075 \times 10^{20}$ ).

*Differences reconciled or flagged.* (i) **POT accounting:** the  $\nu_e$  note quotes Run 1 FHC =  $3.283 \times 10^{20}$  POT (9 846 635 triggers), the figure that includes the open-trigger stream. The beam-on sample processed here is the non-open-trigger dataset,  $2.192 \times 10^{20}$  POT with 5 748 692 triggers, confirmed against the beam bookkeeping and, independently, by the control-region exposure test (Sec. 11.5 of the analysis note); the total FHC exposure is  $7.766 \times 10^{20}$  over Runs 1, 2, 4 and 5. (ii) **Run 4:** the Run4 overlay is identical to Run4c+Run4d combined (same 859,368 events and  $2.834 \times 10^{21}$  POT); only one may be used to avoid double counting. (iii) **Pion threshold:**  $\nu_e$  uses  $KE_\pi > 40$  MeV ( $p_\pi > 0.113$  GeV/c) versus our  $p_\pi > 0.175$  GeV/c (Table 7 of the analysis note). (iv) **Opening-angle cut:**  $\nu_e$  excludes  $\theta_{e\pi} > 170^\circ$  (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses  $\theta_{\mu\pi} < 2.6$  rad ( $149^\circ$ ). (v) **Regularisation:** the  $\nu_e$  analysis smooths on the first derivative, this analysis on the second (§7.1). (vi) **Scope:** both analyses cover Runs 1–5 in FHC and RHC; the present measurement reports FHC, RHC, and their combination.

## 10.2 Is the combined configuration the combination of its own inputs?

The integrated cross section extracted from a given configuration is not a configuration-independent number: it depends on the additional smearing matrix  $A_C$ , which differs per configuration and per observable, so comparisons across configurations are made observable by observable and against the fake-data truth rather than between measurements. Two tests establish that the combined configuration is what it should be.

**The combined truth against the exposure-weighted average of its inputs.** A configuration spanning both horn currents should predict a cross section between those of its two inputs. Comparing the combined fake-data truth with the POT-weighted mean of the FHC and RHC truths, observable by observable, gives a mean ratio of 1.027 with a range of 0.958–1.071 for the inclusive family and 1.020 (0.989–1.052) for the proton-tagged family (measured on one fake-data pass; the ratio is a property of the truth and the normalisation, not of the throw). The test is sensitive: it is what requires the two per-mode detector-variation sets of the combined configuration to be pooled as one sample scaled once to the data exposure, since scaling each set to the full exposure and summing would double the detector covariance, over-regularise the combined extraction and push its truth ratio to 0.91 with a worst observable at 0.77. With the pooled treatment the combined detector term lies between or close to those of its inputs (analysis note, Tables 17–19: 21.4% against 24.7% and 21.5% for  $p_\mu$ ).

**The RHC deficit is in the input, not the extraction.** The one-bin totals of the release give an RHC/FHC ratio of the injected tune of 0.901 (inclusive) and 0.878 (proton-tagged), and the extracted one-bin ratios are 0.901 and 0.873 (Table 22 of the analysis note). Normalisation is not implicated: the exposures sum exactly ( $7.766 \times 10^{20} + 1.1082 \times 10^{21} = 1.8848 \times 10^{21}$  POT) and the per-configuration flux values reproduce the POT-weighted combination to 0.01%. The deficit is therefore a property of the  $\bar{\nu}_\mu$ -dominant sample, and the number to be compared with data at unblinding is the tune ratio.

**External reproduction of the released curves.** The transformation  $(d\sigma/dx)_i = w_i^{-1} \sum_j (A_C)_{ij} p_j$  applied to the released truth-level predictions reproduces every released

generator curve to a worst per-bin deviation of  $1.2 \times 10^{-10}$ . This is part of the release procedure because it is the one check that an internal closure cannot make: the unfolded data and the truth can both be right while the model comparison is wrong, and only an external user’s arithmetic exposes it.

## Part III

# Control regions

## 11 Control regions: the opening request and its sign-off<sup>[INCLUSIVE]</sup>

This section is the self-contained case for the one decision requested of the collaboration at this stage: permission to inspect beam-on data in four reconstruction-level control regions of the inclusive  $\text{CC } 1\mu 1\pi X p$  selection while the signal region stays blind. Everything here is expanded in Sec. 11.6 of the analysis note; the numbers are those of the frozen release identified in the box below.

**Analysis state.** Repository tag `cr-freeze-2026-09-06`; fake data: one Poisson throw per run of the central-value prediction with the unified weight rule (Sec. 5 of the analysis note); binning: the released edges of Appendix A of the analysis note (five-bin forward-split  $\cos\theta_\pi$ , two-bin  $p_\pi$ ); normalisation: per-run exposure, per-run beam-off gate ratios, per-configuration flux; detector variations: native Run-4 FHC and RHC samples; data release: `report/data_release/`, checksum manifest `MANIFEST.sha256` (b8ebc69e3bc91434). Every table marked “current frozen release” is generated from this state.

**Request.** We request approval to inspect beam-on data in four control regions:  $\text{CC}0\pi$ , multi- $\pi$ ,  $\pi^0$  and large muon–pion opening angle (called “cosmic” below for brevity: it is cosmic-*enriched*, 34–41% beam-off, not cosmic-pure). Throughout, “beam-off”, “EXT” and “cosmic data” name the same data-driven component, the beam-off trigger sample scaled by gate counts; “signal contamination” means the inclusive signal unless the proton-tagged selection is named. Each region is anchored by the inversion of one nominal signal requirement; requirements that become undefined once that inversion is made, or that would sculpt the targeted control sample, are dropped as listed in Table 21. Each region is disjoint from the signal region at reconstruction level (demonstrated event by event, §12.1); their expected signal contaminations are 8–24%. The signal region remains blind. Before the regions are opened, the selections, the run-dependent normalisation, the beam-off scaling, the displayed observables and binning, the covariance construction, the goodness-of-fit tests and the permitted responses to a discrepancy are frozen as written in §12.5. Control-region data will be compared with the complete prediction (neutrino MC + beam-off + dirt) separately in FHC and RHC, with the combined configuration as a cross-check. Any discrepancy beyond the pre-declared criteria requires a documented treatment and renewed approval before the signal region is opened. The proton-tagged control regions are *not* part of this request (§12.6).

**What each region can and cannot do.** The physics roles, which the transfer factors of §12.3 decide, are these.

- **$\text{CC}0\pi$**  is the high-statistics validation of the dominant neutrino background class:  $\text{CC}0\pi$  events with a proton or other track taken as the pion are 244 of the 637 FHC neutrino-background events (38%). The region has 72% target purity (65% of the full stack) and 0.6% raw statistical precision. It validates the normalisation and kinematics of the underlying  $\text{CC}0\pi$  population, *not* the proton-to-pion misidentification probability that transfers that

population into the signal region: the transfer carries a 31–34% uncertainty dominated by the detector variations. It is therefore a validation test rather than a per-cent-level constraint, and it is conditionally eligible for a fitted normalisation correction under §12.5. The direct test of the misidentification probability is an inverted pion-PID region, which does not exist yet and is listed among the prerequisites for opening the signal region.

- $\pi^0$  tests the shower veto and the CC+NC  $\pi^0$  background. Its target purity is 50.1% (FHC) and 51.0% (RHC) of the neutrino MC, its transfer factor 0.02 with a 17–19% uncertainty set by the Monte-Carlo-statistical floor of the detector-variation samples. **For this opening the region is diagnostic only.** Approval to inspect it does not authorise a fitted correction; a proposal to fit the CC+NC  $\pi^0$  normalisation would require a separate documented review after the pre-fit result has been recorded, with the fit model, the leverage of the fixed components, the transfer uncertainty and its floor, and the expected effect on the signal-region prediction.
- **Multi- $\pi$**  is diagnostic only. Its transfer factor is  $\simeq 2.5$  (the signal region holds more of the target class than the region does) and its target purity 47%; it is not used to fit or constrain the signal-region normalisation, and exposes gross modelling failures and shape discrepancies.
- **Cosmic** validates the beam-off normalisation and shape to  $\sim 50\%$  once the prediction uncertainties of its neutrino component are included (17% from data statistics alone). That is sufficient because beam-off is 1.9% of the selected signal-region prediction: a 50% uncertainty on it is an effect of order 1% on the selected sample. It is not eligible for a fitted correction.

About 29% of the FHC neutrino background has no region of its own: NC and  $\nu_e$ CC without a  $\pi^0$ , out-of-fiducial-volume events and CC1 $\pi^\pm$  outside the measured phase space (§12.3). These components keep their model uncertainties unchanged; the CC0 $\pi$  and  $\pi^0$  distributions carry indirect sensitivity to the first two through the shared reconstruction, and each is small enough individually that no dedicated region is justified at this exposure.

**What the test detects, and its blind direction.** The pass/fail statistic is a global normalisation  $\chi^2$  over the eight (region, mode) totals with the full pre-fit covariance (§12.5). It detects a discrepancy confined to one region beyond 12% of the CC0 $\pi$  or  $\pi^0$  total and beyond 50–62% of the multi- $\pi$  or cosmic total, an FHC/RHC inconsistency, and, through the normalisation-marginalised shape tests, a shape difference; a *common* normalisation shift of all regions fails it only beyond 58%, because the correlated flux, cross-section and detector terms absorb it. That blind direction is acceptable because it is the direction the signal-region measurement already carries as its flux and cross-section uncertainty: the control regions test the background model region by region, not the overall normalisation.

**What the present pseudo-data studies do and do not show.** The sideband machinery has been exercised on Poisson-thrown central-value pseudo-data. Those studies validate the selection logic, the per-run exposure scaling, the central-value weighting and, in the full-stack variant of §12.2, the exact bookkeeping path that beam-on data will follow. They do *not* validate the background model: the pseudo-data are drawn from the same simulation that supplies the prediction, so agreement is guaranteed up to statistics. The background model is the largest untested assumption in the analysis, and testing it is what the control regions are for. The control-region comparison is made at reconstruction level, so it does not depend on the unfolding or on the alternative-generator closure that is required before the signal region is opened (Sec. 15 of the analysis note).

Table 21: Control-region definitions in terms of the selection stages. The common preselection *pre* is software trigger, reconstructed vertex in the fiducial volume, topological score  $> 0.67$  and a qualifying muon candidate (Cuts 1–4 of Sec. 8 of the analysis note); *shower* is the shower veto (Cut 8), *final* the non-proton multiplicity cut (Cut 10), *contained/gap* the pion containment and three-plane requirements (Cuts 5–7), *opening* the  $\theta_{\mu\pi} < 2.6$  rad cut (Cut 9), and  $n_\pi$  the number of qualifying contained pion candidates. The signal region is *pre*, *contained*, *gap*, *shower*, *opening* and *final* with  $n_\pi = 1$ . “Dropped” lists nominal requirements not applied in the region; “target” the signal-region background the region is meant to constrain. Each region is stored as a flag alongside the signal-region flag when the events are processed.

| Region       | Definition (branch level)                                                                              | Dropped vs. signal                                | Target background                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------|
| CC0 $\pi$    | <i>pre</i> , <i>shower</i> , <i>final</i> ; $n_\pi = 0$                                                | contained, gap, opening (no pion to define them)  | CC0 $\pi$ events with a proton or other track taken as the pion; pion absorption feed-in |
| multi- $\pi$ | <i>pre</i> , <i>final</i> ; $n_\pi \geq 2$                                                             | shower, contained/-gap of the extra pion, opening | multi-pion events with one pion lost                                                     |
| $\pi^0$      | <i>pre</i> , shower veto failed; $n_{\text{shower}} \geq 1$                                            | contained, gap, opening, final                    | CC/NC $\pi^0$ with both photons missed; shower-veto efficiency                           |
| cosmic       | <i>pre</i> , <i>contained</i> , <i>gap</i> , <i>shower</i> , <i>final</i> ; opening-angle cut inverted | opening only                                      | residual beam-off (cosmic) in the signal region; its normalisation                       |

The  $\pi^0$  region is the furthest from a single inversion: besides the pion requirements, which have no meaning once no pion candidate is demanded, it also drops the final multiplicity cut. The cut was dropped when the region was defined to avoid a plausible topology-dependent sculpting (photon showers reconstructed as additional track-like primaries). The dedicated comparison of §12.4 subsequently found no material change in target purity, transfer uncertainty or the frozen leading-shower-energy shape. The no-cut definition is retained because it provides 11–14% more events, not because the study demonstrated a material bias from the cut.

**Points the collaboration is asked to ratify.** (i) The  $\pi^0$  definition without the final multiplicity cut (Table 21); the comparison with the version retaining it is in §12.4 and will *not* be used to modify the region after beam-on inspection. (ii) The  $\pi^0$  target class CC+NC  $\pi^0$ , adopted by the analysers (the NC $\pi^0$  events the shower veto is also meant to remove had been counted under “NC and  $\nu_e$ CC”;  $\nu_\mu$ CC  $\pi^0$  alone is 46% of the region), together with the region’s diagnostic-only status for this opening. Purity is defined throughout as the target class’s share of the *neutrino-MC* prediction in the region, beam-off and dirt being separately constrained components; both the class definition and the denominator were settled after the purities were known, which is one reason the region is diagnostic-only. (iii) The roles of the other three regions as stated above, and the frozen procedure of §12.5.

**Approval checklist.**

| Item                                                                                                                                        | Status / evidence                                                                                                                                                                  | Sign-off     |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <i>Required before the control regions are opened</i>                                                                                       |                                                                                                                                                                                    |              |
| Control-region definitions frozen                                                                                                           | Table 21                                                                                                                                                                           |              |
| Signal-region overlap zero, event by event                                                                                                  | §12.1                                                                                                                                                                              |              |
| Plot list and binning frozen                                                                                                                | §12.5 item 1                                                                                                                                                                       |              |
| Pre-fit covariance, global statistic and thresholds frozen                                                                                  | §12.5 items 2–3; null calibration Fig. 9                                                                                                                                           |              |
| Response to a discrepancy frozen                                                                                                            | §12.5 item 4                                                                                                                                                                       |              |
| $\pi^0$ definition and target class ratified; diagnostic-only status recorded                                                               | this section, (i)–(ii)                                                                                                                                                             | conveners    |
| Flux and detector-systematics contacts have reviewed their parts                                                                            | pending                                                                                                                                                                            | contacts     |
| Repository tag and release checksum recorded                                                                                                | analysis-state box above                                                                                                                                                           |              |
| <i>Required later, before the signal region is opened</i>                                                                                   |                                                                                                                                                                                    |              |
| Control-region results reviewed; any discrepancy treated under §12.5                                                                        | pending                                                                                                                                                                            | conveners    |
| Model-distortion fake-data closure (reweighted GENIE, FHC)                                                                                  | done, Sec. 13 of the analysis note                                                                                                                                                 |              |
| Model-distortion fake-data closure, proton-tagged family (FHC)                                                                              | done, Sec. 4 of the proton-tagged note; low-imbalance $\delta p_T/p_n$ bins biased 1.1–1.2 $\sigma$ under GENIE u545                                                               |              |
| Independent-generator (NuWro or GiBUU) fake-data closure                                                                                    | pending: the existing NuWro overlay failed validation against the standalone prediction and is not used, Sec. 15 of the analysis note                                              |              |
| Flux constants confirmed as PPFX-corrected central values                                                                                   | pending, Sec. 4.4 of the analysis note                                                                                                                                             | flux contact |
| MCS muon-momentum scale ( $\pm 5\%$ , data side) evaluated on $p_\mu$ in FHC, RHC and combined and released as <code>cov.MCscale.txt</code> | done, Sec. 15 of the analysis note; at most 0.35 $\sigma$ (FHC, combined) and 0.85 $\sigma$ (RHC open top bin); not yet inside the framework covariance, other observables pending |              |
| Final empty-cache reproduction of the release to be unblinded                                                                               | pending, §14                                                                                                                                                                       |              |
| Inverted pion-PID region for the CC0 $\pi$ misidentification transfer                                                                       | pending                                                                                                                                                                            |              |
| Inverted-proton-PID region before any proton-tagged result                                                                                  | pending, §12.6                                                                                                                                                                     |              |

**Expected yields and precision.** Table 24 of the analysis note gives the expected composition per region and beam. The raw statistical precision of the data samples is 0.6% (CC0 $\pi$ ), 0.8% ( $\pi^0$ ) and 6% (multi- $\pi$ ) for the combined exposure. The neutrino-MC prediction in each region carries the flux, cross-section and re-interaction uncertainties of Table 23, including the detector variations. The regions carry a few-per-cent sensitivity to the signal normalisation in principle (the CC0 $\pi$  region holds 1 124 signal events, 9% of its neutrino MC, so a 30% signal excess would appear as a 2.8% excess), which is inside the 9–24% detector-variation uncertainty on the region prediction and so invisible to every pull of §12.5; that is what keeps the regions blind to the signal, and the response tree holds the signal at its central value in any case.

## 12 Control-region machinery

[INCLUSIVE]

## 12.1 Region overlaps

The four regions are disjoint from the signal region *by construction*: CC0 $\pi$  and multi- $\pi$  require  $n_\pi = 0$  and  $n_\pi \geq 2$  against the signal region's  $n_\pi = 1$ ; the  $\pi^0$  region requires the shower veto to fail; the cosmic region requires the opening-angle cut to fail. This was also checked event by event on the sideband-instrumented files: the number of events carrying both the signal-region flag and any control-region flag is exactly zero in the Monte Carlo and in the pseudo-data of every configuration. The regions are *not* mutually exclusive: multi- $\pi$  does not apply the shower veto, so events with two pion candidates and a failed shower veto belong to both multi- $\pi$  and  $\pi^0$  (47 of the 124 FHC multi- $\pi$  MC events, 68 of 159 RHC); all other pairs are disjoint. Any simultaneous fit of the regions must account for that overlap; the per-region comparisons of §12.5 do not need to.

## 12.2 Full-stack pseudo-data test

Table 23 of the analysis note compares the neutrino-MC throw with the neutrino MC alone. To exercise the exact path that beam-on data will take, the same comparison was run with pseudo-data consisting of the neutrino-MC throw *plus* an independent Poisson throw of the beam-off and dirt expectations, compared with the complete prediction. The data/prediction ratios are 0.996, 1.047, 0.992, 1.000 (FHC; CC0 $\pi$ , multi- $\pi$ ,  $\pi^0$ , cosmic), 0.992, 1.122, 1.012, 1.080 (RHC) and 0.995, 1.079, 1.006, 1.027 (combined), each within its statistical uncertainty (0.6–10%). This tests bookkeeping, not physics. Figures 6 and 7 show, for every region and both beam modes, the frozen distributions exactly as they will be drawn at unblinding, with the same full-stack pseudo-data in place of the beam-on points.

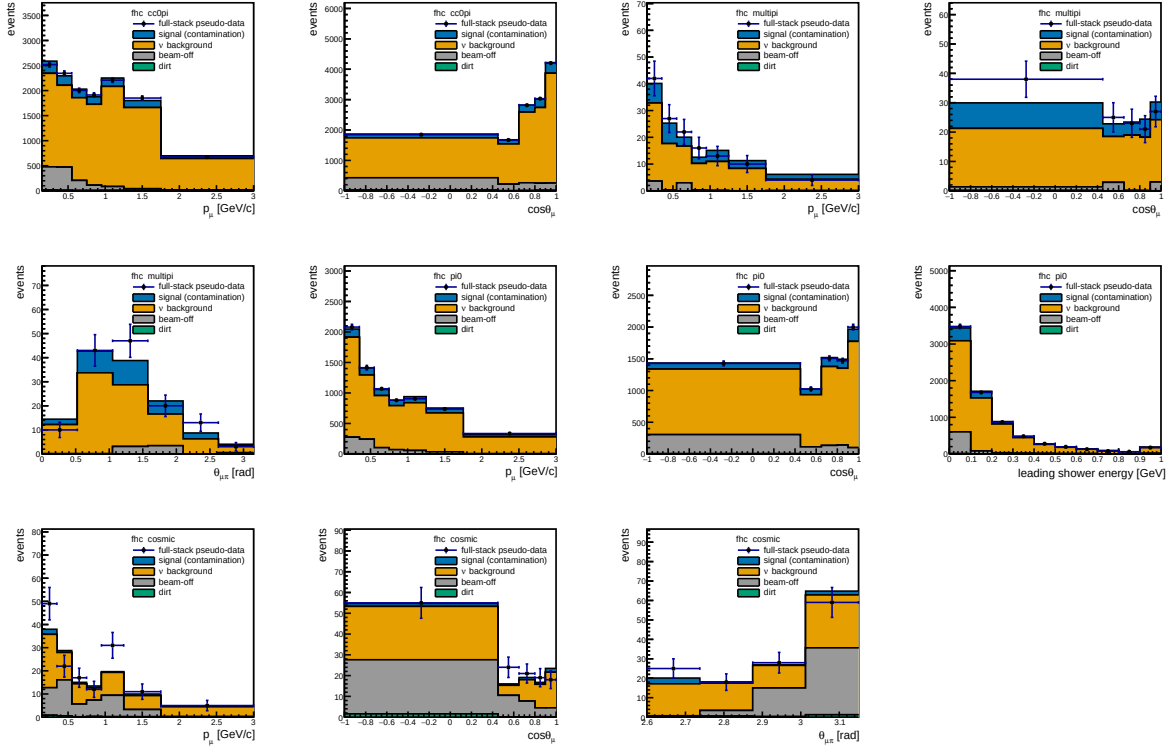


Figure 6: FHC control-region distributions at the frozen binning (the frozen plotting list): by row, CC0 $\pi$ , multi- $\pi$ ,  $\pi^0$  and cosmic;  $p_\mu$ ,  $\cos \theta_\mu$  and, where defined, the muon-pion opening angle or the leading shower energy. Stack: signal contamination, neutrino background, beam-off (per-run scaled) and dirt; points: full-stack pseudo-data (neutrino-MC throw plus a Poisson throw of the beam-off and dirt expectations). These are the plots the collaboration will see with beam-on points after opening; overflow is in the last bin.



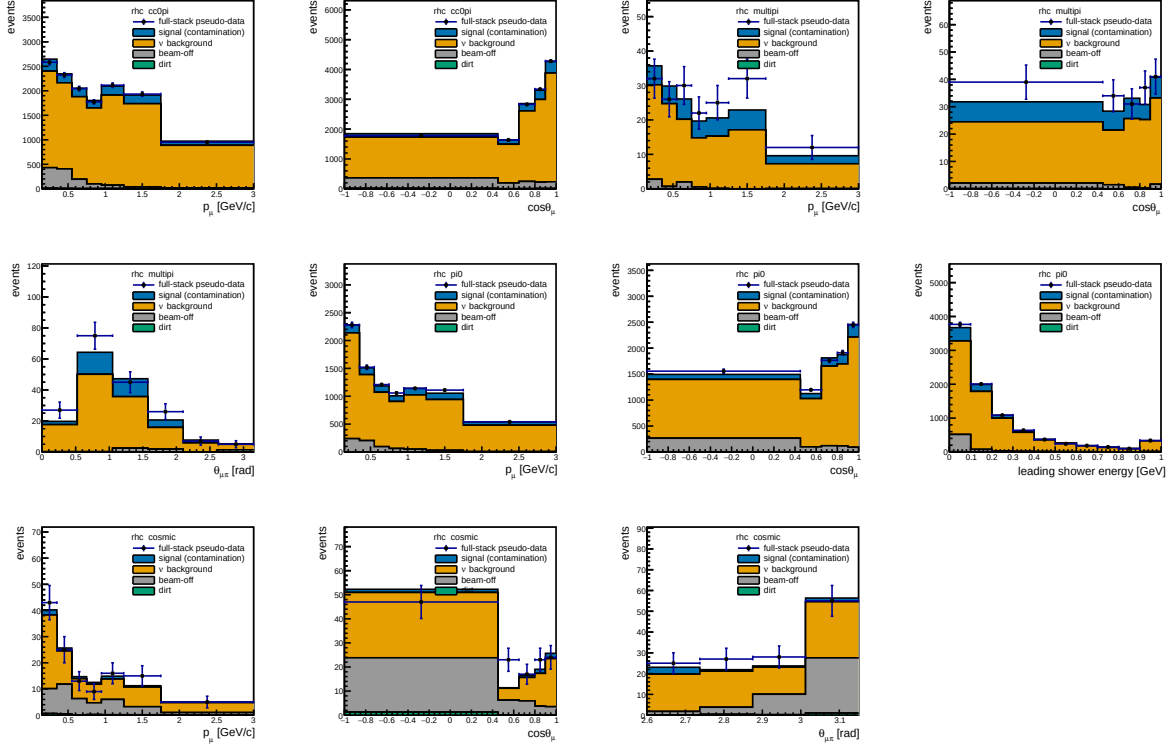


Figure 7: As Fig. 6, for RHC.

### 12.3 Transfer factors and control-region uncertainties

Whether a control-region discrepancy can be propagated into the signal region depends on the transfer factor  $T_{b,c} = N_b^{\text{SR}}/N_b^{\text{CR}_c}$  for each background class  $b$ : a region constrains a class only where  $T$  is small and the ratio is stable against the model variations. Table 22 gives the composition of each region by background class and the transfer factors.

Table 22: Composition of the signal region (SR) and of each control region by background class (CV-weighted neutrino MC, per-run POT-scaled; beam-off and dirt excluded), and the transfer factors  $T = N^{\text{SR}}/N^{\text{CR}}$  of each class. Classes are built from the selection’s truth counters:  $\nu_\mu\text{CC } 0\pi$  (no charged or neutral pion above threshold),  $1\pi^\pm$  outside the measured phase space, any  $\pi^0$ ,  $\geq 2\pi^\pm$  without  $\pi^0$ , out-of-fiducial-volume, and NC plus  $\nu_e\text{CC}$ . A region constrains a class only where  $T$  is small: CC0 $\pi$  constrains the 0 $\pi$  class ( $T \simeq 0.03$ ) and  $\pi^0$  the  $\pi^0$  class ( $T \simeq 0.02$ ; the adopted target class CC+NC  $\pi^0$ , last row of each block, is 50–51% of the region’s neutrino MC); the multi- $\pi$  region contains *fewer* multi-pion events than the signal region ( $T \simeq 2.5$ ) and is a weak constraint; the cosmic region constrains only the beam-off component (not listed: 53 EXT events in the region against 30 in the signal region,  $T = 0.57$ , FHC).

| Class                                                                                               | SR    | CC0 $\pi$ | multi- $\pi$ | $\pi^0$ | cosmic | $T_{\text{CC0}\pi}$ | $T_{\text{multi}}$ | $T_{\pi^0}$ |
|-----------------------------------------------------------------------------------------------------|-------|-----------|--------------|---------|--------|---------------------|--------------------|-------------|
| <i>FHC</i>                                                                                          |       |           |              |         |        |                     |                    |             |
| signal (contamination)                                                                              | 943   | 1 124     | 29.7         | 662     | 6.0    | 0.84                | 32                 | 1.43        |
| $\nu_\mu\text{CC } 0\pi$ (proton or other track as pion)                                            | 244   | 8 781     | 2.6          | 1 011   | 23.2   | 0.03                | 95                 | 0.24        |
| $\nu_\mu\text{CC } 1\pi^\pm$ outside the measured phase space                                       | 52.7  | 629       | 2.7          | 361     | 14.8   | 0.08                | 19                 | 0.15        |
| $\nu_\mu\text{CC}$ with a $\pi^0$                                                                   | 58.0  | 408       | 20.2         | 3 070   | 2.5    | 0.14                | 2.87               | 0.02        |
| $\nu_\mu\text{CC } \geq 2\pi^\pm$ , no $\pi^0$                                                      | 145   | 222       | 57.8         | 658     | 3.9    | 0.65                | 2.51               | 0.22        |
| out of fiducial volume                                                                              | 47.4  | 480       | 3.3          | 253     | 18.5   | 0.10                | 14                 | 0.19        |
| NC and $\nu_e\text{CC}$                                                                             | 89.5  | 549       | 7.1          | 632     | 6.4    | 0.16                | 13                 | 0.14        |
| total $\nu$ -MC                                                                                     | 1 581 | 12 192    | 124          | 6 646   | 75.3   |                     |                    |             |
| <i>CC+NC with a <math>\pi^0</math> (adopted <math>\pi^0</math> target; overlaps two rows above)</i> | 64.5  | 444       | 21.8         | 3 331   | 3.2    | 0.15                | 2.96               | 0.019       |
| <i>RHC</i>                                                                                          |       |           |              |         |        |                     |                    |             |
| signal (contamination)                                                                              | 1 015 | 1 206     | 34.9         | 780     | 6.4    | 0.84                | 29                 | 1.30        |
| $\nu_\mu\text{CC } 0\pi$ (proton or other track as pion)                                            | 257   | 8 981     | 2.4          | 1 117   | 25.7   | 0.03                | 108                | 0.23        |
| $\nu_\mu\text{CC } 1\pi^\pm$ outside the measured phase space                                       | 49.1  | 678       | 3.4          | 384     | 16.7   | 0.07                | 14                 | 0.13        |
| $\nu_\mu\text{CC}$ with a $\pi^0$                                                                   | 75.5  | 445       | 29.0         | 3 756   | 2.7    | 0.17                | 2.61               | 0.02        |
| $\nu_\mu\text{CC } \geq 2\pi^\pm$ , no $\pi^0$                                                      | 176   | 238       | 71.8         | 885     | 4.1    | 0.74                | 2.45               | 0.20        |
| out of fiducial volume                                                                              | 54.5  | 511       | 4.9          | 302     | 17.2   | 0.11                | 11                 | 0.18        |
| NC and $\nu_e\text{CC}$                                                                             | 108   | 599       | 12.5         | 857     | 8.9    | 0.18                | 8.59               | 0.13        |
| total $\nu$ -MC                                                                                     | 1 735 | 12 658    | 159          | 8 080   | 81.8   |                     |                    |             |
| <i>CC+NC with a <math>\pi^0</math> (adopted <math>\pi^0</math> target; overlaps two rows above)</i> | 81.0  | 491       | 31.1         | 4 118   | 3.6    | 0.17                | 2.61               | 0.020       |

The neutrino-MC prediction in each region carries the flux, cross-section and hadron re-interaction uncertainties evaluated directly from the multisim weights on the sideband-instrumented files with the same weight conventions as the extraction and the detector-variation term from the Run-4 detector-variation samples, which carry the region flags. Table 24 gives the uncertainty on the transfer factor of each region’s target class, computed as a ratio in every universe and variation.

Table 23: Normalisation uncertainty (%) of the neutrino-MC prediction in the signal region and in each control region: flux (PPFX), cross-section (GENIE multisim) and hadron re-interaction multisims evaluated on the sideband-instrumented files with the extraction’s weight conventions, and the detector term from the Run-4 detector-variation samples, which carry the region flags (eight variations against the detector-variation central value, summed in quadrature as in the extraction). “Total” is the quadrature sum. The last column is the correlation, across the flux universes, between the region’s yield and the signal-region yield: the flux term is a common normalisation and cancels in a control-region-to-signal-region transfer (Table 24). The  $\nu$ -MC yields apply the safe-weight rule of Eq. 4 of the analysis note, as do Tables 10 of the analysis note and 24 and the pseudo-data throws; the yields agree between the tables.

| Beam | Region        | $\nu$ -MC yield | flux | cross section | re-interaction | detector | total | corr. with SR (flux) |
|------|---------------|-----------------|------|---------------|----------------|----------|-------|----------------------|
| FHC  | signal region | 1 581           | 18.0 | 16.9          | 2.7            | 20.7     | 32.3  | 1.00                 |
| FHC  | CC0 $\pi$     | 12 192          | 16.7 | 11.9          | 1.1            | 21.8     | 30.0  | 1.00                 |
| FHC  | multi- $\pi$  | 124             | 20.0 | 20.5          | 5.5            | 18.5     | 34.6  | 0.99                 |
| FHC  | $\pi^0$       | 6 646           | 18.8 | 15.8          | 2.4            | 21.8     | 32.9  | 1.00                 |
| FHC  | cosmic        | 75              | 15.9 | 13.2          | 1.8            | 24.3     | 31.9  | 0.98                 |
| RHC  | signal region | 1 735           | 16.2 | 16.2          | 2.7            | 14.2     | 27.1  | 1.00                 |
| RHC  | CC0 $\pi$     | 12 658          | 15.1 | 11.8          | 1.1            | 8.7      | 21.1  | 0.99                 |
| RHC  | multi- $\pi$  | 159             | 17.0 | 19.3          | 6.1            | 19.4     | 32.8  | 0.99                 |
| RHC  | $\pi^0$       | 8 080           | 16.3 | 15.4          | 2.8            | 7.3      | 23.7  | 1.00                 |
| RHC  | cosmic        | 82              | 15.2 | 12.8          | 2.5            | 11.9     | 23.3  | 0.99                 |

Table 24: Transfer factor  $T = N^{\text{SR}}/N^{\text{CR}}$  of each control region’s target class and its fractional uncertainty (%), with  $T$  recomputed as a ratio in every universe so that the numerator–denominator covariance is included. The flux term, 15–20% on either yield alone (Table 23), cancels to 2–3% in the ratio, as the yield correlations anticipated; the cross-section term does not cancel, because it changes the composition of the two regions differently, and the detector term is the largest. The detector entries carry the finite statistics of the detector-variation samples (a few thousand events per region), so those for the smaller classes are upper limits rather than measurements; the multi- $\pi$  transfer’s detector entry is below its own Monte-Carlo-statistical floor (§12.3). The “all neutrino MC” row uses the total neutrino MC of the CC0 $\pi$  region as a reference. The Monte-Carlo-statistical floor of the detector entries, estimated from two disjoint halves of the variation samples, is given in the text; for the adopted CC+NC  $\pi^0$  class it is quoted in parentheses after the detector entry and, where it exceeds the entry, it replaces it in the total.

| Beam | Region       | Target class                 | $T$    | flux | cross section | re-int. | MC stat | detector    | total |
|------|--------------|------------------------------|--------|------|---------------|---------|---------|-------------|-------|
| FHC  | CC0 $\pi$    | $\nu_\mu$ CC 0 $\pi$         | 0.0278 | 3.3  | 11.7          | 6.7     | 2.5     | 27.1        | 30.5  |
| FHC  | multi- $\pi$ | $\nu_\mu$ CC $\geq 2\pi^\pm$ | 2.5094 | 3.0  | 4.7           | 2.6     | 5.1     | 18.5        | 20.2  |
| FHC  | $\pi^0$      | $\nu_\mu$ CC with $\pi^0$    | 0.0189 | 2.6  | 6.1           | 1.8     | 4.4     | 15.1        | 17.1  |
| FHC  | CC0 $\pi$    | all neutrino MC              | 0.1296 | 1.7  | 13.2          | 2.6     | 1.0     | 9.7         | 16.7  |
| FHC  | $\pi^0$      | CC+NC with $\pi^0$ (adopted) | 0.0194 | 2.2  | 5.8           | 1.9     | 4.2     | 14.9 (12.1) | 16.8  |
| RHC  | CC0 $\pi$    | $\nu_\mu$ CC 0 $\pi$         | 0.0286 | 3.0  | 11.5          | 6.6     | 2.0     | 30.6        | 33.6  |
| RHC  | multi- $\pi$ | $\nu_\mu$ CC $\geq 2\pi^\pm$ | 2.4527 | 1.9  | 4.0           | 2.8     | 3.9     | 21.3        | 22.3  |
| RHC  | $\pi^0$      | $\nu_\mu$ CC with $\pi^0$    | 0.0201 | 1.8  | 6.3           | 1.8     | 3.3     | 11.5        | 13.7  |
| RHC  | CC0 $\pi$    | all neutrino MC              | 0.1371 | 2.0  | 12.5          | 2.6     | 0.8     | 9.8         | 16.3  |
| RHC  | $\pi^0$      | CC+NC with $\pi^0$ (adopted) | 0.0197 | 1.7  | 6.1           | 1.6     | 3.1     | 11.8 (18.0) | 19.4  |

**Monte-Carlo-statistical floor of the detector term.** The detector entries of Tables 23 and 24 are quadrature sums of eight differences between finite samples, so part of each is the statistics of the variation samples rather than a detector effect. That floor was estimated by

evaluating every shift on two disjoint halves of the generated events (split on the true neutrino energy, which is identical across the central-value and variation samples): half the difference of the two half-sample shifts estimates the statistical noise of the full-sample shift, and its quadrature over the eight knobs is the floor. For the region yields the floors are small: 1.4% (FHC) and 1.7% (RHC) against signal-region detector terms of 20.7% and 14.2%, and 1–2% for the  $\text{CC}0\pi$  and  $\pi^0$  regions. For the transfer factors they are not: the  $\text{CC}0\pi$  target-class term (27% FHC, 31% RHC) sits well above its floor (9%, 6%) and is a genuine detector sensitivity, but the  $\pi^0$  target-class term (15% FHC, 11% RHC) is comparable to or smaller than its floor (13%, 17%), so it is an upper limit set by the variation-sample statistics rather than a measurement (the same holds for the adopted  $\text{CC}+\text{NC}$   $\pi^0$  class: 15% against a 12% floor in FHC, 12% against 18% in RHC), and the multi- $\pi$  target-class term (18–21%) sits below its floor (25% FHC, 23% RHC); the half-sample estimator itself becomes unstable for the smallest classes, where floors of several hundred per cent appear. Any fitted correction under item 4(ii) of the procedure must therefore carry the *larger* of the quoted transfer uncertainty and its floor; for the  $\pi^0$  class that is the floor.

**Two limits of the control-region set.** About 29% of the FHC neutrino background has no region: NC and  $\nu_e\text{CC}$  without a  $\pi^0$  ( $\approx 83$  events), out-of-fiducial-volume (47) and  $\text{CC}1\pi^\pm$  outside the measured phase space (53), the last subtracted from the same generator with a reconstruction bias of +30–66% for below-threshold pions (Table 7 of the analysis note). And the  $\text{CC}0\pi$  region constrains the  $\text{CC}0\pi$  *rate*, not the proton-as-pion misidentification that carries it into the signal region ( $T = 0.03$ , detector-dominated); an inverted pion-PID, proton-enriched region would test the misidentification without touching the signal region and is not part of this request.

## 12.4 The $\pi^0$ region with and without the final multiplicity cut

The frozen  $\pi^0$  definition drops the final multiplicity cut (Sec. 11). The comparison with the definition that retains it was produced from a reprocess of the Monte Carlo, the dirt and the per-run beam-off samples into a separate set of processed files that carry the alternative region flag and the two multiplicities, the Run-4 detector-variation samples included (nineteen reprocessed, the eighteen the detector term uses); no beam-on file was touched, and the frozen flag re-evaluated on the reprocessed files reproduces the yields of Table 24 of the analysis note and the uncertainties of §12.3. Table 25 gives the two definitions side by side.

Table 25: The  $\pi^0$  control region without (frozen) and with the final multiplicity cut, full exposure, CV-weighted and per-run scaled. Purity and contamination are shares of the neutrino-MC prediction; the target share is of the full stack. The transfer-factor uncertainty is the multisim (flux, cross section, reinteraction) and MC-statistical total of §12.3; the detector term is the full-sample quadrature over the eight knobs, as in Table 24, with its MC-statistical floor from the split-sample method, and the total takes the larger of the two as there. The non-target row is all neutrino background other than the target class, the leverage denominator of Sec. 11; the CC0 $\pi$  line is part of it.

|                                                                                     | FHC        |            | RHC        |            |
|-------------------------------------------------------------------------------------|------------|------------|------------|------------|
|                                                                                     | frozen     | with cut   | frozen     | with cut   |
| neutrino MC                                                                         | 6646       | 5889       | 8080       | 6000       |
| target CC+NC $\pi^0$                                                                | 3331       | 2974       | 4118       | 3600       |
| signal                                                                              | 662        | 598        | 780        | 600        |
| non-target neutrino background                                                      | 2654       | 2318       | 3183       | 2400       |
| of which $\nu_\mu$ CC 0 $\pi$                                                       | 1011       | 953        | 1117       | 1000       |
| beam-off                                                                            | 778        | 733        | 678        | 600        |
| dirt                                                                                | 12         | 11         | 9          | 10         |
| target purity (of $\nu$ -MC)                                                        | 50.1%      | 50.5%      | 51.0%      | 51.0%      |
| target share (of full stack)                                                        | 44.8%      | 44.8%      | 47.0%      | 47.0%      |
| signal contamination                                                                | 10.0%      | 10.1%      | 9.6%       | 10.0%      |
| beam-off share (of full stack)                                                      | 10.5%      | 11.1%      | 7.7%       | 8.0%       |
| overlap with multi- $\pi$ ( $\nu$ -MC events)                                       | 46.9       | 46.9       | 68.1       | 60.0       |
| leverage: non-target / target                                                       | 0.80       | 0.78       | 0.77       | 0.75       |
| $T$ (target)                                                                        | 0.0194     | 0.0217     | 0.0197     | 0.0200     |
| $T$ uncertainty, multisim+MC-stat                                                   | 7.8%       | 7.6%       | 7.3%       | 7.5%       |
| $T$ detector term / MC-stat floor                                                   | 14.9/12.1% | 15.7/12.5% | 11.8/18.0% | 11.4/17.0% |
| $T$ uncertainty, total (larger of term and floor added in quadrature)               | 16.8%      | 17.4%      | 19.4%      | 18.0%      |
| $E_{\text{shr}}$ shape, events kept vs removed by the cut, unit-area $\chi^2$ /bins | 10.4/10    |            | 121/10     |            |
| largest per-bin change of the $E_{\text{shr}}$ shape, frozen vs with cut            | 0.2%       |            | 0.7%       |            |

The cut removes 11% (FHC) and 14% (RHC) of the region, and it removes every component in nearly the same proportion: the target class keeps 89/87%, the signal 90/90%, the beam-off 94/93%. The composition is therefore unchanged to within half a percent in purity and identical in the target’s share of the full stack, and the overlap with multi- $\pi$  is identical because multi- $\pi$  requires the cut anyway. The leading-shower-energy distribution the comparison will be made on is tested between the events the cut keeps and the events it removes, which are disjoint samples (the two definitions themselves are nested, so a test between them would understate the difference): in FHC the removed events have the same shape ( $\chi^2$  10.4/10); in RHC they do not (121/10; the removed 14% has a softer shower spectrum, 36% against 41% in the first bin). Because the removed fraction is small, the frozen distribution differs from the with-cut one by at most 0.7% of unit area in any bin (RHC; 0.2% FHC), well inside the shape test’s resolution. The transfer factor rises by 12/15% with the smaller denominator, and its uncertainties are unchanged within their precision. What the cut changes is the multiplicity distributions themselves (Fig. 8): the frozen region extends to five and more primary tracks and four and more non-proton-like tracks, the topologies the multi-shower argument of Sec. 11 anticipated, and they are 11–14% of the region, not a dominant population. One limitation of the frozen definition follows from it: the 11–14% of the region that fails the signal-region multiplicity cut cannot populate the signal region, so a data/MC discrepancy confined to those events would appear in the region total and be attributed to the target class under item 4 of §12.5. That is accepted for the larger sample; the stored multiplicities allow the reconstruction-failure investigation of item 4(i) to locate such a discrepancy if one appears.

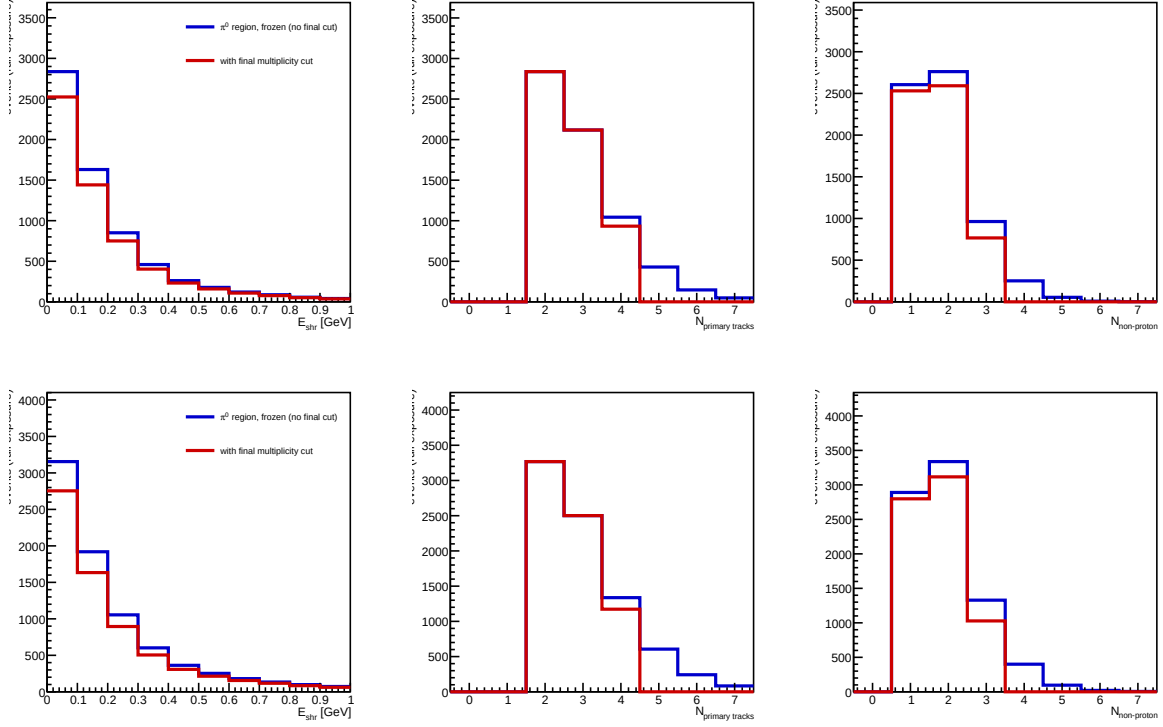


Figure 8: The  $\pi^0$  region without (blue) and with (red) the final multiplicity cut, FHC (top) and RHC (bottom), neutrino MC at full exposure: the leading shower energy, the number of primary tracks and the number of non-proton-like tracks.

The frozen definition is kept: it is the larger and more inclusive sample, and the study shows that the choice is immaterial to the region's composition, purity and transfer uncertainty, and changes the frozen shape by less than a percent, so nothing about the eligibility question of §12.5 item 4(ii) turns on it. The premise that retaining the cut would sculpt the region towards single-shower events is not borne out at the level of the composition; it holds only for the multiplicity distributions, which are not compared. The collaboration is asked to ratify the frozen definition with this comparison in view.

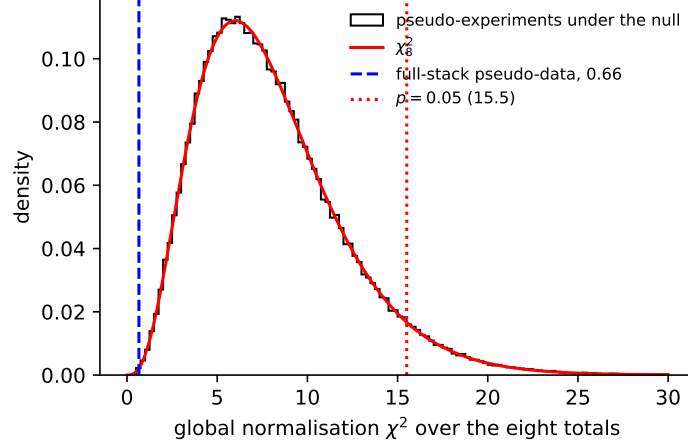


Figure 9: Null calibration of the global normalisation  $\chi^2$  over the eight (region, mode) totals:  $2 \times 10^5$  pseudo-experiments with the prediction drawn from its pre-fit covariance and Poisson-fluctuated data, against the  $\chi^2_8$  density. The full-stack pseudo-data value 0.66 is marked; it is low because the pseudo-data is a Poisson throw of the same prediction and carries none of the prediction’s systematic variation.

## 12.5 Frozen procedure for opening the control regions

The following is the analysers’ proposed procedure, to be ratified with the approval; once ratified it is frozen and any departure from it is itself a change requiring approval.

1. **What is examined.** For each of the four regions: the event total, and the reconstructed distributions of  $p_\mu$ ,  $\cos\theta_\mu$  and the muon–pion opening angle, and for the  $\pi^0$  region the leading shower energy (the primary-track and non-proton multiplicities are stored by the frozen skim, §12.4, but are not in the pre-declared list and are not tested; they are available to the reconstruction-failure investigation of item 4(i) only), at the binning of the released observables where one exists and otherwise at the binning fixed in the frozen plotting list and shown, with full-stack pseudo-data, in Figs. 6 and 7. FHC and RHC are examined separately; the combined configuration is a cross-check only, and within each mode the event total is also compared per run period (FHC Runs 1, 2, 4, 5; RHC Runs 1, 2, 3, 4), which is what makes a single-run excursion detectable. Data are compared with the complete prediction, neutrino MC + beam-off + dirt, with the run-period beam-off scaling of Sec. 5 of the analysis note.
2. **Statistics.** Three quantities are computed for each region and beam mode, all against the *pre-fit* prediction. (a) The normalisation pull,  $(N_{\text{data}} - N_{\text{pred}})/\sigma$ , with  $\sigma^2 = N_{\text{data}} + \mathbf{1}^\top V_{\text{pred}} \mathbf{1}$  and  $V_{\text{pred}}$  the summed covariance of the complete prediction (neutrino MC, beam-off and dirt: their finite statistics, flux, cross-section, re-interaction and detector terms, with all correlations among the prediction components). Because  $V_{\text{pred}}$  carries the 21–35% prediction systematics of Table 23, this pull cannot fail a discrepancy smaller than about 40–60%; it is therefore reported *together with* the pull against the statistical and detector terms alone, whose resolution (9–24%) is set by the detector term. (b) A shape  $\chi^2$  in the pre-declared distributions with the normalisation marginalised: the prediction is scaled to the data total and the covariance is projected onto shape (the normalisation mode removed), so that the common-mode flux and cross-section terms do not make the test powerless, as the closure  $\chi^2$  of Sec. 10.8 of the analysis note shows they otherwise would. (c) A *global* normalisation  $\chi^2$  over the eight (region, mode) totals with the full  $8 \times 8$  pre-fit covariance, which is the single pass/fail statistic. All three are implemented in the protocol



implementation (flux, cross-section and re-interaction multisims index-matched across the two modes, the eight detector knobs as correlated unisims, MC, beam-off, dirt and data statistics) and were exercised on the full-stack pseudo-data of §12.2: the global  $\chi^2$  is 0.66/8 ( $p = 1.00$ ), the sixteen pulls lie within  $\pm 0.5$  and all twenty-two shape tests have  $p > 0.47$ . The same run gives the power of the test: a discrepancy confined to one region fails the set at 95% confidence beyond 12% of the CC0 $\pi$  or  $\pi^0$  total and beyond 50–62% of the multi- $\pi$  or cosmic total, while a *common* normalisation shift of all eight totals, which the 40–91% correlated flux, cross-section and detector terms absorb, fails it only beyond 58%. That is the test’s blind spot, stated: it is a test of the background model region by region, not of the overall normalisation, which the signal-region measurement carries as its flux and cross-section uncertainty.

3. **Acceptable agreement.** The control regions pass as a set if the global  $\chi^2$  of item 2(c) has  $p > 0.05$ ; the expected false-failure rate is then 5% by construction, and it has been calibrated:  $2 \times 10^5$  pseudo-experiments under the null (prediction drawn from its pre-fit covariance, data Poisson-fluctuated,  $\chi^2$  with the nominal covariance) fail the nominal  $p < 0.05$  threshold in 5.1% of cases and  $p < 0.01$  in 1.1% (Fig. 9). The calibration treats the covariance as known; the finite-universe and detector-unisim sampling noise is not re-thrown, which is the residual approximation in the declared threshold. The per-region pulls and shape tests of items 2(a) and 2(b) are diagnostic: a region with  $|\text{pull}| > 2$  (complete prediction covariance) or shape  $p < 0.05$  is flagged for the response tree of item 4 even when the set passes, but the eight independent flags are not themselves pass/fail criteria (with sixteen pulls and twenty-two shape tests at these thresholds about two false flags are expected, and at least one with 85% probability). A single-bin excursion beyond  $3\sigma$  with a passing total is recorded and examined for a reconstruction pathology but does not by itself flag the region.
4. **Permitted responses and the rule for choosing between them.** The response tree is entered when the global test of item 3 fails *or* a region is flagged; a flagged region within a passing set may receive at most response (b) below, never a fitted correction. Regions are treated in the order CC0 $\pi$ ,  $\pi^0$ , then the two diagnostic regions, because the CC0 $\pi$  class is 15% of the  $\pi^0$  region and a CC0 $\pi$  treatment changes the  $\pi^0$  prediction before it is judged. (i) *Localised reconstruction failure.* If one run period’s total in the region pulls beyond  $3\sigma$  (statistical+detector) while the other periods pass, or one bin of a pre-declared distribution exceeds  $3\sigma$  while the rest of the distribution passes, the excursion triggers a reconstruction-quality investigation, not a model correction and not an automatic exclusion: a  $3\sigma$  excursion is expected to occur by chance somewhere among the periods, regions and bins examined, and a control-region discrepancy alone does not show that the signal region of that period is unusable. The period is excluded only if an independent detector or processing defect is established (detector-quality monitors, an independent inclusive sample, reproduction in more than one pre-declared variable, or a known production or calibration boundary); the exclusion then removes the period from *both* the control-region comparison and the measurement with its exposure removed from the normalisation, changes the flux weighting, the detector covariance and the response, and therefore requires a new frozen release and explicit approval. If no defect is established the period stays in the analysis and the discrepancy is treated through the model responses below. No cut or binning may be changed and a single bin is never removed. (ii) *Fitted normalisation correction (c).* Permitted only for CC0 $\pi$  and  $\pi^0$  (target-class purity above 50% of the neutrino-MC prediction; target classes  $\nu_\mu$ CC 0 $\pi$  and CC+NC  $\pi^0$ , adopted, ratification requested in Sec. 11) and only if the region’s shape test 2(b) passes, so that the discrepancy is consistent with a pure normalisation shift: a one-parameter fit of the target class, to FHC and RHC simultaneously, applied if it improves the fit by a likelihood-ratio test at  $p < 0.01$ . The  $\pi^0$  region meets the purity rule narrowly (50.1/51.0%) and with a class and denominator

settled after the purities were known (Sec. 11), so its eligibility is *not* pre-authorised by this document: a fitted  $\pi^0$  correction may be *proposed* only after the pre-fit control-region result is documented, and the fit model, the leverage of the fixed components, the transfer-factor uncertainty and its Monte-Carlo-statistical floor are reviewed by the collaboration before the correction is propagated to the signal region.  $\text{CC0}\pi$  (72%) is pre-authorised under the same rule. The two fitted regions are disjoint by construction ( $\text{CC0}\pi$  requires the shower veto to pass,  $\pi^0$  requires it to fail), so a simultaneous fit counts no event twice; the multi- $\pi/\pi^0$  overlap (47/124 FHC, 68/159 RHC events) concerns a diagnostic region that is never fitted, and the global test of item 2(c) carries it as a covariance between totals. What the fit updates is stated: the normalisation of the target class only, re-evaluated in every systematic universe so that its correlation with the flux and cross-section terms of the signal-region prediction is unchanged (the  $1 + B/S$  coherence of Sec. 14.2 of the analysis note is retained, with  $B$  rescaled); the fitted uncertainty below replaces the class's prior normalisation uncertainty and enters as an independent term; the signal/background joint covariance is not otherwise modified. Its uncertainty is the transfer-factor uncertainty of Table 24 or its Monte-Carlo-statistical floor, whichever is larger (§12.3), added in quadrature to the leverage of the components held fixed in the fit: the prediction uncertainties of the non-target neutrino background, the beam-off and the signal contamination, each multiplied by its share of the target class (0.80, 0.23 and 0.20 in the  $\pi^0$  region; 0.26, 0.15 and 0.13 in  $\text{CC0}\pi$ ; Sec. 11). (iii) *Added uncertainty (b)*. Otherwise the central prediction is left unchanged and the full observed discrepancy, as a fraction of the target class in the region, is assigned as a one-standard-deviation normalisation uncertainty on that class in the signal region through its transfer factor. (iv) *Model revision (d)*. If the post-treatment global  $\chi^2$ , computed with the fitted normalisation or the added uncertainty in place and reported separately from the pre-fit value of item 3, still has  $p < 0.05$ , or a region remains flagged, the background model is revised and renewed collaboration approval is required before the signal region is opened. The pass/fail test of item 3 is always the pre-fit one; the post-treatment value is labelled as such wherever it is quoted. The response letters used here and in Sec. 11 are (a) no action, (b) added uncertainty, (c) fitted normalisation correction, (d) model revision. FHC and RHC share one background model: a class that passes in one mode and fails in the other is treated under (b), never corrected in one mode alone. The signal contamination of each region (1 124 events in  $\text{CC0}\pi$  and 662 in  $\pi^0$  for FHC, more than the 943 signal events of the signal region) is held at the central-value tune in any fit, which is what keeps the regions blind to the signal: a 30% signal excess would move the  $\text{CC0}\pi$  total by 2.8%, well inside the 9–24% detector-limited resolution of every pull. If more than one region is flagged, no independent corrections are combined without accounting for shared events and correlated systematic variations: in particular, multi- $\pi$  and  $\pi^0$  share events (§12.1) and are treated jointly, or multi- $\pi$  is designated diagnostic only, which its transfer factor already argues for.

5. **Prohibited responses.** Changing any signal-selection cut; changing the binning of any released observable; choosing a generator or tune after inspecting the regions; bin-by-bin reweighting; and any use of the beam-on signal-region sample.
6. **Sign-off.** The analysis conveners, the flux and detector-systematics contacts, and the collaboration review committee, on the basis of a short report containing the comparisons of item 1 and the response chosen from item 4.

## 12.6 Scope: inclusive regions only

The request of Sec. 11 covers the four inclusive regions. The proton-tagged regions of §12.7 apply the proton tag rather than inverting it and therefore do not constrain the  $\approx 13\%$  proton mis-tag background that most affects the hadronic observables; the inverted-proton-PID region

that would do so has not yet been implemented. The proton-tagged control-region programme is consequently *not* ready for equivalent approval, and opening the inherited proton-tagged regions, while useful diagnostically, would not validate the proton-tagged background model.

## 12.7 Control regions for the proton-tagged subsample

[PROTON-TAGGED]

The proton-tagged selection inherits its cut logic from the inclusive selection, and with it the four sideband definitions of Sec. 11.6 of the analysis note: the reprocessed proton-tagged files already carry the four region flags. No new definitions or reprocessing are required to exercise the background model of the W/TKI measurement; the same four regions can be evaluated either as inherited (the inclusive phase space) or with a reconstructed tagged proton additionally required, which places them in the same phase space as the measurement itself. Table 26 gives both.

Three things follow from the numbers. First, all four regions are *more* signal-depleted for the proton-tagged signal than they were for the inclusive one: 5.8–7.2% signal contamination in  $\text{CC0}\pi$  and  $\pi^0$ , against 9.2–10.0% for the same regions in the inclusive phase space and a 52% purity in the signal region, because requiring a proton does not make a pion-less or  $\pi^0$  event look like signal. Second, the proton requirement costs about a third of the statistics ( $\text{CC0}\pi$ : 12 181  $\rightarrow$  8 386 pseudo-data events for FHC) but suppresses the beam-off contribution by roughly a factor six (1 358  $\rightarrow$  225, FHC  $\text{CC0}\pi$ , Table 26): cosmic-induced events rarely contain a reconstructed proton above threshold, so the proton tag is itself a strong cosmic rejector. This is worth noting for the systematic budget; the EXT normalisation matters far less in the proton-tagged measurement than in the inclusive one. Third, the pseudo-data to neutrino-MC ratios of the two large regions stay at 0.987–1.007 with the tag, so the per-run bookkeeping holds in the proton-tagged phase space as well; the small regions scatter more widely (0.85–1.52), which on 37–62 pseudo-data events is Poisson scatter of one to two standard deviations, not a normalisation effect.

**What these regions cannot do.** All four *apply* the proton tag rather than invert it, so none of them constrains the proton *mis-tag* background; the  $\sim 13\%$  of tagged protons that are really a pion or another track, which smears  $W_{\pi p}$  and the TKI variables directly. Testing that requires a region defined by inverting the proton PID (selecting candidates that pass the momentum and containment requirements but fail the LLR cut), which does need a new flag in the proton-tagged selection and a reprocess. Slicing a measured W/TKI observable is *not* an alternative: because the signal definition is inclusive in those variables, cutting on them splits the signal region rather than depleting it. Explicitly, on the Run-1 FHC overlay the signal region has 53.7% purity and the slices  $W_{\pi p} > 1.5$  GeV,  $W_{\pi p} < 1.15$  GeV,  $p_n > 0.8$  GeV/c and  $\delta p_T > 0.7$  GeV/c give 50.0%, 48.9%, 47.3% and 52.0% respectively; no depletion at all. Even the kinematically forbidden region  $W_{\text{had}} < m_\pi + m_p$  retains 53.4% signal, because the calorimetric mass is systematically under-reconstructed and true signal populates it through resolution.

Table 26: Sideband yields for the proton-tagged subsample at full exposure, evaluated with the inherited definitions both as-is and with a tagged proton additionally required. “Sig.” is the proton-tagged signal contamination of the neutrino MC.  $N_{\text{data}}/N_{\nu\text{MC}}$  compares the Poisson-thrown pseudo-data with the CV-weighted neutrino MC; EXT and dirt are listed for scale.

| Beam                                                           | Region       | $N_{\text{data}}$ | Signal | $\nu$ bkg | EXT   | Sig. frac. | $N_{\text{data}}/N_{\nu\text{MC}}$ |
|----------------------------------------------------------------|--------------|-------------------|--------|-----------|-------|------------|------------------------------------|
| <i>inherited definitions (inclusive phase space)</i>           |              |                   |        |           |       |            |                                    |
| FHC                                                            | CC0 $\pi$    | 12 181            | 705    | 11 487    | 1 358 | 5.8%       | 0.999                              |
| FHC                                                            | multi- $\pi$ | 125               | 30     | 94        | 7.2   | 23.9%      | 1.012                              |
| FHC                                                            | $\pi^0$      | 6 628             | 466    | 6 181     | 778   | 7.0%       | 0.997                              |
| FHC                                                            | cosmic       | 82                | 3.5    | 72        | 53    | 4.6%       | 1.089                              |
| RHC                                                            | CC0 $\pi$    | 12 590            | 731    | 11 927    | 1 214 | 5.8%       | 0.995                              |
| RHC                                                            | multi- $\pi$ | 175               | 34     | 125       | 5.9   | 21.5%      | 1.101                              |
| RHC                                                            | $\pi^0$      | 8 176             | 524    | 7 556     | 678   | 6.5%       | 1.012                              |
| RHC                                                            | cosmic       | 81                | 3.5    | 78        | 42    | 4.3%       | 0.990                              |
| Comb                                                           | CC0 $\pi$    | 24 771            | 1 436  | 23 414    | 2 572 | 5.8%       | 0.997                              |
| Comb                                                           | multi- $\pi$ | 300               | 64     | 219       | 13    | 22.6%      | 1.062                              |
| Comb                                                           | $\pi^0$      | 14 804            | 990    | 13 736    | 1 455 | 6.7%       | 1.005                              |
| Comb                                                           | cosmic       | 163               | 6.9    | 150       | 94    | 4.4%       | 1.037                              |
| <i>with a tagged proton required (measurement phase space)</i> |              |                   |        |           |       |            |                                    |
| FHC                                                            | CC0 $\pi$    | 8 386             | 486    | 7 886     | 225   | 5.8%       | 1.002                              |
| FHC                                                            | multi- $\pi$ | 59                | 9.8    | 46        | 0.2   | 17.7%      | 1.064                              |
| FHC                                                            | $\pi^0$      | 4 011             | 293    | 3 769     | 189   | 7.2%       | 0.987                              |
| FHC                                                            | cosmic       | 39                | 1.7    | 24        | 0.1   | 6.5%       | 1.517                              |
| RHC                                                            | CC0 $\pi$    | 8 621             | 500    | 8 127     | 197   | 5.8%       | 0.999                              |
| RHC                                                            | multi- $\pi$ | 62                | 14     | 58        | 1.3   | 19.8%      | 0.852                              |
| RHC                                                            | $\pi^0$      | 4 902             | 333    | 4 534     | 162   | 6.8%       | 1.007                              |
| RHC                                                            | cosmic       | 37                | 2.3    | 29        | 0.5   | 7.5%       | 1.187                              |
| Comb                                                           | CC0 $\pi$    | 17 007            | 985    | 16 013    | 422   | 5.8%       | 1.000                              |
| Comb                                                           | multi- $\pi$ | 121               | 24     | 104       | 1.5   | 18.9%      | 0.944                              |
| Comb                                                           | $\pi^0$      | 8 913             | 626    | 8 303     | 351   | 7.0%       | 0.998                              |
| Comb                                                           | cosmic       | 76                | 4.0    | 53        | 0.6   | 7.1%       | 1.336                              |

## Part IV

# Selection, reconstruction and infrastructure

## 13 Selection studies and supporting material

[INCLUSIVE]

The studies in this section are referred to from the analysis note; they are kept here so that the note carries only the current-release argument.

### 13.1 Selection-optimisation study

**Why the phase space and the opening-angle cut are what they are.** The measured phase space and the  $\theta_{\mu\pi} < 2.6$  rad cut were chosen on the Run-1 FHC sample with a reference phase space of  $p_\mu, p_\pi > 0.1$  GeV/c; the numbers in this paragraph and in Table 27 are that study and document the choice. The full-exposure yields, efficiency and purity of the released measurement are those of Table 10 of the analysis note above and are quoted from there.

- Signal MC (POT-weighted): 324.7 events
- Total background: 243.6 ( $\approx 199.7$   $\nu$ -bkg + 40.1 EXT + 3.8 dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- **Signal purity: 57.1%**

- **Signal efficiency: 12.1–12.6%**
- **Measured phase space (reported result): efficiency 15.6%, purity 54.9%** (reference phase-space study: selected signal 311.9, generated 1999, background 256.4; Table 27)

*Per-run scaling.* The yields above and in Table 10 of the analysis note use the per-run POT factors of Table 4 of the analysis note (each run scaled by its own  $D_{\text{run}}/\text{MC}_{\text{run}}$ ), and the EXT by the per-run trigger ratios (per-run gate ratios, Sec. 5 of the analysis note). A single per-mode factor would differ from the per-run scheme by about 2% in the FHC final-cut signal, a small and nearly uniform re-weighting of the run mixture that moves efficiency and purity by  $< 0.5\%$ ; the per-run scheme is used because it is what real beam-on data require. The fake data are thrown from the same per-run-scaled prediction, so the closure tests the bookkeeping of the scheme, not its correctness against data.

**Measured phase space and efficiency turn-on.** The reference selection above has near-dead efficiency at low momentum and at the large opening-angle extreme: from the fine truth distributions, the per-bin efficiency is only  $\sim 2\text{--}5\%$  for  $p_\mu, p_\pi < 0.15 \text{ GeV}/c$  (the 10/20 cm track-length cuts) and the region above the reco  $\theta_{\mu\pi} = 2.6 \text{ rad}$  cut is dominated by cosmics ( $\sim 7\%$  purity). The measured phase space  $p_\mu > 0.15, p_\pi > 0.175 \text{ GeV}/c, \theta_{\mu\pi} < 2.6 \text{ rad}$  (Sec. 6 of the analysis note) raises and flattens the efficiency to  $\sim 13\text{--}20\%$  (Fig. 19 of the analysis note). Table 27 compares the selection across the iterations. Two conclusions: (i) the 2.6 rad *upper* cut is kept: on the full exposure the events it removes are exactly those of the cosmic control region (Table 24 of the analysis note), in FHC 6.0 signal, 69 neutrino background, 53 beam-off and 2.2 dirt, a signal purity of 4.6%; dropping it would lower the signal-region purity from 58.5% to 54.5%, raise its beam-off content from 30 to 83 events and recover only 2.7% of the generated signal ( $2055 \rightarrow 1999$  in the table), and the region it removes is more useful as the beam-off control region; (ii) a *lower* opening-angle cut is counterproductive, the small-angle region is high-purity ( $\sim 58\%$ ), so cutting it lowers the selected signal without improving purity, which is why no lower cut is used.

Table 27: Selection-optimisation study across the phase-space iterations, on the Run-1 FHC sample. Every entry is Monte-Carlo derived (GENIE overlay signal and background, EXT, dirt); the thresholds chosen here are those of the full-exposure selection of Table 10 of the analysis note. The beam-off column uses the pooled beam-off scaling of that study, so its absolute values are not those of the full exposure; the ranking of the options is unaffected. “no upper cut” drops the  $\theta_{\mu\pi} < 2.6 \text{ rad}$  cut; “[0.314, 2.6]” adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

|                     | Reference<br>( $p > 0.1$ ) | no upper cut<br>( $p > .15/.175$ ) | [0.314, 2.6]<br>(lower cut) | <b>Final</b><br>( $\theta < 2.6$ ) |
|---------------------|----------------------------|------------------------------------|-----------------------------|------------------------------------|
| Generated signal    | 2684                       | 2055                               | 1916                        | 1999                               |
| Selected signal     | 324.7                      | 320.3                              | 301.8                       | 311.9                              |
| Efficiency          | 12.1%                      | 15.6%                              | 15.8%                       | <b>15.6%</b>                       |
| EXT (cosmic)        | 40.1                       | 136.2                              | 38.0                        | 40.1                               |
| Total bkg           | 243.6                      | 385.4                              | 252.3                       | 256.4                              |
| <b>Purity</b>       | <b>57.1%</b>               | 45.4%                              | 54.5%                       | <b>54.9%</b>                       |
| sig $\times$ purity | 185.6                      | 145.4                              | 164.4                       | <b>171.2</b>                       |

### 13.2 Cosmic-direction cuts considered and rejected

**A directional cut against residual cosmics: considered and rejected.** Part of the surviving beam-off sample is thought to be broken cosmic muon tracks, where the break is

reconstructed as a vertex and the track direction bears no relation to it. That leaves a directional signature, and it is present in the data: among selected events the beam-off sample is enriched at backward reconstructed muon angles by nearly an order of magnitude relative to signal.

| reco $\cos \theta_\mu$ | EXT (beam-off) | MC signal | enrichment  |
|------------------------|----------------|-----------|-------------|
| $< 0$                  | 31.8%          | 4.1%      | $7.8\times$ |
| $< -0.5$               | 9.1%           | 1.4%      | $6.4\times$ |

The enrichment is real but the arithmetic does not support a cut. At the final cut the FHC sample is 943 signal, 30 EXT and 1612 predicted in total (Table 10 of the analysis note). A  $\cos \theta_\mu < 0$  requirement would remove  $\approx 32\%$  of the EXT, about 10 events, and  $\approx 4\%$  of the signal, about 39 events. Because the beam-off sample is subtracted from the data directly, and that subtraction is unbiased by construction, removing part of it buys only a reduction in the subtraction's *statistical* uncertainty, which is dominated by the 8 raw Run-1 events scaled by 2.1 ( $\approx 6$  events for the 30), to  $\approx 5$  events. Spending 39 signal events to gain  $\approx 1$  event of uncertainty is a net loss.

The same bound applies to any cosmic cut, not just this one: EXT is 1.9% of the selected sample and is already subtracted without bias, so removing *all* of it for *zero* signal loss would gain about 6 events of statistical uncertainty on a sample of 1612. Any cut costing more than about 1% of the signal is a net loss regardless of how efficiently it tags cosmics. The residual beam-off contribution is therefore left in place and handled by subtraction.

A more surgical variant remains untested. The CC0 $\pi$  analysis tags mis-directed tracks with a calorimetric direction-preference flag in coincidence with a poor forward-muon fit,  $\chi_\mu^2 > 6$ , which fires on tracks whose  $dE/dx$  profile disagrees with the assigned direction rather than on backward-going tracks as such. That would not cut genuine backward muons, which outnumber mis-reconstructed ones here by roughly 8:1. Both quantities exist in the input ntuples but are not carried into the processed files. It is bounded by the same  $\approx 6$ -event ceiling on the EXT statistical uncertainty, so it is worth measuring but cannot become a significant improvement.

### 13.3 Combined-configuration reconstruction-level spectra

The selected and background-subtracted spectra of the combined configuration, companions to the FHC and RHC ones of the analysis note.

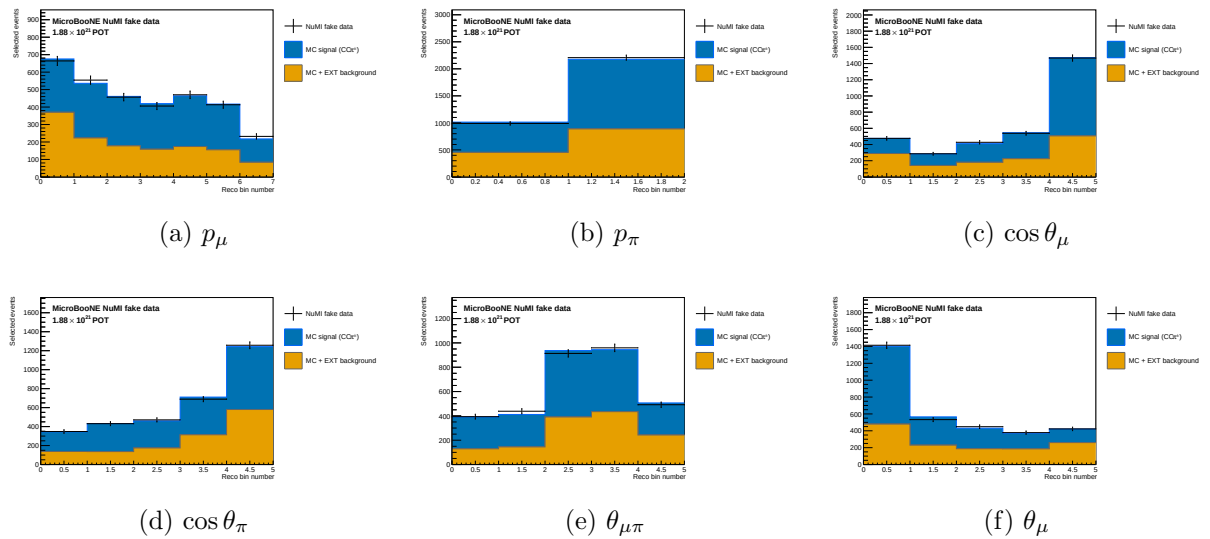


Figure 10: Stage 1, combined: selected reconstructed spectra at the combined FHC+RHC exposure.



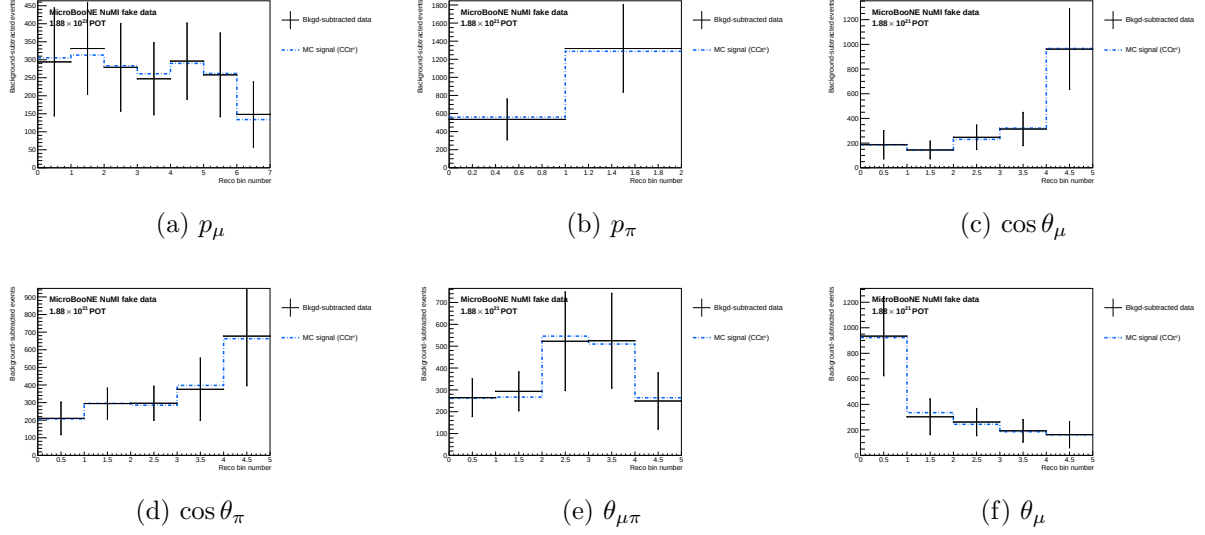


Figure 11: Stage 2, combined: background-subtracted data compared with the central-value MC signal.

### 13.4 The rejected five-bin $p_\pi$ response

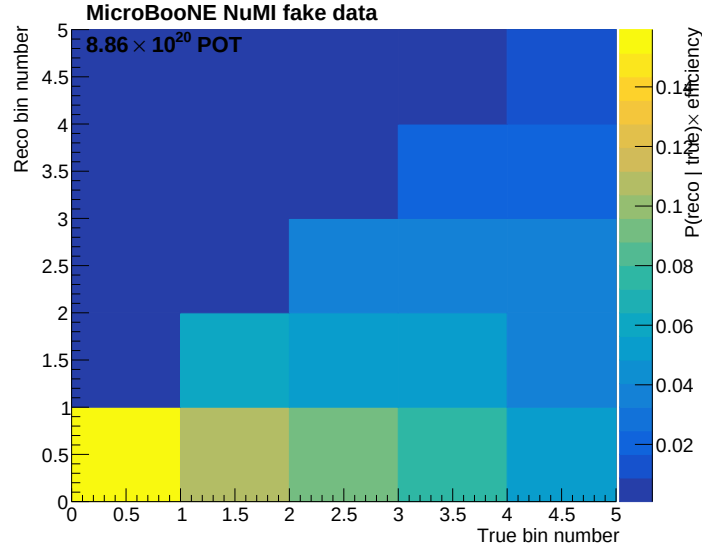


Figure 12: Smearceptance matrix of the rejected five-bin  $p_\pi$  scheme (FHC): reconstructed events pile into the lowest bin from every true bin, the diagnostic that motivated the two-bin scheme of §2.1.6; the released two-bin response is in the analysis note.

### 13.5 $A_C$ row sums of every released extraction

Row sums of the released additional-smearing matrices, inclusive and proton-tagged, and the two matrices with a negative row.



| Observable           | FHC       | RHC        | combined  |
|----------------------|-----------|------------|-----------|
| <i>Inclusive</i>     |           |            |           |
| $p_\mu$              | 0.66–0.91 | 0.58–0.99  | 0.68–1.01 |
| $p_\pi$ (2-bin)      | 0.35–0.54 | 0.40–0.51  | 0.29–0.54 |
| $\cos \theta_\mu$    | 0.17–0.39 | 0.05–0.45  | 0.09–0.43 |
| $\cos \theta_\pi$    | 0.68–0.77 | 0.70–0.84  | 0.71–0.83 |
| $\theta_{\mu\pi}$    | 0.52–1.03 | 0.58–1.15  | 0.63–1.21 |
| $\theta_\mu$         | 0.58–0.78 | 0.22–1.16  | 0.53–0.67 |
| <i>Proton-tagged</i> |           |            |           |
| $p_\mu$              | 0.46–1.00 | 0.55–0.89  | 0.65–0.96 |
| $p_\pi$ (2-bin)      | 0.34–0.84 | 0.41–0.45  | 0.47–0.48 |
| $\cos \theta_\mu$    | 0.61–0.70 | −0.08–0.57 | 0.28–0.46 |
| $\cos \theta_\pi$    | 0.66–0.77 | 0.62–0.77  | 0.65–0.76 |
| $\theta_{\mu\pi}$    | 0.37–0.87 | 0.31–0.75  | 0.35–0.78 |
| $W_{\pi p}$          | 0.45–1.22 | −0.14–1.26 | 0.32–1.24 |
| $W_{\text{had}}$     | 0.54–0.81 | 0.30–0.86  | 0.18–0.87 |
| $\delta p_T$         | 0.54–0.86 | 0.45–0.82  | 0.47–0.87 |
| $\delta \phi_T$      | 0.65–0.78 | 0.53–0.72  | 0.28–0.67 |
| $\delta \alpha_T$    | 0.59–0.98 | 0.57–0.93  | 0.60–0.96 |
| $p_n$                | 0.71–0.86 | 0.61–0.82  | 0.59–0.86 |

Three features matter for interpretation.

The suppression is large: inclusive  $\cos \theta_\mu$  has a row sum as low as 0.05 in RHC, a strong suppression of that output mode for a flat input. It also differs strongly between observables, which is precisely why the integrated cross section obtained by summing a differential result varies from observable to observable while the underlying physical total cannot. The D’Agostini cross-check, whose integral is flat across observables, is what establishes that this spread is the smearing rather than a physical or acceptance effect.

Two extractions have *negative* row sums, and the pattern identifies their cause. In  $W_{\pi p}$  the top row carries negative off-diagonal terms in *all three* configurations (summing to −0.14 FHC, −0.30 combined, −0.61 RHC against diagonals of 0.73, 0.62 and 0.47); only RHC tips negative overall (row sum −0.14), because its diagonal element is the smallest of the three. The same happens in the first  $\cos \theta_\mu$  bin of the proton-tagged RHC sample, whose diagonal is 0.079 against off-diagonal terms summing to −0.16 (row sum −0.08).

So this is not an RHC-specific defect but the regularisation pushing back against the least-determined bin of each observable — for  $W_{\pi p}$ , the same top bin that is only 2% diagonal in the six-bin response and that already led to withdrawing the differential shape. A negative row is not forbidden, since  $A_C$  is a regularisation operator rather than a probability matrix, but a prediction smeared through one is being told that raising the truth in that bin *lowers* the measured content, which is a statement about the regularisation and not about the detector. Treat those two rows with corresponding caution.

### 13.6 Ensemble study: reference construction and full-covariance pulls

**Ensemble coverage: are the uncertainties the right size?** The closure test above uses a single throw. It can show that the extraction reproduces its input, but a single residual has no width, so it cannot say whether the quoted uncertainty is correctly sized. To answer that, up to one hundred statistically independent Poisson realisations of the data of each configuration (six extractions:  $p_\mu$  and  $\cos \theta_\mu$  in FHC,  $\cos \theta_\mu$  in RHC, the two-region  $p_\pi$  in the combined configuration, the FHC and RHC one-bin totals; 100 members each) were each put through the *whole* chain — thrown, universes rebuilt from that throw, unfolded — and the per-bin pull

$$p_i = \frac{\hat{x}_i - (A_C x_{\text{CV}})_i}{\sigma_i}$$

formed against the fixed central-value truth smeared by that member’s own  $A_C$ . (The per-throw *fake-data* truth is not a valid reference: it follows the Poisson draw, so numerator and reference fluctuate together and the width comes out too small.)

The throws follow the weight rule of Eq. 4 of the analysis note, and the extractions are chosen to bracket the conditioning and the configurations: the well-behaved  $p_\mu$ ;  $\cos \theta_\mu$ , whose  $A_C$  row sums fall as low as 0.17 in FHC (0.050 in RHC) so that the regularisation removes up to 83% (95%) of a bin’s normalisation; the two-region  $p_\pi$  of the combined configuration; and the one-bin totals, which have no regularisation at all. If the uncertainties were mis-sized anywhere, the second is where it would show; if the extraction carried a bias independent of the regularisation, the third is where it would show.

| Reference covariance                     | pull mean | pull width | 68% / 95% coverage |
|------------------------------------------|-----------|------------|--------------------|
| <i>statistical only</i>                  |           |            |                    |
| $d\sigma/dp_\mu$ , FHC                   | −0.08     | 0.98       | 67.9% / 96.4%      |
| $d\sigma/d \cos \theta_\mu$ , FHC        | −0.08     | 1.01       | 68.0% / 95.2%      |
| $d\sigma/d \cos \theta_\mu$ , RHC        | −0.07     | 1.07       | 66.6% / 93.6%      |
| $d\sigma/dp_\pi$ (two regions), combined | −0.09     | 0.92       | 74.0% / 95.5%      |
| one-bin total, FHC                       | −0.10     | 0.97       | 67.0% / 95.0%      |
| one-bin total, RHC                       | −0.21     | 0.99       | 64.0% / 94.0%      |
| <i>full</i>                              |           |            |                    |
| $d\sigma/dp_\mu$ , FHC                   | −0.02     | 0.34       | 99.1% / 100.0%     |
| $d\sigma/d \cos \theta_\mu$ , FHC        | −0.02     | 0.35       | 99.4% / 100.0%     |
| $d\sigma/d \cos \theta_\mu$ , RHC        | −0.01     | 0.33       | 99.0% / 100.0%     |
| $d\sigma/dp_\pi$ (two regions), combined | −0.02     | 0.22       | 100.0% / 100.0%    |
| one-bin total, FHC                       | −0.01     | 0.11       | 100.0% / 100.0%    |
| one-bin total, RHC                       | −0.02     | 0.11       | 100.0% / 100.0%    |
| expected                                 | 0         | 1          | 68% / 95%          |

Against the statistical covariance all six widths are consistent with unity — 0.92–1.07, each determined to about  $\pm 7\%$  ( $1/\sqrt{2} \times 100$ , more for the multi-bin extractions, whose bins are correlated) — and the interval coverage is correct within the same statistics. **The statistical uncertainties are correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one.** That the severe  $A_C$  suppression in  $\cos \theta_\mu$  does not distort its uncertainties is the more informative of the two results, since  $p_\mu$  passing was the expected outcome.

Two further things the ensembles show. Against the *full* covariance the pull width is 0.21–0.35 for the differential results and 0.12 for the totals, and coverage is essentially 100%: the quoted uncertainty is about three times the throw-to-throw spread, because the ensemble varies only the data statistics while the denominator carries flux, detector and cross-section terms that are identical in every member. That is the same effect as the closure  $p$ -values clustered near unity, now measured rather than argued.

Second, the ensemble means sit below the reference in all six cases, by 0.4–1.1% ( $0.5$ – $2.0\sigma$  on the mean; largest in significance for the RHC total,  $-0.81\%$  at  $2.0\sigma$ , and in size for FHC  $\cos \theta_\mu$ ,  $-1.10\%$  at  $1.3\sigma$ ), with per-bin pull means of  $-0.07$  to  $-0.10$  and  $-0.21$  for the RHC total (Table above; `data_release/ensemble_2026-09-20.tsv`). The extractions of one configuration share the same throws and are correlated, so this is a few measurements of a small negative offset, each consistent with zero at the present ensemble size. The throw itself is unbiased: the mean selected count over the 32 members of the earlier FHC ensemble (same throw rule) is 1578.7 against the neutrino-MC expectation of 1580.6 (ratio 0.999), so whatever offset remains is in the extraction, not in the pseudo-data. The offset is at most two standard deviations of the ensemble mean and is largest where the regularisation is strongest, but the one-bin totals, which have no regularisation, also sit low, so the excess in  $\cos \theta_\mu$  over the common  $\approx 1\%$  is the only part

that can be attributed to  $A_C$ . No correction is applied and no uncertainty is assigned; a larger ensemble would settle whether the common offset is real.

Ten of the 600 members were first built from a truncated read of their fake data: two ensembles sharing a throw number re-threw the same file at the same time, and a member that opened it mid-write saw part of the tree. Four RHC-total members (Run 2 read at 47–72%) and one FHC  $\cos\theta_\mu$  member (43%) were affected materially; they widened the RHC-total pull width to 1.28. Every member was checked (the number of events univmake read against the entries on disk; 2791 of 2800 inputs agree exactly, and the five remaining one-entry differences are one out-of-range truth event in the one-bin binning, which does not enter the reco-level data), and the affected members were re-run with the throw written to a temporary file and renamed on completion. The table is the corrected result.

The study covers the inclusive sample in all three configurations. Extending it to the proton-tagged family would test whether the same holds where the statistics are sparser; the machinery is in place.

## 14 Analysis Pipeline and Automated Guards

[SHARED]

Every released number is produced by the XSECANALYZER chain in four stages that are re-runnable from the raw PeLEE ntuples:

1. the event processing applies the selection(s) to each PeLEE ntuple and writes the processed event files, for the neutrino overlays, the dirt overlay and the per-run beam-off samples;
2. the universe builder constructs, per observable and configuration, the true and reconstructed histograms of every systematic universe from the files listed in that configuration;
3. the unfolders sums the universes with the per-run exposure scaling, builds the covariance, unfolds, and writes the closure sidecar, the systematics dump and the matrix tables;
4. the export step writes the data release (curves,  $A_C$  matrices, covariance decompositions); the figure macros draw from the closure sidecars; the one-bin total uses the same chain with the single-bin configurations; the count-level tables (cut flow, cut-and-count check, control-region yields) are produced with the same weights and scalings.

Five automated guards protect the normalisation, the currency of released artefacts, the result tables and the blinding (the change log records the incident that motivated each):

- A normalisation manifest reads the authoritative configuration files and asserts that exposure, flux, target count and beam-off scaling agree across all configurations, and that every declared systematic is backed by a file.
- A coupling assertion in the unfolders refuses a configuration whose declared detector variations are not present in its file list.
- A staleness check refuses to ship any released artefact or result figure older than the extraction it depicts, and compares the integrals printed on the two-bin figures with their sidecars.
- A table check parses every result table in this note and fails if a value disagrees with the released results table, if a table does not sit under its own label, or if a row's column count differs from its header.
- The blinding guard: a beam-on file without MC truth is real data and is refused unless an explicit unblinding switch is set; the same detection selects the beam-off treatment (added to pseudo-data, subtracted from real data), so no manual switch exists to forget.

The release was built from the processed ntuples with every cached universe deleted, in a single pass, so no preserved intermediate exists; a clean-cache rebuild of the same kind reproduced an earlier release bin by bin to  $10^{-9}$  (the comparison record is in the change log), and repeating that comparison on the release to be unblinded is on the pre-unblinding list (Sec. 15 of the analysis note).

## Part V

# Physics background

## 15 Physics background and prior measurements (extended)

[SHARED]

### 15.1 Production Mechanisms

At neutrino energies of 0.5–5 GeV, charged-current single charged-pion production proceeds through three mechanisms.

**Baryon resonance production** dominates at  $E_\nu \lesssim 2$  GeV. The neutrino excites the struck nucleon to a baryon resonance, most prominently the  $\Delta(1232)$  isobar, which then decays to a nucleon and pion:

$$\nu_\mu n \rightarrow \mu^- \Delta^{++} \rightarrow \mu^- \pi^+ p, \quad \nu_\mu p \rightarrow \mu^- \Delta^+ \rightarrow \mu^- \pi^0 p \text{ (or } \pi^+ n). \quad (3)$$

The standard model is due to Rein and Sehgal [10], which uses the Feynman–Kislinger–Ravndal (FKR) quark model to compute form factors for all baryon resonances up to  $W \lesssim 2$  GeV. The modern implementation used in GENIE for resonance (RES) production is the Kuzmin–Lyubushkin–Naumov Berger–Sehgal (KLN-BS) model [11, 12, 13, 14], which combines the lepton-mass-corrected vector and axial form factors of Kuzmin *et al.* [11] with the updated  $\Delta(1232)$  electromagnetic form factors of Graczyk and Sobczyk [12] and the improved treatment of the  $\Delta$  line shape and the lepton-mass effects of Berger and Sehgal [13]. (Berger and Sehgal also discuss the interference between the resonant and non-resonant amplitudes; that interference term is not implemented in the GENIE v3 configuration underlying the MicroBooNE tune [2], where the non-resonant contribution is added incoherently.) Higher resonances ( $N(1440)P_{11}$ ,  $N(1520)D_{13}$ , etc.) are included up to  $W \lesssim 2$  GeV and contribute at  $E_\nu \gtrsim 1.5$  GeV.

**Non-resonant and DIS backgrounds.** A non-resonant continuum contributes at all  $W$ ; in a full amplitude-level treatment it interferes (constructively or destructively by isospin) with the resonant amplitude, although the generators used here add it incoherently. Deep-inelastic scattering (DIS) becomes significant above  $W \approx 1.6$  GeV. Together these processes represent  $\sim 20\%$  of the  $\text{CC}\pi^\pm$  rate in the energy range of this analysis and are modelled in GENIE via the Bodek–Yang formalism.

**Coherent pion production.** The neutrino can scatter coherently off the entire nucleus [7, 8],  $\nu_\mu A \rightarrow \mu^- \pi^+ A^*$ , leaving the nucleus in its ground state. The cross section is small ( $\lesssim 10\%$  of the total) below 5 GeV. The signal definition of Sec. 6 of the analysis note is a final-state topology (one muon, one charged pion, no other mesons) and carries no truth-level veto on the interaction mode, so coherent  $\text{CC}\pi^\pm$  events that satisfy it are *included in the signal*; they are a small signal component, not a background.

### 15.2 Nuclear Effects and Final-State Interactions

On a heavy nucleus such as argon ( $A = 40$ ), two classes of nuclear correction modify the  $\text{CC}\pi^\pm$  rate and kinematics significantly.

**Initial-state effects.** Fermi motion of the struck nucleon broadens kinematic distributions, and Pauli blocking suppresses transitions to low-momentum nucleon final states near threshold. Nuclear binding also shifts the effective nucleon mass relative to free-nucleon predictions.

**Final-state interactions (FSI).** The produced pion must traverse the nuclear medium before being detected. Intranuclear rescattering occurs via:

- *Pion absorption* ( $\pi A \rightarrow NN \dots$ ): probability  $\sim 25\text{--}30\%$  on Ar; converts  $\text{CC}\pi^\pm$  events into  $\text{CC}0\pi$  topology.
- *Charge exchange* ( $\pi^+ n \rightarrow \pi^0 p$ ): produces a  $\text{CC}\pi^0$  event.
- *Quasi-elastic scattering*: pion survives but changes direction; modifies  $p_\pi$  and  $\theta_\pi$  distributions.
- *Inelastic production* ( $\pi N \rightarrow \pi\pi N$ ): rare below  $p_\pi \approx 400 \text{ MeV}/c$ .

FSI are modelled in GENIE via the INTRANUKE ( $hA$ ) cascade. Because absorption and charge exchange mix  $\text{CC}\pi^\pm$ ,  $\text{CC}\pi^0$ , and  $\text{CC}0\pi$  topologies at the  $\sim 30\%$  level, the signal in any  $\text{CC}\pi$  analysis is necessarily defined by the final-state particle content observed in the detector, not by the underlying production process (Sec. 6 of the analysis note).

### 15.3 Generator Predictions and Known Tensions

Major neutrino event generators reproduce the inclusive CC cross section to 10–15% but show systematic spread in the  $\text{CC}\pi^\pm$  rate, owing to differences in the resonance model, the treatment of the transition region between resonance and DIS, and the FSI cascade.

**GENIE v3** [1, 3]. For resonance production GENIE implements the KLN-BS model [11, 12, 13, 14] for the full tower of 18 baryon resonances up to  $W \lesssim 2 \text{ GeV}$ , including the  $\Delta(1232)P_{33}$ ,  $N(1440)P_{11}$  (Roper),  $N(1520)D_{13}$ ,  $N(1535)S_{11}$ ,  $\Delta(1620)S_{31}$ ,  $\Delta(1700)D_{33}$ ,  $N(1680)F_{15}$ ,  $\Delta(1950)F_{37}$ , and their isospin partners. The hadronic invariant mass range  $1.0 < W < 1.7 \text{ GeV}$  is bridged by a resonance–DIS transition using the AGKY (Andreopoulos–Gallagher–Kehayias–Yang) model, which applies KNO scaling in the transition region and the Bodek–Yang DIS model above  $W \approx 1.7 \text{ GeV}$ . Non-resonant single-pion production is included through the low- $W$  extension of the Bodek–Yang DIS model and is added incoherently to the resonant contribution; no resonant/non-resonant interference is implemented. Coherent  $\pi^+$  production uses the Rein–Sehgal coherent model. FSI are handled by the INTRANUKE  $hA$  empirical model, in which each hadron undergoes at most one effective interaction whose probability and outcome are tuned to hadron–nucleus data; it is the alternative  $hN$  microscopic cascade, also available in GENIE, that propagates hadrons step-by-step using free-nucleon  $\pi N$  and  $NN$  cross sections. GENIE has been reported to over-predict the  $\text{CC}\pi^\pm$  rate by 20–35% relative to bubble-chamber and MiniBooNE data in several comparisons when normalised to the BNL dataset; the tension is reduced but not eliminated in the AR23 tune.

**NuWro** [4]. NuWro models resonance production via the  $\Delta(1232)$  only; no higher baryon resonances are included. The  $\Delta$  is treated with the Paschos–Yu formalism for the  $\Delta$  excitation amplitude, retaining the full angular structure of the pion decay. A smooth analytic interpolation connects the single- $\Delta$  resonance region to the DIS regime at higher hadronic invariant mass ( $W \gtrsim 1.4 \text{ GeV}$ ), using Bodek–Yang structure functions for the deep inelastic contribution; the transition avoids a hard boundary between resonance and continuum. Non-resonant single-pion production is included as a separate amplitude. FSI are implemented using the microscopic Salcedo–Oset model, which treats pion propagation in nuclear matter through the pion self-energy in the random-phase approximation, naturally accounting for medium modifications such as  $\Delta$  broadening and pion absorption via two-body processes ( $\pi NN \rightarrow NN$ ) that are absent in purely cascade-based approaches. NuWro has historically shown better agreement with MiniBooNE

and bubble-chamber data than GENIE, partly attributable to the absence of overestimated higher-resonance contributions.

**NEUT** [15, 6]. NEUT, the generator used by T2K and Super-Kamiokande, implements resonance production via a modified Rein–Sehgal model including 18 resonances up to  $W \lesssim 2$  GeV, with Graczyk–Sobczyk [12] electromagnetic form factors for the  $\Delta(1232)$ . The axial mass  $M_A^{\text{RES}}$  is a tunable parameter; T2K applies a data-driven tune that reduces the overall  $\text{CC}\pi^\pm$  normalisation by  $\sim 10\text{--}20\%$  relative to the default. The transition to DIS is handled similarly to GENIE, using GRV98 LO parton distribution functions with Bodek–Yang corrections above  $W \approx 2$  GeV. Coherent pion production follows the Rein–Sehgal coherent model with a Berger–Sehgal-style correction for finite lepton mass. FSI are modelled by a custom NEUT intranuclear cascade that tracks each produced hadron through the nucleus using step-by-step interactions sampled from  $\pi N$  cross sections, with nuclear density and mean-free-path corrections derived from the optical potential.

**GiBUU** [5]. The Giessen Boltzmann–Uehling–Uhlenbeck (GiBUU) generator takes a fundamentally different approach: instead of a probabilistic step-by-step cascade, it solves a coupled set of kinetic transport equations for the full hadronic system inside the nuclear medium. Resonance production includes the  $\Delta(1232)$  and higher resonances whose in-medium spectral functions are broadened by collisional damping; the  $\Delta$  width in the nuclear medium is substantially larger than its free-space value of  $\Gamma_0 \approx 118$  MeV, suppressing threshold production. Pion propagation after production is governed by a mean-field potential (the pion optical potential), which shifts pion momenta as they traverse the nucleus – an effect absent in cascade models and particularly important for low-momentum pions ( $p_\pi \lesssim 200$  MeV/ $c$ ) near the pion production threshold. The transition to DIS uses PYTHIA for string fragmentation above  $W \approx 1.6$  GeV. GiBUU predictions for pion angular distributions and for the backward-hemisphere  $d\sigma/d\cos\theta_\pi$  can differ substantially from cascade-based generators, making it a valuable cross-check of FSI systematics.

Because the present results are extracted on central-value fake data, no data/MC deficit is measured here. What the analysis does provide is the spread of the standalone generators in the measured phase space, a factor of  $\sim 1.6$  between GENIE and NEUT (Table 16), which brackets the injected MicroBooNE tune; the over-prediction observed for several generator configurations in parts of the relevant phase space in the measurements surveyed below is the context for that spread, not a result of this note.

## 15.4 Previous Experimental Measurements

Table 28 summarises published  $\text{CC}\pi^+$  measurements on nuclear targets ordered chronologically.

Table 28: Selected  $\text{CC}\pi^+$  cross-section measurements on nuclear targets.  $\langle E_\nu \rangle$  is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

| Experiment                      | Target        | $\langle E_\nu \rangle$<br>(GeV) | Type | Key result / reference                                                                                                              |
|---------------------------------|---------------|----------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------|
| ANL<br>12-ft [16]               | D             | 1.0                              | T    | Bubble chamber; foundational<br>$\sigma$ vs $E_\nu$                                                                                 |
| BNL 7-ft [17]                   | D             | 1.5                              | T    | Higher statistics; 15%<br>normalisation tension with<br>ANL                                                                         |
| K2K [18]                        | C             | 1.3                              | T    | First accelerator $\text{CC}\pi^+$ on<br>carbon near detector                                                                       |
| MiniBooNE [19]                  | $\text{CH}_2$ | 0.7                              | D    | $Q^2$ , $W$ , $p_\mu$ , $\theta_\mu$ ; 20% excess over<br>NUANCE                                                                    |
| MINERvA [20]                    | CH            | 3.6                              | D    | $p_\pi$ , $\theta_\pi$ , Adler angles; NuMI<br>medium-energy                                                                        |
| T2K<br>ND280 [23]               | CH            | 0.85                             | D    | $p_\mu$ , $\cos\theta_\mu$ ; off-axis beam similar<br>to this work                                                                  |
| ArgoNEUT [24]                   | Ar            | 9.6                              | T    | First $\text{CC}\pi^+$ on Ar; 115 $\nu$ +<br>158 $\bar{\nu}$ events ( $1.25 \times 10^{20}$ POT)                                    |
| MicroBooNE<br>BNB [26]          | Ar            | 0.7                              | D    | Full BNB dataset<br>( $1.11 \times 10^{21}$ POT); $\sigma =$<br>( $3.75 \pm 0.80$ ) $\times 10^{-38}$ $\text{cm}^2/\text{Ar}$       |
| MicroBooNE<br>NuMI $\nu_e$ [25] | Ar            | 0.73                             | D    | First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar;<br>$\sigma = (0.93 \pm 0.28) \times$<br>$10^{-39}$ $\text{cm}^2/\text{nucleon}$ |

### Deuterium bubble chambers: ANL and BNL (1979–1986)

The ANL 12-foot bubble chamber [16] and BNL 7-foot bubble chamber [17] provided the first high-statistics measurements of  $\nu_\mu d \rightarrow \mu^- \pi^+ pp$  in the 1–3 GeV range. These measurements are important because the deuterium target is close to a free nucleon, minimising nuclear corrections relative to heavier targets. The ANL and BNL total cross sections disagree by  $\sim 15\%$  in absolute normalisation, a discrepancy long attributed to flux uncertainties; the BNL flux is now believed to be better constrained. This ANL–BNL tension directly propagated into the spread of generator predictions: models tuned to ANL tend to predict lower rates than those tuned to BNL. The MINERvA experiment [22] subsequently resolved much of this tension by simultaneously constraining the flux and the cross section.

### K2K near detector (2008)

The K2K experiment [18] made the first accelerator-based total  $\text{CC}\pi^+$  cross-section measurement on a carbon target at  $\langle E_\nu \rangle = 1.3$  GeV using its fine-grained near detector. The result was consistent with NEUT predictions and established the nuclear-target correction from deuterium to carbon at intermediate energies.

### MiniBooNE ( $\nu_\mu$ on $\text{CH}_2$ , BNB beam, 2011)

The MiniBooNE detector (818-tonne mineral-oil Cherenkov detector) published differential  $\text{CC}\pi^+$  cross sections on  $\text{CH}_2$  in the Booster Neutrino Beam ( $E_\nu^{\text{peak}} \approx 0.7$  GeV) [19]. MiniBooNE was the first experiment to provide high-statistics differential distributions as a function of  $Q^2$ , the invariant hadronic mass  $W$ , and the outgoing muon and pion kinematics. The flux-integrated total cross section was measured to be  $\sigma \approx 3.4 \times 10^{-38}$   $\text{cm}^2$  per  $\text{CH}_2$  molecule, or equivalently



$\sim 2.0 \times 10^{-38}$  cm<sup>2</sup> per carbon nucleus at  $\langle E_\nu \rangle \approx 1$  GeV [26]. The measured rate exceeded the NUANCE prediction by  $\sim 20\%$ , i.e. that model *under*-predicted the data; the “generator over-prediction” discussed in the later literature refers to the opposite tension, in which GENIE-based predictions lie above subsequent measurements in parts of the phase space, and is generator- and phase-space-dependent rather than universal. The hydrogen content of CH<sub>2</sub> complicates the extraction of the carbon-only cross section, and the spherical Cherenkov geometry makes it difficult to measure low-energy pions or reconstruct proton recoils. These limitations drove subsequent adoption of finer-grained tracking detectors and liquid-argon targets.

#### **MINERvA ( $\nu_\mu$ on CH, NuMI medium energy, 2015–)**

The MINERvA experiment [20] uses a finely-segmented scintillator-strip tracking calorimeter in the NuMI medium-energy beam ( $\langle E_\nu \rangle \approx 3.6$  GeV). It has published the most kinematically complete CC $\pi^+$  measurements to date, reporting differential cross sections as a function of: pion momentum  $p_\pi$ , pion angle  $\theta_\pi$ , muon momentum  $p_\mu$ , muon angle  $\theta_\mu$ , and the full set of Adler angles describing the pion in the hadronic centre-of-mass frame. MINERvA also measured CC $\pi^+$  on iron and lead targets [21], providing direct constraints on nuclear  $A$ -dependence at the same beam energy. GENIE consistently over-predicts MINERvA by 20–35% across all kinematic variables in the  $\Delta(1232)$  peak, the same qualitative tension seen at lower energies. The higher beam energy at MINERvA means that the  $p_\pi > 0.5$  GeV/ $c$  region is well-populated, making it complementary to the MicroBooNE NuMI result which is statistics-limited at high pion momentum.

#### **T2K ND280 ( $\nu_\mu$ on CH, off-axis beam, 2020)**

T2K measured CC $\pi^+$  at its near detector ND280 in the off-axis T2K beam (OAB), which peaks at  $\langle E_\nu \rangle \approx 0.85$  GeV [23]. The T2K OAB spectrum closely overlaps with the MicroBooNE NuMI off-axis spectrum in the kinematically-sensitive 0.3–1.5 GeV range, making this the most directly comparable prior measurement to the present work. T2K reported differential cross sections in  $p_\mu$  and  $\cos \theta_\mu$ , finding a data/MC deficit of  $\sim 30\%$  relative to NEUT. The carbon target ( $A = 12$ ) differs from argon ( $A = 40$ ), but the overlapping beam spectrum enables direct comparison of nuclear-mass dependence at fixed neutrino energy.

#### **ArgoNEUT ( $\nu_\mu + \bar{\nu}_\mu$ on Ar, NuMI, 2018)**

ArgoNEUT [24] was a 175-litre LArTPC placed upstream of the MINOS near detector in the NuMI low-energy beam ( $\langle E_\nu \rangle \approx 9.6$  GeV). Using  $1.25 \times 10^{20}$  POT, ArgoNEUT selected 115  $\nu_\mu$  and 158  $\bar{\nu}_\mu$  CC $\pi^+$  candidate events after background subtraction [24], publishing the first CC $\pi^+$  total cross section on argon as a function of neutrino energy. The result was consistent with GENIE predictions at these high energies, where sensitivity to the  $\Delta(1232)$  resonance is suppressed. The LArTPC technology and argon target make ArgoNEUT the direct predecessor to MicroBooNE, but the factor  $\sim 10$  higher beam energy means that extrapolation to the  $E_\nu < 2$  GeV regime relevant for oscillation experiments requires generator assumptions and limits the constraining power on FSI models in the resonance region.

#### **MicroBooNE BNB channel: inclusive precedent (2022)**

The same MicroBooNE detector used in this analysis published the first measurement of energy-dependent inclusive  $\nu_\mu$  charged-current cross sections on argon in the Booster Neutrino Beam [9] ( $E_\nu^{\text{peak}} \approx 0.7$  GeV), establishing the detector’s absolute cross-section capability on argon together with the BNB flux modelling and reconstruction performance relevant to the present work. The dedicated exclusive CC $\pi^+$  measurement in the same beam followed on the full BNB dataset

(below), and it is that exclusive result to which the present NuMI CC $\pi^+$  measurement is most directly compared.

### MicroBooNE BNB channel: full dataset (2026)

Reference [26] reports CC $\pi^+$  differential cross sections using the complete MicroBooNE BNB dataset of  $1.11 \times 10^{21}$  POT [26], selecting 6816 events after background subtraction and measuring differential cross sections in muon momentum, muon angle, pion momentum, and pion angle – the same four observables as the present analysis. The flux-integrated total cross section is

$$\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$$

at mean neutrino energy  $\langle E_\nu \rangle \approx 0.8$  GeV. Most generators reproduce the data within uncertainties; a notable exception is the muon angle with respect to the beam, where all tested generators overestimate the cross section at very forward angles ( $\cos \theta_\mu \rightarrow 1$ ) [26]. The pion momentum differential cross section reported in this analysis represents the first such measurement on an argon target.

The BNB and NuMI analyses probe complementary phase space: BNB is on-axis with a narrow  $\nu_\mu$  spectrum peaked at  $\sim 700$  MeV and negligible  $\bar{\nu}_\mu$  content, while NuMI FHC at 135 mrad off-axis provides a softer, broader spectrum with a  $\sim 35\%$   $\bar{\nu}_\mu$  admixture and extends sensitivity to  $E_\nu \sim 0.1\text{--}0.5$  GeV.

### MicroBooNE NuMI $\nu_e + \bar{\nu}_e$ CC $\pi^+$ (2025)

Reference [25] presents the first measurement of  $\nu_e + \bar{\nu}_e$  CC single charged-pion differential cross sections on argon, using the full MicroBooNE NuMI dataset ( $2.0 \times 10^{21}$  POT, FHC and RHC combined). The observable set – electron energy  $E_e$ , electron angle  $\theta_e$ , pion angle  $\theta_\pi$ , and electron–pion opening angle  $\theta_{e\pi}$  – mirrors the muonic observables of the present analysis. The flux-integrated total cross section is measured to be

$$\sigma = (0.93 \pm 0.13_{\text{stat}} \pm 0.27_{\text{syst}}) \times 10^{-39} \text{ cm}^2/\text{nucleon}$$

at a mean  $\nu_e$  energy of 730 MeV, in good agreement with all tested generator predictions (GENIE, NuWro, NEUT, GiBUU) within the combined uncertainties. Prior to this measurement, CC $\pi^+$  production by electron neutrinos on argon had never been measured; the result is a key input to the DUNE  $\nu_e$ -appearance analysis, where CC $\pi^+$  events are a background to the  $\nu_e$  quasi-elastic signal.

## Glossary

**Signal definition** The set of particles leaving the nucleus that counts as the process being measured, with momentum thresholds and a fiducial requirement. It is deliberately *not* a statement about the interaction mechanism, because final-state interactions mix mechanisms and only the final state is observable.

**Fake data / pseudo-data** A simulated dataset used in place of beam-on data while the analysis is blind. Here it is a Poisson throw: each simulated event is replicated a Poisson-distributed number of times with mean equal to its weight at data exposure, so the sample has the statistical fluctuations of a real dataset.

**Blind** The signal region has not been looked at in beam-on data. Every plotted “data” point in these documents is fake data.

**Beam-on, beam-off (EXT), dirt** Beam-on data are recorded in the beam spill; beam-off data are recorded between spills and measure cosmic-ray activity; “dirt” is simulated neutrino interactions outside the cryostat whose products enter the detector. The prediction is neutrino simulation + beam-off + dirt.

**Gate ratio** The factor scaling beam-off data to the beam-on exposure: the ratio of the number of beam-on to beam-off trigger gates in a running period.

**Tune, central value (CV)** The MicroBooNE tune is the collaboration’s adjusted GENIE model; the central value is the nominal prediction from it. “CV weight” is the per-event weight that turns the raw simulation into the tuned prediction.

**Safe-weight rule** Any event weight that is non-finite, negative or above 30 is replaced by 1. One rule, applied everywhere, so that the prediction and every pseudo-data throw agree.

**Universe** One alternative version of the prediction with a systematic parameter varied. A *multisim* is a set of hundreds of universes drawn randomly from the parameter uncertainties; a *unisim* is a single  $\pm 1\sigma$  variation of one parameter. PPFX universes vary the flux; GENIE universes the interaction model; reinteraction universes the hadron transport.

**Detector variations (detVar)** Dedicated simulation samples with one aspect of the detector response changed — light yield, recombination, space-charge, wire response — used to estimate the detector uncertainty.

**Migration matrix, efficiency, smearceptance** The migration matrix  $M_{ij}$  is the probability that an event in true bin  $j$  is reconstructed in bin  $i$ ; the efficiency  $\varepsilon_j$  is the fraction of true bin  $j$  that is selected at all. Their product  $S_{ij} = \varepsilon_j M_{ij}$  is the response the unfolding inverts; we call it the smearceptance matrix.

**Migration diagonal** The fraction of events reconstructed in a bin that were truly in that bin (the column-normalised diagonal of  $M$ ). A binning is accepted only if every bin’s diagonal exceeds 0.68.

**Unfolding, Wiener-SVD** Solving for the true spectrum given the reconstructed one and the response. A plain inversion amplifies statistical noise; Wiener-SVD regularises it with a filter built from the simulation, trading a small bias for a large reduction in variance.

**Additional-smearing matrix  $A_C$**  The price of regularisation, made explicit: the unfolded result estimates  $A_C T$  rather than the truth  $T$ . A prediction must be multiplied by  $A_C$  before it is compared with the result.  $A_C$  is released with every extraction.

**Not normalisation-preserving** The rows of  $A_C$  do not sum to one, so the integral of an unfolded differential result is not the total cross section and differs between observables of the same sample. The total is measured separately.

**One-bin extraction** The total cross section, extracted with a single true bin covering the whole phase space. Nothing is regularised,  $A_C = 1$ , and the result is the physical total.

**Conditioning,  $s_{\min}/s_1$ , rank-deficient** The ratio of the smallest to the largest singular value of the regularised response. Near zero, some direction in the true spectrum cannot be recovered from the data at all (rank-deficient); the result in that direction is the prior. A binning can pass the migration-diagonal criterion and still be unusable here.

**Closure** Unfolding pseudo-data and checking that the truth comes back. Closure against the  $A_C$ -smeared truth tests the implementation; it cannot test whether the model is right, because the same model is on both sides.

**Covariance** The matrix of uncertainties and their correlations between bins, one per systematic source, propagated through the unfolding. “Nominal configured covariance” means the

sum over the sources implemented in the analysis; the sources known to be missing are listed in the limitations.

**Flux-averaged cross section** The cross section averaged over the neutrino energy spectrum of the beam, per argon nucleus, in the stated phase space. It depends on the flux, which is why each horn configuration is a separate measurement.

**FHC, RHC** Forward and reverse horn current: the two polarities of the NuMI focusing horns, giving a neutrino-dominated and an antineutrino-enriched beam.

**Beam frame** All angles are measured about the neutrino direction, which at MicroBooNE’s off-axis position is  $28^\circ$  from the detector  $z$  axis.

**Control region, sideband** A sample selected by inverting one requirement of the signal selection, so that it is disjoint from the signal region and enriched in one background. It can be inspected on beam-on data while the signal region stays blind.

**Transverse kinematic imbalance (TKI)** Variables built from the momenta transverse to the neutrino direction of the lepton and the hadronic system:  $\delta p_T$ ,  $\delta \alpha_T$ ,  $\delta \phi_T$  and the inferred nucleon momentum  $p_n$ . They isolate nuclear effects from the unknown neutrino energy.

**LLR particle identification** The log-likelihood-ratio score built from track  $dE/dx$  that separates protons from muons and pions.

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